



USER GUIDE

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DRAFT FOR PUBLIC REVIEW



Building Resilience Index is an innovation of IFC, a member of the World Bank Group.

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OVERVIEW OF BUILDING RESILIENCE INDEX

1. Introduction

An innovation of IFC, a member of the World Bank Group, Building Resilience Index (BRI <http://www.resilienceindex.org/>) is a web-based hazard mapping and resilience assessment framework for the building sector. It is designed to facilitate access to location-specific natural hazard and climate information, provide resilience measures to manage applicable risks, improve transparency for rating and disclosing a building's resilience information between sector stakeholders. BRI makes it easy for the building sector value chain, including construction developers, financial institutions, insurers, and governments to assess, improve, and disclose the resilience of buildings.

The objective of BRI is to provide a simple self-assessment tool for owners and developers to identify and address risks for the property, while simultaneously providing a platform for banks and insurance companies to understand the potential risk exposure of the asset. BRI measures risk and resilience across different sectors (all building types) providing consistent definitions for all actors and increased transparency to ultimately support enhanced resilience in developments for any region and reduced risks for all involved. It also aims to suggest resilient strategies that stakeholders can consider to improve their index. BRI offers a relative rating of resilience from higher to lower and to motivate owners and investors to reduce physical risk, rather than an absolute rating against some standard of performance. It is understood that even the highest levels are not 100% resilient, but retain certain residual risks.

When paired with the green buildings certifications, BRI can help to ultimately de-risk projects and achieve cost-effective means in delivering both climate change mitigation (energy and water consumption efficiency) and adaptation (resilient buildings).

BRI can also be of use and benefit to multiple stakeholders:

Construction Developers

- Assess and improve resilience to site-specific natural and climate hazards
- Disclose resilience rating to financiers, insurers, and users
- Differentiate brand for developers of resilient buildings

Banks

- Make informed investments considering natural hazards and climate risks on buildings
- Save time and resources on project evaluation processes
- Reduce property investor risk exposure

Insurance Companies

- Complement catastrophe modeling with a multi-hazard approach
- Inform underwriting with resilience rating information of assets

- Improve the site specific risks data inputs while saving time and resources on project evaluation processes

Governments & Local Authorities

- Create skills and jobs in the market for more resilient construction practices
- Reduce repetitive costs of post-disaster recovery and reconstruction
- Create an enabling environment for mainstreaming resilient buildings

Property Buyers & Owners

- Make informed investment or retrofit decisions
- Learn the resilience value of your investment
- Minimize operational disruptions and insurance costs

Occupants & Lessors

- Choose to live and work in safer buildings
- Minimize operational disruptions
- Reduce risk of operational and financial losses due to natural disasters and climate change

2. Key Principles

The following key principles guide the development of BRI:

Building Level Focus

Hazards are examined from perspective of a building's exposure, rather than human perspective or physical origin. Hence BRI is organized around the four major hazard categories that buildings experience: wind (air motion), water (liquid motion), fire (rapid oxidation), and geo-seismic (ground motion). BRI rates a building's risk reduction measures for physical integrity to withstand these hazards. It also rates a building for operational continuity measure to address during events of these physical hazards, plus enhancing the operational continuity to withstand a fifth major hazard category of climate hazards (extreme precipitation, drought, and heat) and continue to function.

Hazard Scope and Approach

Site development related to flooding, building fire, wind downburst and ground subsidence can occur most anywhere. Even deserts require drainage design to address flooding. These four hazards in BRI require evaluation for all projects, with basic protections applied as specified. Other natural hazard-based wind, flood, wildfire, and geo-seismic hazard exposure varies by geographic locations. These are identified with the BRI application global hazard maps, searchable by address, coordinate lookup, or mouse cursor.

The mapped hazard probability has only indirect bearing on a BRI rating. A hazard is designated as applicable to a particular location based upon mapped exposure exceeding a rational assigned specific minimum intensity threshold. Most of the site hazards exposure identified by BRI are from assigned applicability thresholds using probabilistic maps. Others

are event based or deterministic. For buildings identified to have an applicable hazard, BRI then assigns individual resilience measures with performance criteria independent of any target aggregate hazard probability. In addition, BRI ratings can be aggregated to build portfolios, for asset risk management over long periods. Portfolio risk metrics are useful for decision making in this setting by insurers, portfolio managers, and governments. BRI assessment is of an individual building's performance. A BRI rating indicates a level of protection where all identified applicable local hazards are assigned measures to adequately mitigate the effects, with some conservatism. Each designated resilience rating is of progressively higher performance from B to A to AA, the highest BRI rating. Resilience for human-induced building hazards such as biohazards, security threats, arson, bombs, etc. are not within the scope of the BRI.

Prerequisites

BRI rating is of risk reduction measures: it is important to underline that BRI is a tool to assess the existence of specific risk reduction measures and organize them in a way to assign an overall rating by individual and combined hazards. It is not a direct assessment of the building's performance, though some of the measures have performance criteria. BRI therefore does not predict or model building behavior during natural disasters or climate events,

Compliance with local regulations: BRI does not substitute, but complements most local codes and regulations, to both protect occupants and preserve buildings. BRI does not apply to informal building sector/unplanned developments but could be used to renovate such developments to make them code-compliant. Buildings applying BRI must comply with any local planning or building regulations or codes related to any of the hazards applicable to the site. In addition, designs, such as structures, drainage, or permeable surfaces, must be implemented correctly during the construction stage. Therefore, onsite construction inspections must ensure that the building adheres to specified standards, codes and BRI resilience measure. This is an essential part of the BRI verification process.

Impacts on the surrounding area: BRI does not rate urban or community resilient but is a key building block and contributes towards community resilience given buildings constitute about 80% of the value of the built environment. Thus, BRI can be used as one of the components for rating resilient communities and infrastructure. New developments can impact the surrounding area, and these impacts should be minimized as part of the design. For example, new developments can elevate the risk of urban floods through the changes in land-use and land cover, which increases pressure on drainage systems. Therefore, the impact of the development on existing drainage systems and any alterations in the landscape on the water cycle should be evaluated.

Assessment Scope

All types of buildings can use BRI: residential, commercial, institutional, and public. These can be new buildings in planning or design, buildings that may be renovated or existing buildings. Emergency response and life-saving escape plans are assumed to be part of local building regulations. If a building can maintain its physical integrity in a hazard event, it is

more likely, but not guaranteed, that occupants will be protected. Emergency response and training is a critical part of hazard safety.

Defensive infrastructure outside the property perimeter is not a factor in BRI. Although defensive infrastructure can affect hazard levels, much infrastructure was designed and built using historic standards and may be outdated to withstand current and future hazards. It also may not have been maintained sufficiently. A building needs to withstand applicable hazards, even when defensive infrastructure malfunctions or fails.

Residual Risk / Insurance coverage is not a factor in BRI. Although insurance is recommended for all buildings, in order to cover residual risk, BRI aims to address physical risk. In the case of any residual risk, transfer or sharing the risk with insurance companies should be considered. Buildings with high ratings may benefit from better insurance availability and pricing. In any project, when wind, water, fire, and geo-seismic hazard impacts cannot be fully managed, insurance policies should be purchased to alleviate the associated financial risks.

Insurance policies that can be considered for purchase are:

- **Builders risk insurance** covering the impact of wind, water, fire and geo-seismic hazards at design and construction stage;
- **Long term hazard insurance** covering the applicable hazard impacts to protect against the financial risk of damage of the building.

For insurance companies, a higher BRI rating signals lower risk. Such buildings may qualify for lower premiums. However, insurance providers use their own proprietary methods for pricing, which in some jurisdictions is regulated.

Methodology

BRI assigns hazard applicability to a site considering empirical, code based, and judgement-based hazard intensity thresholds where damaging effects become apparent, as a practical approach. The tool uses a variety of publicly available hazard datasets including climate data developed by the World Bank, standard by IFC, and reliable third-party data. The source of each dataset currently used by BRI is provided in Appendix A and also disclosed on the application Explorer map view. It may evolve overtime, based on further refinement or availability of improved updated data. Assigned applicability thresholds are generally above hazard levels for minimum performance requirements of "low code" design risk categories in many local building codes whose focus is for minimum life safety protection, but not necessarily to preserve the asset. All rated buildings need to be compliant with local codes.

BRI letter ratings are based on metrics related to the quantity and levels of risk reduction measures of the building when exposed to hazard. Earning the "+" in addition to the letter rating requires operational continuity measures to be included, such as backup power, water, telecom, and emergency access and planning.

BRI is a relative index, in other words, buildings having higher indices will be more resilient than those with lower. However, there may be certain conditions when this does not apply. It should not be viewed as an absolute measure against an external performance standard.

The overall BRI rating is based on the 'weakest link' principle. That means that the risk to all applicable local hazards must be reduced in order to achieve overall resilience. The levels of different hazards are not mathematically combined to extract overall resilience. The building's resilience cannot be higher than the weakest level vis-a-vis any applicable hazard. This is also known as the 'min-max rule.' The overall rating will be accompanied by scores for each applicable hazard, some of which may be higher than the Index. To improve its overall rating, a building will need to improve its 'weakest link' measures.

The resilience rating level of a building for any particular hazard in BRI is determined by the applicant self-declaring and then verifying inclusion of risk reduction measures identified by the application. The measures incorporate global empirical best practices, code-based local standards, and good general practices. Some of these may be region-specific or country-specific. BRI also provides information on how the building could reach the next higher level. This enables the owner or representatives to assess the incremental cost of changes needed to achieve a higher rating.

Data Transparency

Given that resilience transparency is important to all of society, BRI verified ratings may be posted and available on a public database maintained by the application, unless otherwise requested by the building owner to maintain the privacy, in which case the building will be considered Not Resilient (NR).

Quality Assurance & Updates

Resilience depends on resilient design features having been properly constructed. Although verifications provide a level of assurance, they cannot reveal all errors, omissions and defects that may have occurred during design and construction. Resilience depends on resilience features being properly maintained over time and occasionally updated. Also, significant changes in the building's use, compared to its design assumptions, and major remodeling or modifications can affect its resilience. Therefore, the Index will need to be renewed every 10 years, or whenever major use, modification, or repair changes or damage events occur. A BRI assessment can be revised up or down, if at any time new information regarding the hazard or building become known, is supplied by the owner and/or developer, or otherwise uncovered, provided it includes the necessary documentation for verifiers.

Liability Risk

The liability of design and construction of buildings will remain with the project owner, design, and development team. BRI terms of use and privacy policy is applicable to all buildings assessed, rated, or verified with the BRI application. The design and construction team including the developer and any construction company (contractor) is responsible of the construction quality. The liability of construction quality stays with them and not with the financial institutions nor IFC, who do not have the liability to the construction quality.

In no event will IFC, its affiliates, business partners, and service providers or any of our and their respective licensors, shareholders, employees, agents, officers or directors, be liable for damages of any kind, under any legal theory, arising out of or in connection with your use, or

inability to use, the application, forms or documents, the site, any websites linked to it, any content on the site or such other websites or any services or items obtained through the site or such other websites, including any direct, indirect, special, incidental, consequential or punitive damages, including but not limited to, personal injury, pain and suffering, emotional distress, loss of revenue, loss of profits, loss of business or anticipated savings, loss of use, loss of goodwill, loss of data, and whether caused by tort (including negligence), breach of contract or otherwise, even if foreseeable. The foregoing does not affect any liability which cannot be excluded or limited under applicable law.

3. Building Resilience Index Methodology

3.1. Hazard Categories and Types

As described in the Key Principles, BRI is organized around the four major hazard categories with physical impacts that can damage exposed buildings: wind (air motion), water (liquid motion), fire (rapid oxidation), and geo-seismic (ground motion). Under these are a total of 13 sub-categories. In addition, BRI has a fifth hazard category for Climate Hazards, with 3 sub-categories affecting a buildings operational resilience (drought, extreme heat, precipitation), as shown in Table 11 below. Most hazards are determined by the location; however, four sub-categories of physical impact hazards can exist anywhere: local floods, downbursts, local (building) fires, and subsidence. Thus these are acknowledged as “Default Hazards” by BRI, requiring design consideration at every site. Hazards can be interrelated and consequential. For example, storms can cause flood and storm surges, and earthquakes, coastal landslides can cause tsunamis, volcanos can cause fires, and precipitation can cause landslides.

Depending on the location of a project, the project may be exposed to multiple hazards and can face a combination of hazards concurrently. Therefore, it is best to identify location-specific hazard at each specific project level. The application searches global hazard maps to determine exposure and applicability of each hazard category at a given site, using publicly available datasets accessible to IFC for use free of charge, including those developed by reliable third parties and those developed by the institution itself.

Most hazards in BRI will be affected in the future by climate change, particularly wind and water hazards, as well as wildfires. The BRI app has scenario assessment capabilities for future climate conditions. For example, current and future climate change heat stress is provided as a temperature hazard layer in BRI with the options to address 2050 and 2080 time horizons from models of projected future conditions. , applicable under the operational continuity measures. It is to be used for designing appropriate efficient building cooling systems to address current conditions, and with adaptive plans for future heat conditions and associated drought. They should be properly maintained over the life of the building and be configured for appropriate backup power supply for emergency needs.

The assessment framework also includes some hazard factors which are not directly categorized under climate change, but are considered high risk for the building sector, including local fires and geo-seismic risks. Some hazards, or the magnitude of hazards, may

not be appropriately identified if only historical climate indices and records are considered, and thus cannot appropriately included in the project design. For example, a location with no flooding history may become a flood-prone zone due to changes in the land-use, precipitation regimes or sea level. Therefore, BRI takes into account available downscaled data providing climate change projections for 2050 and 2080.

By being conservative and incorporating site location and topography into its hazard applicability and risk reduction measures, BRI encourages buildings to invest high performance building mitigations and standards, thus creating a safety margin for climate uncertainty.

Table 1. Summary of hazard categories and subcategories

*Hazard Category	Hazard Type (Subtype)	Description	Affected by	
			Climate	Location
WIND Change in Wind Speed	Downbursts*	Downbursts are short duration/straight line high velocity vertical wind events with thunderstorms. Also consider less severe surface winds i.e. shear, downslope gradients or funneling.	Yes	Yes
	Storms (Cyclone, Typhoon, Hurricane),	Storms are long duration/distance cyclical;	Yes	Yes
	Tornados	Tornados are short duration/distance cyclical;	Yes	Yes
WATER Change in, Water Flow and Level	Local / Urban Flood*	Ste development, grading, basements, retaining walls and storm drainage design can creat local flooding or exacerbate flooding in surrounds, including urban areas..	Yes	Partly
	River / Lake / Flash Flood	Inland flood can result from precipitation, ice and melt. Extreme precipitation causing flood can even occur in deserts.	Yes	Yes
	Coastal Storm Surge / Tidal Flood	Storm surge can result from storm events and worsen by tide and sea level rise in coastal areas and tidal river waters.	Yes	Yes
	Tsunami	Tsunamis can result from geo-seismic events in coastal areas	No	Yes
FIRE Heat, Dryness	Local (Building) Fire*	Local fires can occur anywhere, from internal or external causes.	Partly	Partly
	Wildfire	Wildfires occur near urban wildlife interfaces. Building risk is influenced by precipitation, temperature, and vegetation.	Yes	Yes
GEO-SEISMIC Land motion	Subsidence*	Subsidence is caused by a range of human and natural factors such as extraction of groundwater and underground resources, earthquakes, surface erosion and internal erosion (piping fluids)	Partly	Yes
	Earthquake	Earthquakes occur predominantly in tectonic plate interface areas, but also near geological fault lines	No	Yes
	Volcano	Active volcanos are predominantly in tectonic plate interfaces	No	Yes
	Landslides	Landslides can be caused by earthquakes, extreme precipitation and melting glaciers typically in loosely-vegetated slopes	Partly	Yes
CLIMATE HAZARDS Affect Operational Resilience	Precipitation: Annual, consecutive days, >20mm	Precipitation levels affect building exposure, and associated flood, erosion and temperature. Increasing variability of intensity and fluctuations is associated with future climate projections.	Yes	Yes
	Drought	Lack of precipitation coupled with extreme heat can degrade aquifers and stress reservoirs, affecting water supply of buldings	Yes	Yes
	Extreme Heat: Surface Temp, #days (>35°C)	Temperature can affect a buildings insulation, ventilation, operating system and maintenance. Future climate changes include increased variability and intensity.	Yes	Yes

*4 default hazards, applicable at each site.

3.2. Rating System

BRI uses the rating scale as shown in **Figure 1**. A rating refers to the collective risk reduction measures that the building incorporates and is not a prediction (or much less a guarantee or certification) of precise building behavior to be expected during a natural disaster or climate change event. Each disaster has unique force, duration and combinations of other event characteristics. The rating system uses letters similar to a bond rating, with associated ranges of assigned preliminary estimated probable maximum loss (PML). The assembly of resilience measures for a given rating and hazard is based upon generalized past performance of the individual measures. The ranges for each rating were assigned as a rational scale, noting the boundaries in terms of PML are not absolute for individual measures or combinations, but represent ranges of performance of the majority of data for given measures.

The BRI ratings descriptions in Table 2 below, are computed outputs of the application, further explained in this section, and then with hazard specific criteria in following sections under Physical Integrity and Operational Continuity. Presently the probable maximum losses shown are indicative, as future experience with rated building levels will provide added data.



* Probable Maximum Loss (PML) current replacement cost, including structural and equipment, excluding operational costs

Figure 1. BRI aims to report and improve the relative resilience of buildings

Table 2. BRI rating level descriptions

Level	Probable maximum loss	Definition
NR	>50%	The building fails to incorporate at least one or more recommended resilience measures of BRI at a basic level.
B	~30%-50%	The building incorporates recommended resilience measures of BRI at a basic level.

A	~10%-30%	The building incorporates recommended resilience measures of BRI at a strong level.
AA	~5%-15%	The building incorporates all recommended resilience measures of BRI at a global best practices level.

Note: Probable maximum loss ranges are indicative preliminary estimates. Projects with ratings NR, B and A will have a range of risks of structural damage and residual risks (non-structural, operational, and core functions for business continuity). For the AA rating, a building is expected to have the lowest structural damage, and also lowest residual risks.

3.3. Hazard Category and Building Rating

The following flow chart outlines the process that will be followed to determine the rating per hazard category.

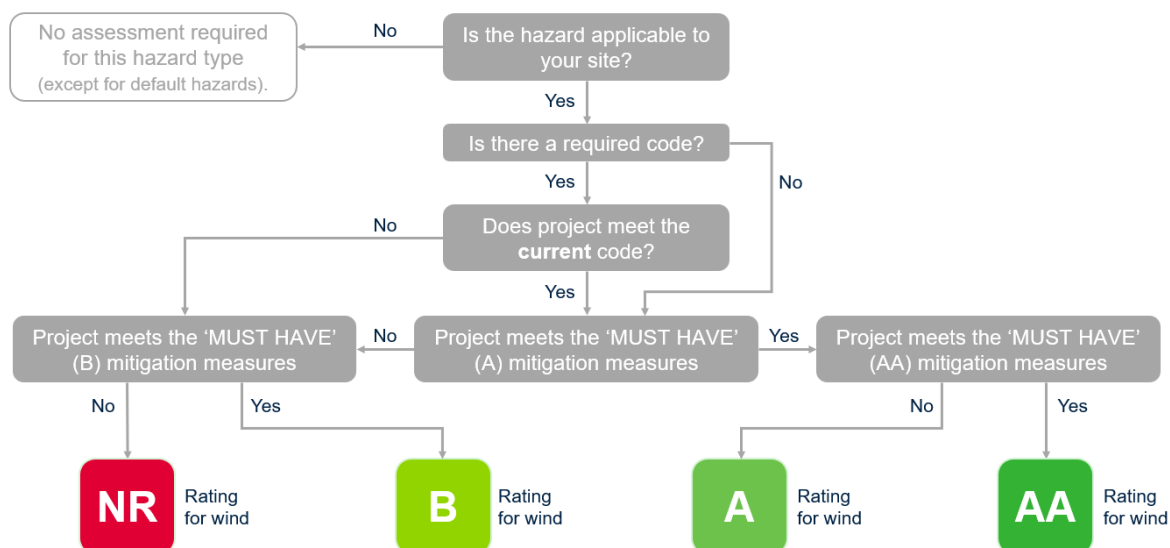


Figure 2. Rating process flow chart per hazard category

Figure 2 in short, requires that both BRI criteria and local building code hazard provisions must be satisfied. Whichever is more stringent prevails. Likewise, where improved hazard modeling or design phase site specific risk analysis (often allowed in building code provisions), shows a hazard exposure greater than BRI, the updated findings must be used for the design basis. If it shows a hazard exposure lower than BRI, then the proposed updated design performance criteria must provide an equivalent conservatism, (i.e. factor of safety or equivalent probability) to the mapped “raw” hazard which the BRI rating threshold provides above for the BRI mapped hazard probability return period.

Figures 3-5 present the “Weakest Link Principle,” where the building rating is determined by the lowest score achieve per hazard category. For example, if three out of the four hazard categories achieve rating of A, but in category the rating is B, then the overall building rating

will be B. A building which achieved a rating above NR on all four hazard categories can obtain a “+” next to the rating, such as A+, by implementing at least three operational continuity measures.

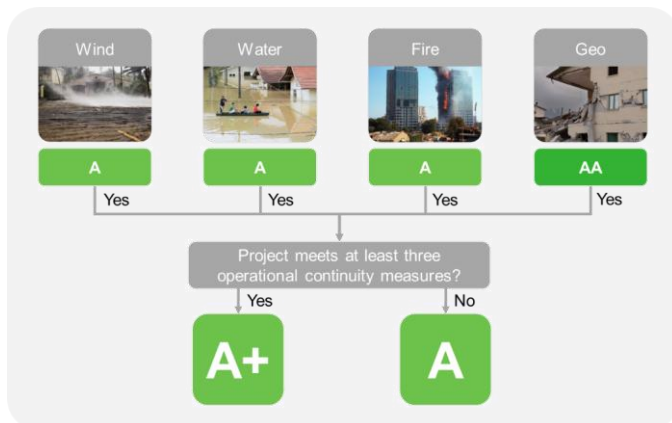


Figure 3. Scenario for a project with operational continuity

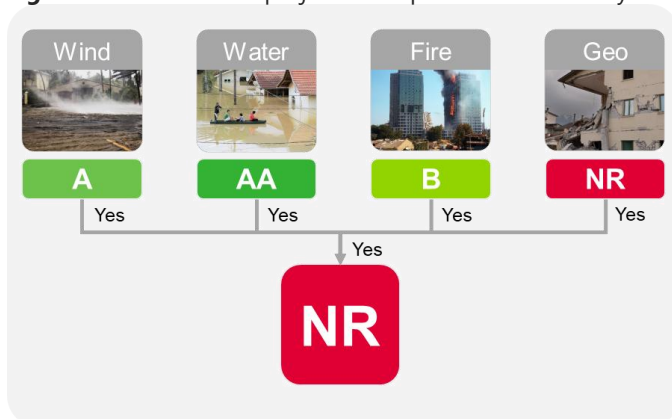


Figure 4. Scenario for NR rated projects

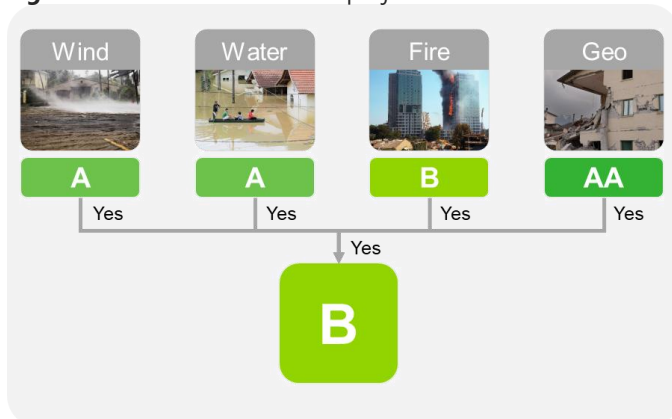


Figure 5. Scenario for B rated projects

3.4. Process

The BRI rating process has two steps, which are sequential:

- **Self-Assessment.** A representative of the building owner or project developer (referred to as the “Assessor”) can initiate the BRI assessment by inputting the building information and indicating which BRI mitigation and continuity measures the building has, as noted in this guide for each measure under ‘Assessor Procedure’. The Assessor can be a member of the building or project’s in-house staff, or its design and code-responsible technical team.

The Assessor should also collect related documents (permits, maps, drawings, product specs, inspection reports, design engineer and contractor contact information, etc) needed to verify the correctness of the inputs and supply them to the Verifiers. If a not-applicable (N/A) reply is provided, the Assessor should clearly note the reason for this.

Planned or under construction buildings will be given a preliminary design rating, with the final rating given following post-construction verification when the building has received its official occupancy permit.

- **Verification.** All buildings and projects must be verified for the BRI rating to become official, after the Assessor completes the assessment. When ready for verification, the owner/developer will select a verification provider company or two individual verifiers (one of which should be a civil or structural engineer), add them at Verifier access level to the project, and request verification. Then upon verification of each risk reduction measure by two individual Verifiers, the Project gets verified (Figure 6).

Verifiers should physically visit the site, in addition to reviewing documents and talking to the Assessor and other building parties. Both Verifiers need to agree on each individual measure and can discuss these with each other to complete the process. They should also verify the correctness of any N/A selections.

When visiting older buildings, verifiers should also note visible structural cracks, corrosion, subsidence and water stains, discuss them with the owner and satisfy themselves that the owner has a plan to properly address them. If a Verifier considers these to be serious, this should be noted and the BRI rating will await completion of any corrective actions.

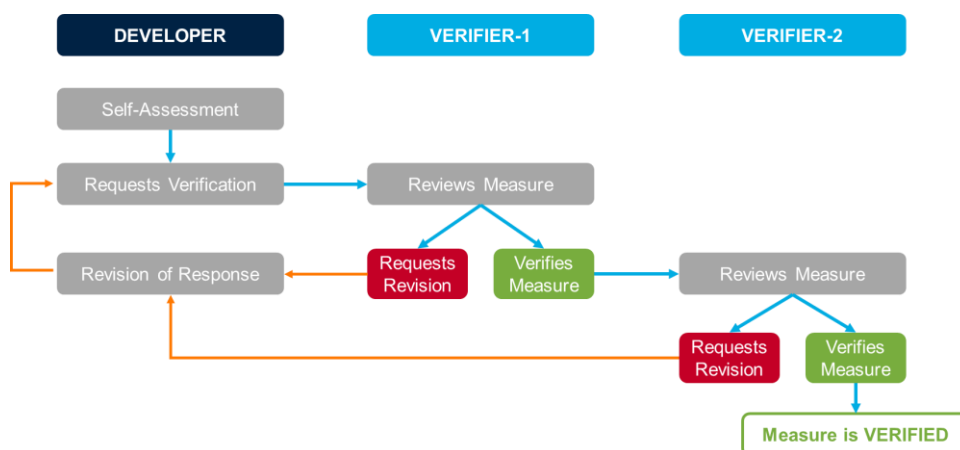


Figure 6. Verification process for each measure

For periodic updates to the BRI platform including assessor and verifier procedural clarifications, see the release notes at the bottom of the BRI website home page::
<https://resilienceindex.org> Qualifications of Assessors and Verifiers are presented in the BRI Governance Protocol, also linked on the BRI website learning page.

PHYSICAL INTEGRITY

1. Wind

1.1. Hazard Types

There are three types of Wind Hazards: Storms (known in different parts of the world as Cyclone, Typhoon, Hurricane), Tornados and Downbursts (also known as Derechos). Each are discussed in detail. Downbursts are considered to be the default wind hazard, meaning they are applicable for all locations.

To be rated AA, buildings need to meet, or exceed the below wind speeds.

Table 3. AA level for wind hazards

Hazard Type	AA Level	B Level
Typhoon, Hurricane or Cyclone	290 km/h (180 mph)	200 km/h (125 mph)
Tornado	250 km/h (155 mph)	200 km/h (125 mph)
Downburst	200 km/h (125 mph)	130 km/h (80 mph)

Buildings in storm and tornado-applicable locations will be rated NR, if they cannot withstand a minimum 200 km/h (125mph) wind speeds, the B level requirement (see Table 3 and Table 5). And all buildings outside storm and tornado-applicable locations must meet 130km/h to meet the minimum B level requirement, and for AA rating, it is 200 km/h (125 mph) this being the criteria for downbursts, the universal default applicability.

1.1.1. Storm (Cyclone, Typhoon, Hurricane)

Definition

Severe winds and storms are referred to by different terms and occur at different times of the year and depending on geography. Hurricanes develop between June and November, in the north Atlantic (Caribbean) and the northeast Pacific Oceans. Typhoons occur from May to November in the northwestern Pacific Ocean. Cyclones develop in the south Pacific and Indian Oceans and the season normally runs from November to April. However, in the Northern Indian Ocean (Bay of Bengal and the Arabian Sea), cyclones do not have an official season but tend to occur from April to June, and September to November.

Storms (tropical) form when air is heated by warm ocean water near the Equator. This hot air rises from near the surface of the sea causing an area of low pressure. Air from surrounding areas then moves to the area of low pressure and in turn becomes warm, moist and rises too. As the air rises it cools and forms clouds and the whole system grows and spins. As the storm starts to spin more quickly it forms an 'eye' in the center which has calm, low pressure conditions. High pressure air which surrounds the eye flows down.

When the wind speed is below 61-63 km/h, they are known as tropical depressions. When the sustained high wind speed is more than 63 km/h, they are known as tropical storms or

cyclones. For more information on naming, please see Table 4 below or the World Meteorological Organization website. When wind speeds reach 119 km/h (74mph) they are termed a tropical cyclone, hurricane or typhoon. For the purpose of this document, they are referred to as 'Storms'.

According to most climate models, it is likely that climate change will increase the severity of tropical cyclones and storms, in terms of wind speed and rainfall, whilst the global average frequency of tropical storms is likely to decrease or remain unchanged¹ However, the IPCC reports a predicted increase in both the frequency and the intensity of the most intense storms with stronger winds and more intense rainfall rates caused by climatic change². The present hazard level in areas currently affected by tropical storms may increase in the long-term. Projects located in such areas should be designed to account for future increases in storms.

Table 4. Summary of the classifications for tropical cyclones used by official warning centers worldwide

Tropical cyclone classifications								
Beaufort Scale	1-minute sustained winds	10-minute sustained winds	NE Pacific & N Atlantic (NHC/CPHC)	NW Pacific (JTWC)	NW Pacific (JMA)	N Indian Ocean (IMD)	SW Indian Ocean (MF)	Australia & S Pacific (BOMS/FMS)
0-7	<32 knots (59 km/h)	<28 knots (52 km/h)	Tropical Depression	Tropical Depression	Tropical Depression	Depression	Zone of Disturbed Weather	Tropical Disturbance
7	33 knots (61 km/h)	28-29 knots (52-54 km/h)				Deep Depression	Tropical Disturbance	Tropical Depression
8-9	34-37 knots (63-69 km/h)	30-33 knots (56-61 km/h)	Tropical Storm	Tropical Storm			Tropical Depression	Tropical Low
9-10	38-54 knots (70-100 km/h)	34-47 knots (63-87 km/h)			Tropical Storm	Cyclonic Storm	Moderate	Category 1
							Tropical Storm	Tropical Cyclone
10-11	55-63 knots (102-117 km/h)	48-55 knots (89-102 km/h)			Severe	Severe	Severe	Category 2
12+	64-71 knots (119-131 km/h)	56-63 knots (104-117 km/h)	Category 1	Typhoon	Tropical Storm	Cyclonic Storm	Tropical Storm	Tropical Cyclone
	72-82 knots (133-152 km/h)	64-72 knots (119-133 km/h)	Hurricane		Typhoon	Very Severe	Tropical Cyclone	Category 3
	83-95 knots (154-176 km/h)	73-83 knots (135-154 km/h)	Category 2 Hurricane			Cyclonic Storm		Severe Tropical Cyclone
	96-97 knots (178-180 km/h)	84-85 knots (156-157 km/h)	Category 3		Very Strong Typhoon			
	98-112 knots (181-207 km/h)	86-98 knots (159-181 km/h)	Major Hurricane			Extremely Severe	Intense	Category 4
	113-122 knots (209-226 km/h)	99-107 knots (183-198 km/h)	Category 4			Cyclonic Storm	Tropical Cyclone	Severe Tropical Cyclone

¹ K.J. Walsh et al., "Tropical Cyclones and Climate Change." *Wiley Interdisciplinary Reviews: Climate Change*, 7(1) (2016): 65-89.

² IPCC, *Climate Change 2013: The Physical Science Basis* (Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2013), <https://www.ipcc.ch/report/ar5/wg1/>

Tropical cyclone classifications								
Beaufort Scale	1-minute sustained winds	10-minute sustained winds	NE Pacific & N Atlantic (NHC/CPHC)	NW Pacific (JTWC)	NW Pacific (JMA)	N Indian Ocean (IMD)	SW Indian Ocean (MF)	Australia & S Pacific (BOMS/FMS)
	123–129 knots (228–239 km/h)	108–113 knots (200–209 km/h)	Major Hurricane		Violent Typhoon			Category 5
	130–136 knots (241–252 km/h)	114–119 knots (211–220 km/h)		Super		Super	Very Intense	Severe Tropical Cyclone
	> 137 knots (254 km/h)	> 120 knots (222 km/h)	Category 5 Major Hurricane	Typhoon		Cyclonic Storm	Tropical Cyclone	

Sources: World Meteorological Organization (WMO); National Hurricane Center (NHC); Central Pacific Hurricane Center (CPHC); Joint Typhoon Warning Center (JTWC); Japan Meteorological Agency (JMA); India Meteorological Department (IMD); Météo-France (MF); Bureau of Meteorology (BOM); Fiji Meteorological Service.

1.1.2. Tornado

Definition

Tornados are a violently rotating column of air between a cloud base (thunderstorm) and the ground.³ Although often depicted as a funnel, they can vary greatly in their appearance, wind speed and duration, for example they can be almost stationary or travel more than 100 km with groundspeeds of 90 km/h.⁴ They can last from a few seconds to over an hour. Most strong tornados develop in thunderstorms called supercells which have a deep and persistent updraft.⁵ Tornados are most common in the USA (with on average around 1,000 per year⁶). However, they can extend to Canada and Mexico, and can occur in many other parts of the world, including South America, Australia, Asia and Europe. Bangladesh and Northern Argentina are particularly prone to tornados. The peak of the season in the USA is May to August, although they can occur all year round.

Larger tornados can be predicted partly with Doppler radar, though smaller ones can be harder to predict. Due to the difficulty in forecasting tornados, a combination of radar, meteorological data analysis and visual reports from storm spotters is used. The severity of tornados is measured by estimating the wind speed using the Enhanced Fujita (EF) scale which ranges from 0 to 5. Wind speed isn't necessarily determined by the size of the tornado and all tornados have the potential to cause property damage⁷. This is particularly the case

³ "Severe Weather 101: Tornado Basics," NOAA, accessed April 2026, <https://www.nssl.noaa.gov/education/svrwx101/tornadoes/>

⁴ "Warm Season Weather Hazards," Government of Canada, last modified Aug 18, 2025, <https://www.canada.ca/en/environment-climate-change/services/seasonal-weather-hazards/warm-season-weather-hazards.html>

⁵ FM Global, *Defy the Twister: Tornado Resilience in the Age of Climate Change* (2018), [https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-\(1\).pdf](https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-(1).pdf)

⁶ FM Global, *Defy the Twister: Tornado Resilience in the Age of Climate Change* (2018), [https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-\(1\).pdf](https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-(1).pdf)

⁷ "Warm Season Weather Hazards," Government of Canada, last modified Aug 18, 2025, <https://www.canada.ca/en/environment-climate-change/services/seasonal-weather-hazards/warm-season-weather-hazards.html>

when multiple tornados occur at the same time or in the same area, and are usually accompanied by hail and heavy rainfall which can add to the damage from wind. Damage can be significant as the building is affected by wind from multiple directions rapidly, and the drop-in pressure caused by the tornado can cause upward suction which makes structural failure of roofs and walls more likely⁸. Flying debris in the vortex can also increase damage to the structure.

1.1.3. Downburst

Definition

A downburst a general term to describe an area of wind hazards at or near the ground caused by a strong downdraft (a column of air which rapidly sinks to the ground), usually in a thunderstorm.⁹ The difference with tornados is that downbursts are not associated with rotation (they are sometimes known as straight-line winds). Downbursts can further be divided into macrobursts and microbursts. Macrobursts are downbursts which have a diameter larger than 4 km (2.5 miles).¹⁰ Microbursts are smaller than 4 km (2.5 miles) and are divided into two types: dry and wet. Wet microbursts are accompanied by strong precipitation.¹¹ Another term for downbursts is a *derecho*, which is more long-lasting and widespread (more than 400 km (c. 250m)). *Derechos* occur when meteorological conditions produce multiple downbursts which in turn can contain several microbursts.¹²



Source: NOAA (<https://www.weather.gov/lmk/derecho>)

Figure 7. Climatology for downburst also known as derecho (only USA)

⁸ FM Global, *Defy the Twister: Tornado Resilience in the Age of Climate Change* (2018), [https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-\(1\).pdf](https://www.fm.com/-/media/Files/FM-Global/Whitepapers/W00623_18-Defy-the-Twister-(1).pdf)

⁹ "Severe Weather 101: Types of Damaging Winds," NOAA, accessed February 1, 2020. <https://www.nssl.noaa.gov/education/svrwx101/wind/types/>

¹⁰ "Microbursts," NOAA, accessed February 1, 2020. https://www.weather.gov/bmx/outreach_microbursts

¹¹ "Downbursts," NOAA, accessed February 1, 2020. <https://www.weather.gov/ohx/downbursts>

¹² "About Derechos," NOAA, last modified May 15, 2018. <https://www.spc.noaa.gov/misc/AbtDerechos/derechofacts.htm>

Downbursts can be difficult to predict and detect as they do not occur for a long time, but Doppler radar and meteorological data analysis can be used.¹⁵ Downbursts are more common than tornados and can cause significant damage, including structural collapse of buildings and impact damage from debris. They are often confused with tornados when they are occurring, but after a downburst event, the debris will be in one direction, as opposed to damage from a tornado which will be in different directions.¹⁶

1.2. Best Practice Standards

Global Best Practice (GBP) for high wind hazards is the South Florida High Velocity Building Code¹⁷. This requires reinforced concrete construction and wind speeds of 290 km/h (180mph). BRI applies this standard for AA rating in storm and tornado risk locations, while recognizing National Codes consistent with this GBP.

Most updated National Codes for lower risk category design today are at BRI's B level, even in storm and tornado areas. Therefore, A and AA ratings require going to high risk category of the code and in some cases above code. BRI expects that future codes will continuously improve and move in the direction of higher wind speed design ratings. Increased frequency and intensity of wind hazards caused by climate change will be a major factor of this.

Below for reference is a sampling of code standards that BRI has considered in developing its wind mitigation measures:

The International Code Council (ICC) is a United States association which developed the International Codes (I-Codes), a family of model building codes. Two current standards relate to wind hazards safety and preparedness¹⁸:

- ICC 500-08 ICC/NSSA Standard for the Design and Construction of Storm Shelters
- ICC 600-08 Standard for Residential Construction in High-Wind Regions

The International Property Maintenance Code (IMPC) provides guidance for maintaining existing buildings through model code regulations. It can be accessed at:

<https://codes.iccsafe.org/content/IPMC2021P2>

1.3. Consequential Hazards

Buildings need to also consider the impact of wind-borne debris that can accompany storms and tornados, and inflict additional damage on structures, openings and exterior equipment.

¹⁵ "Microbursts," NOAA, accessed February 1, 2020. https://www.weather.gov/bmx/outreach_microbursts

¹⁶ "Warm Season Weather Hazards," Government of Canada, last modified August 18, 2025, <https://www.canada.ca/en/environment-climate-change/services/seasonal-weather-hazards/warm-season-weather-hazards.html>

¹⁷ ICC, "Chapter 16: Structural Design," in *Florida Building Code: Building, Seventh Edition* (USA: International Code Council, 2020), <https://codes.iccsafe.org/content/FBC2017/chapter-16-structural-design>

¹⁸ "Hurricane Safety," ICC, accessed March 1, 2020, <https://www.iccsafe.org/professional-development/safety2/hurricane-safety/>

1.4. Risk Reduction Measures

The following risk reduction measures list is in the BRI application and form the basis for the BRI wind rating. Measures marked 'Y' apply, if the building is in a location with that particular hazard type. Buildings that do not obtain an AA, A or B rating are rated NR. Each measure needs to be affirmed by two local, independent IFC-registered Verifiers, one of which should be a civil or structural engineer. Each Verifier will be required to upload their license to confirm their qualification. Verifiers should collect and retain for their files whatever information they consider necessary to conduct their work.

Table 5. List of risk reduction measures for wind hazards.

Wind Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Downburst (Default Hazard)	Tornado	Storm (Cyclone, Typhoon, Hurricane)
WD01. Steel Reinforced Concrete Primary Structure and National Code Applicable to Essential Facilities (both apply)	Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings need to be designed to withstand ultimate wind speeds of at least 290 km/h for storms or tornados, and 200 km/h for downbursts/surface winds. This includes any large exterior cladding to the building.	Y			Y	Y	Y
WD02. Primary Structure to National Code Applicable to Essential Facilities	Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings will need to be designed to withstand ultimate wind speed levels of at least 250 km/h or higher for storms or tornados and 160 km/h or higher for downbursts. This includes any large exterior cladding to the building. . This measure is different from WD01 in that it does not require reinforced concrete construction.		Y		Y	Y	Y
WD03. Primary Structure Meeting Structural Code Requirements	Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings will need to be designed to withstand ultimate wind speed levels of at least 200 km/h or higher for storms or tornados and 130 km/h or higher for downbursts. This includes any large exterior cladding to the building. This measure's difference from WD02 is that it does not require meeting wind speed levels and other code provisions specific for essential/critical service buildings as long as the referenced wind levels are met.			Y	Y	Y	Y
WD04. Steel Reinforced Slender Building Elements	Slender building elements (i.e. chimneys) must be designed to meet Meet A level wind criteria of 250kmh		Y		Y	Y	Y
WD05. Steel Reinforced Parapets	Parapets must contain steel reinforcement designed to meet A level wind criteria of 250kmh.		Y		Y	Y	Y

Wind Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Downburst (Default Hazard)	Tornado	Storm (Cyclone, Typhoon, Hurricane)
WD06. Larger Than 1 m ² Openings are Wind-rated	The structure openings must be designed and rated to sustain wind pressure of 290 kmh for AA and 250kmh for A resilience levels.	Y	Y			Y	Y
WD7. Securely Anchored External Fixtures	External fixtures design such as HVAC equipment, generators, lighting, solar panels, and their anchorage need to be rated and installed to meet AA wind levels of 290 km/hr prescribed in WD01.	Y			Y	Y	Y
WD08. Design & Construction Review for Wind Hazard Risk Management Measures	The building design and occupancy must be submitted to and approved by the approving agency, as prescribed by the governing national or local building code. The construction must be observed and documented at critical milestones by the structural engineer that was responsible for the code compliance at design stage.		Y		Y	Y	Y
WD09. Construction Independently Audited for Wind Hazard Risk Management Measures	A second independent qualified engineer , different from the one assigned from the approving agency or the responsible engineer of record for the project, was engaged to inspect the construction of the building during critical milestones. The construction was/will be inspected and approved by an independent qualified engineer or architect, apart from the project designer and the approving agency.	Y			Y	Y	Y

WD01. Steel Reinforced Concrete Primary Structure and National Code Applicable to Essential Facilities (both apply) (AA Rating)

Criteria: Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings need to be designed to withstand ultimate wind speeds of at least 290 km/h for storms or tornados, and 200 km/h for downbursts/surface winds. This includes any large exterior cladding to the building.

Assessor Procedure: Provide the year of building construction permit and applicable wind code provisions for the permit, including wind speed of the code, confirming conformance with the above criteria.

Or provide the stamped design documents, calculations or as-built drawings, confirming conformance with the above criteria.

Or provide written confirmation from the design structural engineer that the above criteria is satisfied.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: The global best practice for mitigating very high wind hazards is the use of steel reinforced concrete (poured, precast and/or block) for the primary structure¹⁹, which includes roof, columns, beams, load bearing walls, floor framing,

foundations and footings and connected load paths.

To achieve an AA rating this measure applies to all buildings (residential, commercial, offices, retail, hotels, or recreational), regardless of being an essential/critical service building or not.

A properly constructed steel-reinforced concrete structure will likely meet the wind load of 290km/h, even if the national code to which it was built prescribes a lower wind speed, and therefore satisfy this measure.

Wind speeds noted in BRI are 'ultimate wind speeds' and are higher than 'nominal design wind speeds' that are often referenced by National Codes. Ultimate wind speeds closely approximate 3-second wind gusts measured in storms. Nominal design wind speeds are used for engineering wind load calculations.

To obtain the nominal wind speed, multiply the ultimate wind speed by the square root of 0.6, or 0.7746. A 290 km/h ultimate wind speed corresponds to a 225 km/h nominal design wind speed.

Many National Codes, even in storm and tornado applicable areas, may not require the wind speeds specified for BRI's AA and A levels, or require them for critical structures. BRI Assessors and Verifiers should ask the structural design engineer should what the conservative maximum wind speed for the building is.

Note that the roof is the most wind-vulnerable part of a building. High wind

¹⁹ **Structural elements:** The main components of a building such as columns, beams, load bearing walls, floor plates, foundations, trusses, and roof structures. These are

the basic skeleton of a building which withstand the gravitational forces put upon a building. The structure is the frame which all other coverings are applied to.

speeds can result in extensive damage if the roof is not well-connected to the building structure. Uplift forces can pull out and remove cover materials.

Damages may include: covering becoming loose (i.e. gravel) causing damage to windows and doors; metal panels pulled off; loosening of the roof flashing; damage to gable ends; and deck being lifted off the frame. Roof damage can allow water to enter the building.

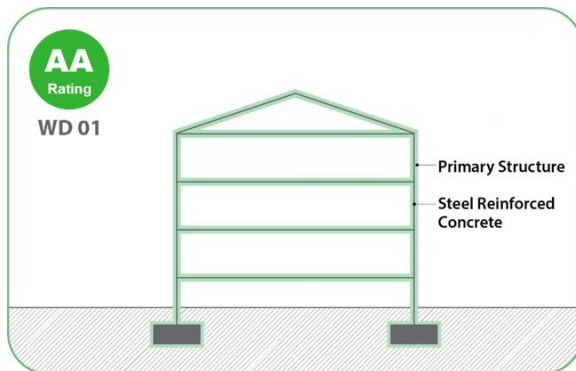


Figure 8. Steel reinforced concrete for the primary structure

The shape and weight of the roof can affect how vulnerable it is to high winds. Low profile hip roofs (all sides slope/inclined and no vertical sections) with narrow overhangs, as well as flat roofs tend to resist better.

Therefore we are labeling measure WD-01A to be the provision of a properly constructed steel-reinforced concrete roof to meet the wind load of 290 km/h to satisfy this measure.

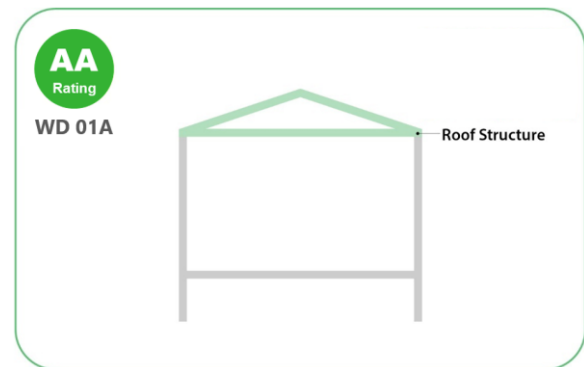


Figure 9. Roof built to 290+ km/h

WD02. Primary Structure to National Code Applicable to Essential Facilities (A Rating)

Criteria: Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings will need to be designed to withstand ultimate wind speed levels of at least 250 km/h or higher for storms or tornados and 160 km/h or higher for downbursts. This includes any large exterior cladding to the building.

This measure is different from WD01 in that it does not require reinforced concrete construction.

Assessor Procedure: Provide the year of building construction permit and applicable wind code provisions for the permit, including wind speed of the code, confirming conformance with the above criteria.

Or provide the stamped design documents, calculations or as-built drawings, confirming conformance with the above criteria.

Or provide written confirmation from the design structural engineer that the above criteria is satisfied.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: This measure is similar to WD01, except that this does not require steel reinforced concrete construction and lowers the wind speed for downbursts to 160 km/h. The primary structure can be built of metal, wood, composite or any other material allowed by the national code. However, to achieve an A rating this measure applies to all buildings

(residential, commercial, offices, retail, hotels, or recreational), regardless of being an essential/critical service building or not.

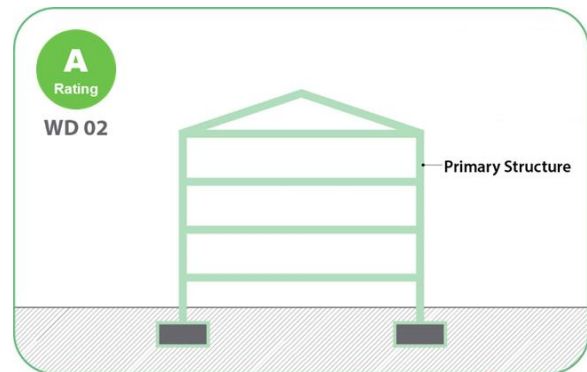


Figure 10. Primary structure built to 250 km/h

Wind hazards can pull buildings apart as they create differences in pressures between the building interior and exterior. Therefore, the structural elements including the columns, beams, foundation, roof, and load-bearing walls must be well connected to each other.

Structures where the roof is well connected to columns and beams, (using internal or external steel or other type of connections), and the columns are well connected to the foundation create a continuous load path and fare better. It is important that this not only be included the design, but also spot checked during construction.

Connections made of ductile materials better respond to wind loads.

This measure includes an additional features WD-02A, being the design of a fully connected structure, emphasizing the importance of structural connections to reduce loss. Assessors and Verifiers should ask the structural design engineer for sample drawings of the design connections, as well as photos from spot-check inspections during construction.

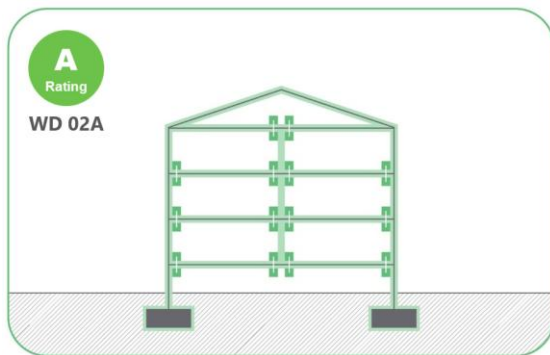


Figure 11. WD 02A - Well-connected structure

WD03. Primary Structure Meeting Structural Code Requirements (B Rating)

Criteria: Primary structural elements, which include columns, beams, load bearing walls, floor framing, and foundations, such as footings will need to be designed to withstand ultimate wind speed levels of at least 200 km/h or higher for storms or tornados and 130 km/h or higher for downbursts. This includes any large exterior cladding to the building.

This measure's difference from WD02 is that it does not require meeting wind speed levels and other code provisions specific for essential/critical service buildings as long as the referenced wind levels are met.

Assessor Procedure: Provide the year of building construction permit and applicable wind code provisions for the permit, including wind speed of the code. Confirm conformance with above criteria.

Or provide the stamped design documents, calculations or as-built drawings, confirm conformance with the above criteria.

Or provide confirmation from design structural engineer that criteria is satisfied.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: Note that this measure is similar to WD02, except the wind levels are at least 200 km/h for storms and tornados, and 130 km/h for downbursts.



Figure 12. Primary structure built to 200 km/h

High winds in storm and tornado events can overturn and collapse unreinforced masonry walls that are over 3 m high.

Brick, concrete or stone masonry walls must be steel reinforced²⁰, or equivalent, to prevent overturn and collapse from 200 km/hr winds. See figure WD 03A

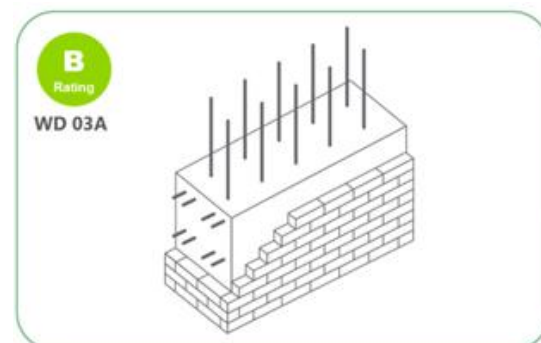


Figure 13. Steel reinforced for main building

Assessors and Verifiers should ask the structural design engineer if there are wall sections (between beams) over 3 m high. If yes, then to request sample drawings of the reinforcement, as well as photos from spot-check inspections during construction.

This measure differs from WD-02 in that it does not need to meet requirements for an essential facility.

²⁰ **Steel Reinforcement:** Known as 'Rebar,' a steel bar or mesh of steel wires used to reinforce/stabilize masonry,

concrete, or stone structures, holding the masonry in tension.

WD04. Steel Reinforced Slender Building Elements (A Rating)

Criteria: Slender building elements must be designed to meet Meet A level wind criteria of 250kmh

Assessor Procedure: Provide the stamped design documents, calculations or as-built drawings, confirming conformance with the above criteria.

Or provide written confirmation from the design structural engineer that the above criteria is satisfied.

Verifier Procedure: Review documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: Slender building elements, for instance chimneys, as well as large pipe vents of mechanical rooms or fireplaces,

can experience very high wind loads, as they sit both above the roofline and sometimes on the outer wall of a building, causing them to potential collapse or topple.

Therefore, slender elements must be reinforced with steel equivalent to resist toppling and collapse forces for the wind loads and be well connected to the wall they are attached to. In case of fabricated metal construction, they should be securely connected to the building structure.

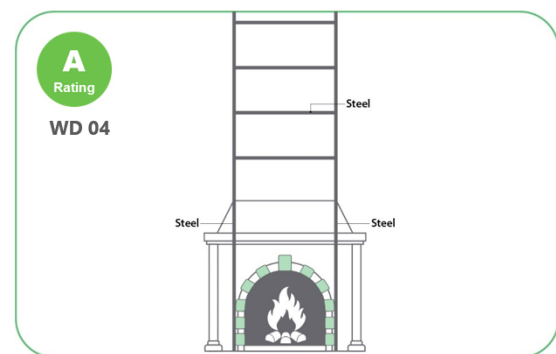


Figure 14. Reinforced fireplaces and chimneys

WD05. Steel Reinforced Parapets (A Rating)

Criteria: Parapets must contain steel reinforcement designed to meet A level wind criteria of 250kmh.

Assessor Procedure: Provide the stamped design documents, calculations or as-built drawings, confirming conformance with the above criteria.

Or provide written confirmation from the design structural engineer that the above criteria is satisfied.

Verifier Procedure: Review documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: Parapets are sections of the building exterior wall that extend above the base of the roof. They act as a barrier for human fall safety and roof debris, as well as for insulation, better roof wind resistance, fire protection and visual aesthetics

Parapets in the building must be constructed with steel reinforcement equivalent to resist design wind loadings, and connected to the main structure.

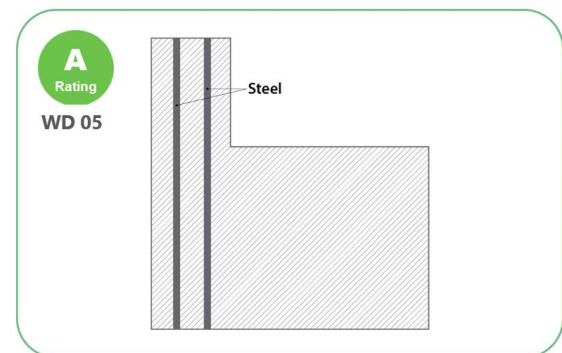


Figure 15. Steel reinforced parapets

WD06. Larger Than 1 m2 Openings are Wind-rated (AA & A Rating)

Criteria: The structure openings must be designed and rated to sustain wind pressure of 290 kmh for AA and 250kmh for A resilience levels.

Assessor Procedure: Provide supplier, or manufacturer and installer documentation of the external windows and doors showing rated wind speed meeting the criteria above and confirmation of proper installation.

Or provide photographic evidence of the manufacturer labeling (etching, watermark, etc) of the window or door design pressure rating to conform to the above wind criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Confirm visual inspection for a representative number of these materials during the site inspection.

Background: Openings in a building structure, such as windows and doors, or façade are points of weakness for wind to enter a building and apply forces in areas not designed to withstand them.

In areas prone to storms and tornados, windows, doors, curtain walls and garage doors, or coverings, must be wind-rated and impact-resistant to withstand the wind levels of 290km/hrprescribed in WD01 for AA rating and 250km/hr WD02 for A rating.

Assessors and Verifiers should ask the building design professionals to provide a sample of manufacturers' ratings, and if not available, spot check the rating information labels on a sample of windows and doors.



Figure 16. Wind-rated / impact resistant openings

WD07. Securely Anchored External Fixtures (AA Rating)

Criteria: External fixtures design such as HVAC equipment, generators, lighting, solar panels, and their anchorage need to be rated and installed to meet AA wind levels of 290 km/hr prescribed in WD01

Assessor Procedure: Provide supplier, or manufacturer and installer documentation of the external fixtures and anchorage showing meeting the wind speed criteria above and confirmation of proper installation.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria. In case of missing required documentation the measure shall not be approved, and returned to the assessor to obtain the missing information.

Background: generators , External fixtures can be dislodged and anchorages fail under strong wind causing damage to the building and windows, and as projectiles also damage nearby buildings.

Therefore, all external fixtures that could become windborne debris have to be easily moved to a secure location, or securely anchored to the building structure, foundation or the ground.

Loose gravel surfaces should be avoided for equipment pads or roof surfacing. .

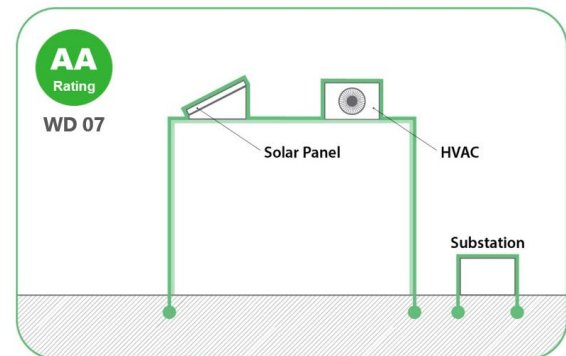


Figure 17. Securely anchored external fixtures

WD08. Design & Construction Review by Government of Wind Hazard Risk Management Measures (A Rating)

Criteria: The building design and occupancy must be submitted to and approved by the approving agency, as prescribed by the governing national or local building code.

The construction must be observed and documented at critical milestones by the structural engineer that was responsible for the code compliance at design stage.

Assessor Procedure: Provide documentation from the approving agency, the certificate of completion or equivalent. Also provide documentation from the structural engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: The construction observation at critical milestones provides a cross-check to the construction contractor QC program, helping to minimize the scope for error and assure that what is constructed was what was initially designed, and will aid verification.

In newer buildings the responsible engineer or architect should be able to provide the assessor with sample photos of inspections during construction milestones that relate to the wind measures herein.

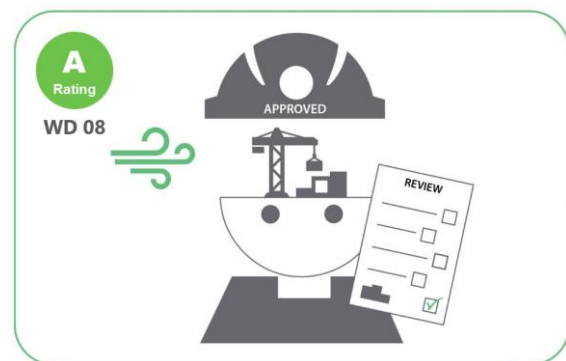


Figure 18. Structural design reviewed by the local approving agency

WD09. Construction Independently Audited for Wind Hazard Risk Management Measures (AA Rating)

Criteria: A second independent qualified engineer , different from the one assigned from the approving agency or the responsible engineer of record for the project, was engaged to inspect the construction of the building during critical milestones.

Assessor Procedure: Provide documentation from the independent qualified engineer attesting to compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: This independent professional should provide the assessor sample photos of milestone inspections and discuss the wind-related recommendations for the building

For new construction the engineer verifier should, when possible, perform the independent construction inspection of critical milestones.



Figure 8. Construction independently audited for wind hazard risk management measures

2. Water

2.1. Hazard Types

There are various types of Water Hazards, Tsunami, Storm Surge and Floods (River/Lake, Coastal/Tidal, Flash and Local/Urban). Each are discussed in detail. A certain degree of water hazard risk is found everywhere, as even deserts can experience local floods. To be rated AA on water section by BRI, the lowest occupied floor and mechanical/electrical equipment of buildings need to be at or above the high-level metric as provided in Table 6.

Table 6. High level (AA) BRI performance criteria for water hazards

Hazard Type	Elevation for BRI High Level (AA)	Comments
Tsunami	15 m (50 ft)	From mean sea level, up to 10 km (6.2 miles) from coast, based on land morphology.
Storm surge	6 m (20 ft)	From mean sea level, up to 2 km (1.2 miles) from coast, based on land morphology.
River/lake flood	6 m (20 ft)	From mean river/lake level, up to 2 km (1.2 miles) from river/lake, based on land morphology.
Flash flood	6 m (20 ft)	2 km (1.2 miles) from low point of terrain, based on land morphology.
Coastal/Tidal flood	6 m (20 ft)	From mean sea level, up to 0.5 km (0.3 miles) from coast, based on land morphology.
Local/Urban flood	6 m (10 ft)	From low point of terrain within 50 m (160 ft) radius, based on land morphology.

For an AA rating the minimum high level for non-tsunami risk areas including local urban flooding is set at 6 m (20 ft). BRI views that all locations are subject to local/urban flooding and any buildings with an occupied level less than the B level rating which is 3 m (10 ft) from low point of terrain within 50 m (160 ft) radius, based on land morphology will be rated NR. See Table 7 for more details.

Building disasters from water hazards have escalated in recent decades, due to more development in risky locations, lagging regulations and climate change. BRI thus takes a conservative approach in both marking applicable locations and mitigation measures, setting these minimum building heights in flood prone areas to include a factor of safety or “freeboard” above mapped flood levels, which increase with higher BRI ratings.

Since water hazards have the greatest climate risk uncertainty, BRI will be periodically reviewed with new information.

2.1.1. Flood

Definition

Local/urban flooding is defined as the inundation of land or property, mainly in developed areas, which is caused by excessive runoff, pervious soils with shallow water tables that become saturated and impervious surfaces which overwhelm the capacity of drainage systems²¹. It can be linked to extreme storms (e.g. cyclones), but also to relatively small amounts of rain falling on impervious surfaces, which overwhelms inadequate or poorly maintained drainage infrastructure found in many urban areas²². Such localized floods have little to do with water bodies (e.g. rivers and coasts) and can occur in areas outside of known floodplains. They are more related to grading and drainage design conditions. In certain topographies these floods can escalate quickly into flash floods²³.

It includes situations when water enters a building through windows or other openings, backs up through pipes and drains, and seeps in through walls and doors. When this occurs, it can damage the building's structure and contents, as well as rusting and damaging equipment. Electronic equipment is particularly vulnerable. Disaster management of local/urban flood is becoming a growing priority as city populations expand and existing drainage infrastructure ages and becomes inadequate. Coupled with this, there is high confidence that climate change will result in more frequent and intense precipitation days and an increase in the number of extreme rainfall events.

River floods are associated with high water in rivers, streams, and canals which occur when the water is no longer contained within the channel. River floods can be caused by heavy rain, snow melt, deforestation, extreme weather events like cyclones and human infrastructure (such as dams). Flood occurs adjacent to the river body, normally on dry land. Urban land use can also increase the risk of flood as this often reduces natural floodplains and increases the area of impermeable surfaces. There is high confidence that the intensity and frequency of precipitation days will increase with climate, which will increase river flood risk in many areas²⁴.

Flash Flood occurs in certain topographies when rainfall is severe and fast, and the ground cannot absorb it fast enough.²⁵ It can also be caused by a dam or levee failure. In contrast to river flooding, which develops slowly, there is very little time between the trigger and the flash flood. It can occur anywhere water tends to flow and depends on a number of factors, including rainfall intensity, topography, how dry the soil is.²⁶ However, it normally occurs

²¹ Anna Weber, "What Is Urban Flooding?" NRDC, last modified January 15, 2019, <https://www.nrdc.org/experts/anna-weber/what-urban-flooding>

²² FM Global, *Flood Data Sheet* (2019), <https://www.fmglobal.com/research-and-resources/fm-global-data-sheets>

²³ Ivan Andjelkovic, *Guidelines on Non-Structural Measures in Urban Flood Management*, (Paris: UNESCO, 2001).

²⁴ IPCC, *Climate Change 2013: The Physical Science Basis* (Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2013), <https://www.ipcc.ch/report/ar5/wg1/>

²⁵ "Surface Water Flooding" U.K. Met Office, updated Dec, 2025, <https://assets.publishing.service.gov.uk/media/69369de26a167b6884b73601/transport-hazard-summary-surface-water.pdf>
<https://www.metoffice.gov.uk/weather/learn-about/weather/types-of-weather/rain/flash-floods>

²⁶ "What is Flash Flooding?" NOAA, accessed July 1, 2020, <https://www.weather.gov/phi/FlashFloodingDefinition>

where locations where the topography is narrow and steep and areas with more impermeable surfaces.²⁷

For BRI, Local/urban flood is considered always applicable and needs to be addressed during design to meet BRI flood criteria on Table 7 for the desired rating. There is no hazard map associated with local/urban flood. River flood and flash flood on the other hand use a common flood hazard layer to determine applicability to require it be addressed in the design.

Location

Locations that are more exposed to this hazard are in close proximity to the coast, water bodies or areas with high impermeable areas, as well as in low topographies where water tends to flow and accumulate. Storm Surge

Definition

Storm surge occurs when low-lying land is inundated with seawater caused by a combination of high tides and severe storms.²⁸ Storm surge is often a greater threat to life and property than wind hazards themselves, as the depths of storm surge do not always correlate with wind speed categories²⁹. An example of coastal flood was the damage that New Orleans suffered following Hurricane Katrina in 2005.

Seawater can flood nearby land directly if it exceeds the elevation of the land or if its waves can breach human (e.g. sea wall) or natural (e.g. sand dunes) barriers. Human activities can increase the risk. For example, engineered protection measures (e.g. sea walls) can interfere with natural processes leading to increased rates of erosion and risk of flood as a result.

Climate change is also likely to increase the intensity and frequency of extreme storm events, which will put some areas at greater risk. Alongside this, coastal regions are growing in population and development, putting a greater number of people and assets at risk³⁰.

Tidal floods / Non-Storm or “Sunny day flooding” can be caused by a very high tide alone without heavy rainfall or a storm.³¹ Normally high tide flood is limited to a small area, but if it occurs in combination with a storm the damage can be amplified. The height of coastal (or tidal) floods can be affected by astronomical factors, like ‘king tides’, and factors such as El Nino.³² Coastal (or tidal) flood is a global problem which affects all coastal areas. The

²⁷ Surface Water Flooding” U.K. Met Office, updated Dec 2025, <https://assets.publishing.service.gov.uk/media/69369de26a167b6884b73601/transport-hazard-summary-surface-water.pdf>

²⁸ S. Muis et al., “A global reanalysis of storm surges and extreme sea levels,” *Nature communications*, 7(1) (2016): 1-12.

²⁹ “The Dangers of Storm Surge”, 07/31/2020 <https://blog.response.restoration.noaa.gov/dangers-storm-surge-and-flooding>

³⁰ R.J. Dawson et al., “Integrated analysis of risks of coastal flooding and cliff erosion under scenarios of long-term change.” *Climatic Change*, 95(1-2) (2009): 249-288.

³¹ “What is high tide flooding?” NOAA, last modified July 25, 2025, <https://oceanservice.noaa.gov/facts/high-tide-flooding.html>

³² “High-Tide Flooding,” U.S. Climate Resilience Toolkit, accessed April 12 2026, <https://toolkit.climate.gov/topics/coastal-flood-risk/shallow-coastal-flooding-nuisance-flooding>

incidence of coastal (or tidal) floods has increased in recent years due to rising sea levels and is projected to increase further both in severity and frequency.

For BRI Tidal floods are incorporated into the Storm Surge coastal mapping for determining applicability as a design consideration.

Location

Locations that are more exposed to this hazard are in storm-prone areas with close proximity to the coast.

2.1.2. Tsunami

Definition

A tsunami is a wave (or series of waves) with high velocity and depth, usually caused by a displacement of a large volume of water triggered by an earthquake under the ocean. However, they can also be caused by events such as landslides or volcanoes. As a tsunami wave travels inland, the height builds up as the depth of the ocean decreases.

There are three ways in which tsunamis cause destruction when they reach the shore: 1. Flood, wave impact on structures, 2. Erosion of soils around foundations, and 3. Debris impact of boats, transport, trees and building components carried by the flowing water.

Very large tsunamis, usually generated along subduction zones between tectonic plates can affect the coastlines of multiple countries and extend several kilometers inland, depending on topography. These large tsunamis are rare but have a potentially high impact. Recent examples include the Indian Ocean (2004) and the Great East Japan (2011) tsunamis.

As sea levels rise due to climate change, the potential impact of tsunamis is also expected to increase. Findings indicate that flood will occur further inland in coastal communities with the tsunami-induced flood increasing by 1.2 to 2.4 times for every 0.5 m increase in sea level rise. This would mean that smaller tsunamis have the potential to cause the same adverse impacts as large tsunamis do today³³.

Location

Locations that are more exposed to this hazard are low elevation areas in close proximity to the coast.

2.2. Best Practice Standards

Global Best Practices (GBP) for flood and storm surge water hazards are considered those of the Netherlands, where the underlying soils are generally non-porous. The flood risk legislation of the Netherlands prevents municipalities from building in flood plains along

³³ L. Li et al., "A modest 0.5-m rise in sea level will double the tsunami hazard in Macau," *Science Advances*, 4(8) (2018).

rivers and coastal zones. Buildings permitted in flood-prone locations are elevated on piles, or adjust to water levels. Wet and dryproofing measures are generally required.³⁴

Tsunami standards in Japan are considered GBP's. Following the Tsunami Evacuation Building Guidelines (Japan Cabinet Office, 2005), the Ministry of Lands, Infrastructure, Transport and Tourism of Japan (MLIT) proposed the '*Interim Guidelines on Structural Requirements for Tsunami Evacuation Buildings*'³⁵. This document considered the Great East Japan Earthquake as fundamental part of the research, where the relationship among structural size, required lateral strength, and tsunami inundation depth is studied and discussed. The tsunami pressure formula is provided and it is expressed by the form of equivalent hydrostatic concept, the pressure can be reduced depending on the presence/absence of tsunami energy dissipation structures and the distance of the building from shoreline or riverbank. The tsunami loads are in general given considering the tsunami inundation depth that appears in the regional hazard maps provided by the local government.

Water hazards in many locations are addressed by community and public defenses outside the building perimeter, such as dams, sea walls, berms, levees and pumps. Such defenses have limits on their design/durability and require proper maintenance and periodic upgrades to remain effective. Therefore, BRI does not consider them sufficient substitutes for building risk reduction measures and does not consider defenses as a building level hazard risk reduction measure.

Most National Codes today regarding water are at BRI's NR level, even in tsunami, storm surge and flash flood areas. Therefore, unless already sited on high ground, even B rating can require going above the code. BRI expects that future codes will continuously improve and move in the direction of its higher ratings. The increased frequency and intensity of water hazards caused by climate change will be a major factor driving this.

Below for reference is a sampling of code standards that BRI has considered in developing its water mitigation measures:

Flooding: The International Code Council (ICC) is a United States association which developed The International Codes (I-Codes), a family of model building codes. The I-Codes which contain flood provisions include:

1. The International Building Code – largely references *Flood Resistant Design and Construction*, adopting flood design standard provisions of ASCE 24, and ASCE 7 Chapter 5 *Flood Loads*.
2. International Residential Code – requirements for flood-resistant design and construction of one or two dwellings and townhomes, also adopting ASCE 24.
3. International Code Council Performance Code – performance-based standards to provide resistance to flood loads and damage.

³⁴ <https://www.stowa.nl/sites/default/files/assets/PUBLICATIES/Publicaties%202014/STOWA%202014-12%20EN.pdf>; General Regulations on Spatial Planning Decree, 2.4, 2.4, 2.11

³⁵ Yoshiaki Nakano, *Structural Design Requirements for Tsunami Evacuation Buildings in Japan* (2011), http://www.jci-net.or.jp/e/publish/others/ACI_JCI_Joint_Seminar/img/ACI-JCI_Joint_01.pdf

The International Property Maintenance Code (IMPC) provides guidance for maintaining existing buildings through model code regulations. It can be accessed at: <https://codes.iccsafe.org/content/IPMC2021P2FM> GLOBAL Property Loss Datasheets provide a set of standards and detailed guidance to reduce property losses due to natural hazards. They are available free of charge and can be accessed at: <https://www.fmglobal.com/research-and-resources/fm-global-data-sheets>. The most relevant for water hazard are:

- 1-40 Property Loss Prevention Data sheet: Flood. Oct 2019 (Ref. FMDS0140)
 - 1-34 Data sheet: Hail Damage. March 2018 (Ref. FMDS0134)
4. **Tsunami:** The ICC model International Building Code (IBC) – references and adopts ASCE 7: *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*, including design standard provisions of Chapter 6, *Tsunami Loads and Effects*.

For community level tsunami hazard planning, the United Nations Educational, Scientific and Cultural Organization (UNESCO) has produced *Tsunami Preparedness Information Guide for Disaster Planners (2018)*. This document is recommended for use by planners to form a strong community tsunami readiness policy and can be accessed at: <https://unesdoc.unesco.org/ark:/48223/pf0000160002>. It provides a framework for dealing with tsunamis, including:

- The hazards associated with tsunamis
- The construction and maintenance of defensive structures
- How current emergency response plans can be improved .

2.3. Consequential Hazards

- **Local (or urban) flood** from grading and drainage measures can increase the risk of landslides.
- Wind storms causing debris blockages of waterways often combine with high rainfall events that can cause **flash flooding**.
- The risk of **river flood** is increased by intense precipitation and wind storms, or a landslide falling on the river stream. River floods increase the risk of other downstream landslides.
- **Coastal flood** is very common in low lying coastal area when storm surges occur
- **Tsunamis** can be triggered by earthquakes and coastal landslides.

2.4. Risk Reduction Measures

The following risk management measures list is in the BRI application and forms the basis for the BRI water hazards resilience rating. Measures marked 'Y' apply, if the building is in a

location with that particular hazard type. Buildings that do not obtain an AA, A or B rating are rated NR.

The location of the site is key when avoiding water hazards. A location assessment should be undertaken prior to commencement with the development. Examining pre-existing flood hazard information in the area can help to assess the likelihood of flood. Assessors and Verifiers are asked to check the elevations of the site, its first occupied floor, and mechanical/electrical equipment. They should also examine topographic maps of the area within a 2 km radius, to assess where excess water will flow. Elevation information is essential for many of BRI's water measures.

For its highest rating BRI takes a critical view of locations that were a floodplain, wetland, swamp, river/lake/seabed or filled creek prior to human development, are near the low point of a valley/gully, or receive water from land sloped towards it, rather than drain water away. For this, historic maps are useful as they reveal water features hidden by subsequent development.

Each risk reduction measure needs to be approved by at two locally licensed engineers or architects for the building's rating to be verified. Each Verifier will be required to upload their license to confirm their qualification. Verifiers should collect and retain for their files whatever information they consider necessary to conduct their work.

Table 7. List of risk reduction measures for water hazards

Water Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local/Urban Flooding (Default Hazard)	River/Lake Flooding	Flash Flooding	Coastal/Tidal Flooding	Storm Surge	Tsunami
WT01. Not a Historically-known Natural Water Area	The proposed site must not be a filled floodplain[1], wetland[2], swamp[3], river, lake, seabed[4] or creek[5]. Also, any river basin and former floodplain area downstream of a man-made dam should be considered a natural water area	Y			Y	Y	Y	Y	Y	
WT02. Not Located at the Low Point of a Valley or Gully	In areas prone to river/lake, local/urban, or flash flooding, the location of a building should not be less than 6m above the lowest point of a valley[1] or gully[2] within a 2km radius.	Y			Y	Y	Y			
WT03. More than 60% of the Site Perimeter is Sloped Away From Building	The building should be located on a site which has more than 60% of the site perimeter sloping away from the building	Y			Y		Y			
WT04. Lowest Occupied Floor and Equipment Elevated at 6 m minimum	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 6 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek. Nearby means within a 2km radius of the property. If the building is located downstream of a man-made earthen dam, the lowest occupied floor and mechanical and electrical equipment should at least be 12 m above the mean water level of the nearby river.	Y				Y	Y	Y	Y	
WT05. Lowest Occupied Floor and Equipment Elevated at 5 m minimum	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 5 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek. Nearby means within a 2km radius of the property. If the building is located downstream of a man-made earthen dam, the lowest occupied floor and		Y			Y	Y	Y	Y	

Water Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local/Urban Flooding (Default Hazard)	River/Lake Flooding	Flash Flooding	Coastal/Tidal Flooding	Storm Surge	Tsunami
	mechanical and electrical equipment should at least be 9 m above the mean water level of the nearby river.									
WT06. Lowest Occupied Floor and Equipment Elevated at 3 m minimum	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 3 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek, independent of any public flood management infrastructure. Nearby means within a 2km radius of the property. If the building is located downstream of a man-made earthen dam, the lowest occupied floor and mechanical and electrical equipment should at least be 5 m above the mean water level of the nearby river.			Y	Y	Y	Y	Y	Y	
WT07. Lowest Occupied Floor Elevated at 15 m (Tsunami)	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 15m above the mean sea water level. I	Y								Y
WT08. Lowest Occupied Floor Elevated at 10 m (Tsunami)	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 10m above the mean sea water level.		Y							Y
WT09. Lowest Occupied Floor Elevated at 5 m (Tsunami)	The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 5m above the mean sea water level.			Y						Y
WT10. Unoccupied Ground-Floor Flood Resistant Materials	If an unoccupied ground-floor is elevated less than 5m above a nearby water body or dry creek measures WT04 or WT05 are satisfied for the lowest occupied floor, but the ground floor is below 6m and 5m, respectively, then flood waters should be able to freely flow in and out of the ground level.		Y		Y	Y	Y	Y	Y	

Water Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local/Urban Flooding (Default Hazard)	River/Lake Flooding	Flash Flooding	Coastal/Tidal Flooding	Storm Surge	Tsunami
WT11. Roof is Sealed with Waterproof Membrane	Entire roof is sealed with waterproof membrane.			Y	Y					
WT12. Sealed Openings	Exterior windows and doors are sealed to prevent water intrusion.			Y	Y					
WT13. Backflow Management Measures	If the ground floor elevation is less than 5 m from sea level or nearby water body, then backflow management systems, such as backflow prevention valves, should be installed to prevent sewage backflow during flooding		Y		Y	Y	Y	Y	Y	Y
WT14. Onsite Water Retainage for 200-year Precipitation Scenario	Onsite water retainage areas designed/built to contain allow a peak precipitation for a 200-year local rain scenario.	Y			Y		Y			
WT15. Onsite Water Retainage for 100-year Precipitation Scenario	Onsite water retainage areas designed/built to contain allow a peak precipitation for a 100-year local rain scenario. Or provide WT16 as an alternative.		Y		Y		Y			
WT16. Permeable Outdoor Surrounding Surfaces	At least 80% of outdoor surfaces should be designed/built to be water permeable. Or provide WT 15 as an alternative.		Y		Y	Y	Y	Y	Y	
WT17. Water Resistant Materials	Exterior and interior building materials that may come in contact with flood water need to be water resistant.	Y			Y	Y	Y	Y	Y	Y

Water Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local/Urban Flooding (Default Hazard)	River/Lake Flooding	Flash Flooding	Coastal/Tidal Flooding	Storm Surge	Tsunami
WT18. Resistance to 160 Kilonewton Impact from Water-Borne Debris	If the ground elevation is less than 5 m above sea/lake/river level, the building structure must be strong enough to resist damage from water-borne debris with an impact force of 160 KN.		Y			Y	Y	Y	Y	Y
WT19 Roof Drains Sized for 100-year Precipitation Scenario	The drainage system should be sized and maintained for a 100-year precipitation scenario. The roof drains are designed/built for 100-year precipitation scenario and ensure water doesn't accumulated around the building and is directed away.		Y		Y	Y	Y	Y	Y	Y
WT20 Design Review for Water Hazard Risk Management Measures	The building design and occupancy must be submitted to and approved by the approving agency, as prescribed by the governing national or local building code.		Y		Y	Y	Y	Y	Y	Y
WT21. Construction Independently Audited for Water Hazard Risk Management Measures	A second independent qualified engineer or architect, different from the one assigned from the approving agency or the responsible engineer or architect for the project, iwas engaged to inspect the construction of the building during critical milestones.	Y			Y	Y	Y	Y	Y	Y

WT01. Not a Historically-known Natural Water Area (AA Rating)

Criteria: The proposed site must not be a filled floodplain³⁶, wetland³⁷, swamp³⁸, river, lake, seabed³⁹ or creek⁴⁰.

Also, any river basin and former floodplain area downstream of a man-made dam should be considered a natural water area.,

Assessor Procedure: Provide documentation evidence that the site meets the criteria, including current and historic maps (preferably over 50 years), photographs, geologic information, etc,

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: The location of a site is important in the prevention of water-related dangers. Climate change is amplifying hazards such as sea levels rising and intense rainstorms during varied seasons.

Nature leaves enduring physical imprints on a site and water seeks where it once flowed and accumulated. If the site has been a natural water area during the past 100 years, or has flooded at least once since being developed, it will very likely flood again in the future. Altered by human development, areas upstream to the site that once acted as buffers, may no longer do so.

The site visits should include careful observation of features such as irregular topography, drainage conditions, erosion, site changes, and confirm proximity and elevation of natural and man made water features.

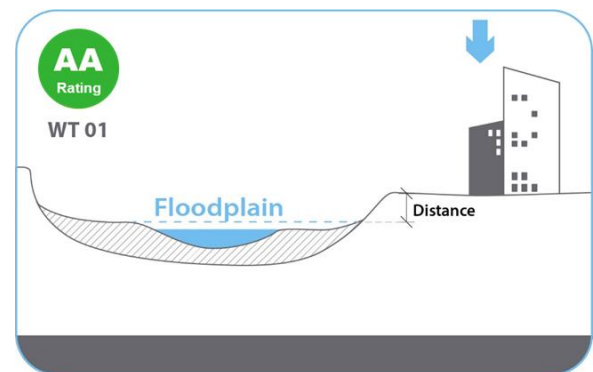


Figure 90. Site was never a natural water area

³⁶ **Floodplain:** An area of land adjacent to a stream or river which stretches from the banks of its channel to the base of the enclosing valley. It is prone to flooding during periods of high discharge caused by storms or heavy rainfall.

³⁷ **Wetland:** A distinct ecosystem that is flooded by water, either permanently or seasonally. The areas are dense in aquatic vegetation which are adapted to the unique hydric soil.

³⁸ **Swamp:** A forested wetland which have more mature vegetation than a wetland and are considered to be transition zones between land and water. They vary in size and can contain fresh water, brackish water, or sea water.

³⁹ **Seabed:** The base of a sea or ocean, it contains subsoil and is a solid surface, reaching many depths.

⁴⁰ **Filled creek:** A narrow area of water that flows into the land from a body of water such as the sea or a lake which has been filled in due to human water management.

WT02. Not Located at the Low Point of a Valley or Gully (AA Rating)

Criteria: In areas prone to river/lake, local/urban, or flash flooding, the location of a building should not be less than 6m above the lowest point of a valley⁴¹ or gully⁴² within a 2km radius.

Assessor Procedure: Identify on a topographical map the site elevation and elevation of any valley/creek low point within a 2km radius. Investigate the site flow routing, flood history and development history. Provide documentation evidence that the site meets the criteria, including maps, photographs, geologic information, etc,

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Water flowing with the force of gravity from high to low, carves pathways in the terrain that it uses repeatedly or seasonally. Therefore, these pathways will become filled with water again and will overflow and flood, causing destruction to any buildings within their vicinity.

Be mindful that many former valleys or gullies have been filled or paved during recent and historic development. These are locations still vulnerable to flooding hence are included under this measure to be investigated and a focus of the site visits. For purposed of measuring the 6m vertical elevation select the point in the middle of the valley or gully that is closest to the property.

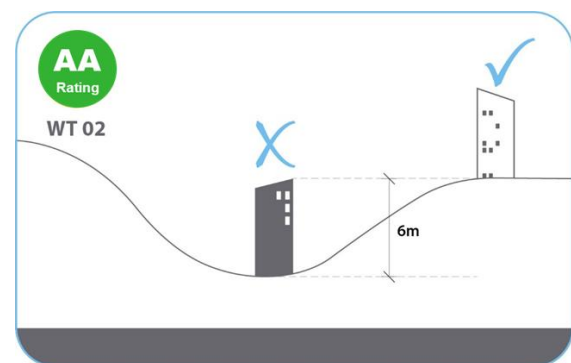


Figure 210. Location not near the lowest point of valley/gully

⁴¹ **Valley:** An elongated low area often running between hills or mountains which typically contain a river or stream that form part of the larger tributary water table network. Most valleys have been formed by erosion of the land surface by rivers, streams of melting glaciers and their subsequent debris, over a very long period of time.

⁴² **Gully:** A landform created by running water eroding sharply into the soil, typically located on a hillside. They resemble large ditches or small valleys, with meters or tens of meters in width and depth.

WT03. More than 60% of the Site Perimeter is Sloped Away From Building (AA Rating)

Criteria: The building should be located on a site which has more than 60% of the site perimeter sloping away from the building

Assessor Procedure: Identify on the topographical map used for WT02 the elevations at 45 degree intervals of the land adjacent to the property (8 points). At least 5 of these should be lower than the property elevation.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: In areas prone to local/urban or flash flooding, buildings should be located on higher ground to allow water to flow away from, rather than towards the building.

The requirement is to have more than 60% of the site perimeter sloping⁴³ away from the building and to have the steeper gradient of land below the building. That way the water will flow faster away from the building. This often requires grading the surrounding landscape to improve sloping. For large parcels such as a campus the intent is to limit the evaluation to the developed perimeter.

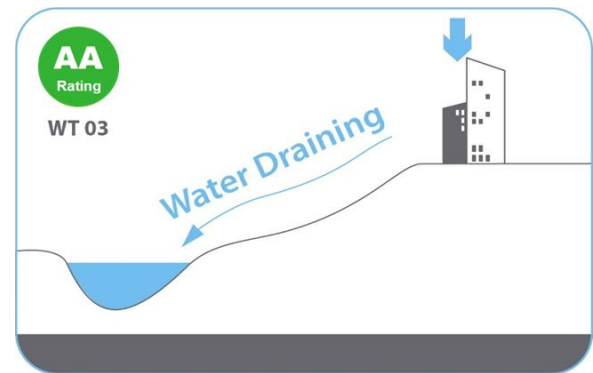


Figure 11. Location sloped away from and not towards building

⁴³ **Slope:** The surface on the side of a mountain, hill or valley which is at an angle, making one end higher than the other.

WT04. Lowest Occupied Floor and Equipment Elevated at 6 m minimum (AA Rating)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 6 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek. Nearby means within a 2km radius of the property.

If the building is located downstream of a man-made earthen dam, the lowest occupied floor and mechanical and electrical equipment should at least be 12 m above the mean water level of the nearby river.

Assessor Procedure: Using the topographical map from WT02/03 identify the elevations of water bodies and dry creeks within a 2km radius. Show on the topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or pictures.

Confirm that they meet the criteria.

If water elements on elevated slopes more than 0.5km away do not meet the criteria, provide drainage/water management studies, maps and other materials that show they do not flow towards the property.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Flooding can cause damage to a building's structure, interior fittings, furnishings, products, and mechanical and electrical equipment. AA-rated protection from storm surge, river/lake/coastal/tidal flood- ing and storm surge requires elevation as primary flood protection.

Dams are prone to failure and risk releasing their reservoirs to downstream sites at low elevations causing severe flash floods.

Public water risk reduction infrastructure (e.g., dam, levee, and other flood defense) near the building is not a substitute for building elevation. Such infrastructure may be have been designed to lower hazard levels and suffer from insufficient maintenance and partial obsolescence, thus unable to provide the expected flood protection when needed.

The lowest occupied floor does not directly reflect ground floor of a building. It refers to a floor that serves the core functions of the building, depending on how a building is used it can even be a basement floor. In residential buildings it relates to living areas, in offices to work spaces, in schools to classrooms, in hospitals to patient and procedure rooms etc. Ground level, garages, receptions, storage spaces, recreation rooms and commercial stores can be considered non-occupied, if their users are aware that these may flood.

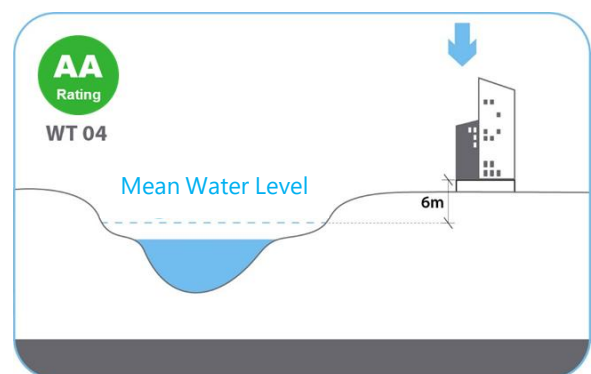


Figure 12. Lowest occupied floor and equipment elevated at 6 m minimum

WT05. Lowest Occupied Floor and Equipment Elevated at 5 m minimum (A Rating)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 5 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek. Nearby means within a 2km radius of the property.

If the building is located downstream of a man-made earthen dam, the lowest occupied floor and mechanical and electrical equipment should at least be 9 m above the mean water level of the nearby river.

Assessor Procedure: Using a topographical map identify the elevations of water bodies and dry creeks within a 2km radius. Show on the topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or pictures.

Confirm that they meet the criteria.

If water elements on elevated slopes more than 0.5km away do not meet the criteria, provide drainage/water management studies, maps and other materials that show they do not flow towards the property.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Flooding can cause damage to a building's structure, interior fittings, furnishings, products, and mechanical and electrical equipment. A-rated protection from storm surge, river/lake/coastal/tidal flooding and storm surge requires

elevation as primary flood protection. Dams are prone to failure and risk releasing their reservoirs to downstream sites at low elevations causing severe flash floods.

Public water risk reduction infrastructure (e.g., dam, levee, and other flood defense) near the building is not a substitute for building elevation. Such infrastructure may have been designed to lower hazard levels and suffer from insufficient maintenance and partial obsolescence, thus unable to provide the expected flood protection when needed.

The lowest occupied floor does not directly reflect ground floor of a building. It refers to a floor that serves the core functions of the building, depending on how a building is used it can even be a basement floor. In residential buildings it relates to living areas, in offices to work spaces, in schools to classrooms, in hospitals to patient and procedure rooms etc. Ground level, garages, receptions, storage spaces, recreation rooms and commercial stores can be considered non-occupied, if their users are aware that these may flood.

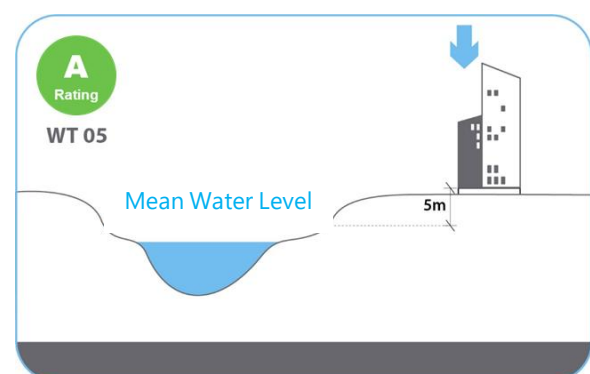


Figure 13. Lowest occupied floor and equipment elevated at 5 m minimum

WT06. Lowest Occupied Floor and Equipment Elevated at 3 m minimum (B Rating)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 3 m above the mean water level of any nearby water body (sea, lake, river, creek, canal) or bottom of a dry creek, independent of any public flood management infrastructure. Nearby means within a 2km radius of the property.

If the building is located downstream of a man-made earthen dam, the lowest occupied floor and mechanical and electrical equipment should at least be 5 m above the mean water level of the nearby river.

Assessor Procedure: Using a topographical map identify the elevations of water bodies and dry creeks within a 2km radius. Show on the topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or pictures.

Confirm that they meet the criteria. Public flood management infrastructure near the building, such as stormwater drainage, dams, floodwalls, levees etc are not a substitute for building elevation.

If water elements on elevated slopes more than 0.5km away do not meet the criteria, provide drainage/water management studies, maps and other materials that show they do not flow towards the property.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Flooding can cause damage to a building's structure, interior fittings, furnishings, products, and mechanical and electrical equipment. B-rated protection from storm surge, river/lake/coastal/tidal flooding and storm surge requires elevation as primary flood protection. Dams are prone to failure and risk releasing their reservoirs to downstream sites at low elevations causing severe flash floods.

Public water risk reduction infrastructure (e.g., dam, levee, and other flood defense) near the building is not a substitute for building elevation. Such infrastructure may have been designed to lower hazard levels and suffer from insufficient maintenance and partial obsolescence, thus unable to provide the expected flood protection when needed.

The lowest occupied floor does not directly reflect ground floor of a building. It refers to a floor that serves the core functions of the building, depending on how a building is used it can even be a basement floor. In residential buildings it relates to living areas, in offices to work spaces, in schools to classrooms, in hospitals to patient and procedure rooms etc. Ground level, garages, receptions, storage spaces, recreation rooms and commercial stores can be considered non-occupied, if their users are aware that these may flood.

Infrastructure like dams, stormwater drains, floodwalls and levees may have been designed for lower hazard levels, not maintained sufficiently and partly obsolete, thus unable to provide the expected flood protection when needed. Furthermore,

impact from vehicles, marine vessels and other floating objects/debris can compromise such infrastructure.

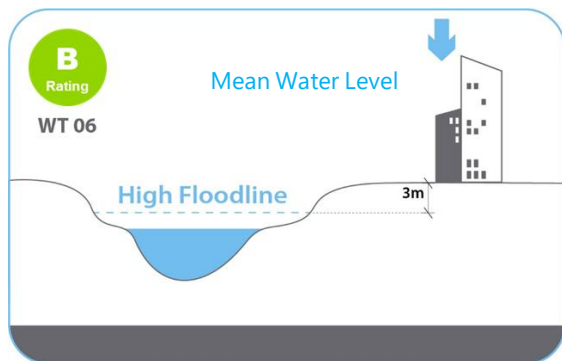


Figure 14. Lowest occupied floor and equipment elevated at 3 meters minimum

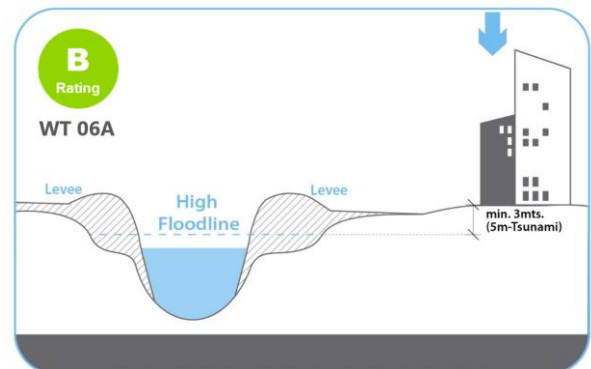


Figure 25A. Lowest occupied floor and equipment elevated at 3 meters minimum.

WT07. Lowest Occupied Floor Elevated 15 m (Tsunami) (Rating AA)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 15m above the mean sea water level.

Assessor Procedure: Show on a topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or photos.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Also, structures or areas above 15 m height can potentially be used as vertical evacuation shelters subject to proper occupancy load and use planning⁴⁴.

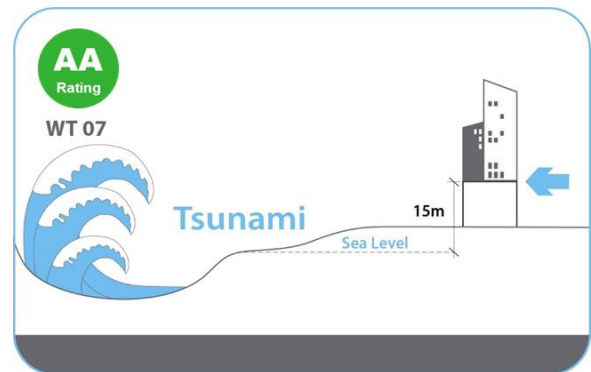


Figure 15. Lowest occupied floor elevated 15 m (tsunami)

⁴⁴ **Vertical evacuation shelters:** A building, structure, or earthen mound that has sufficient height above the level of tsunami inundation, is designed and constructed with the strength and resiliency needed to resist the effects of

tsunami waves and located in a way that people can take refuge within the time available between warning and tsunami inundation.

WT08. Lowest Occupied Floor Elevated 10 m (Tsunami) (A Rating)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 10m above the mean sea water level.

Assessor Procedure: Show on a topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or pictures.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

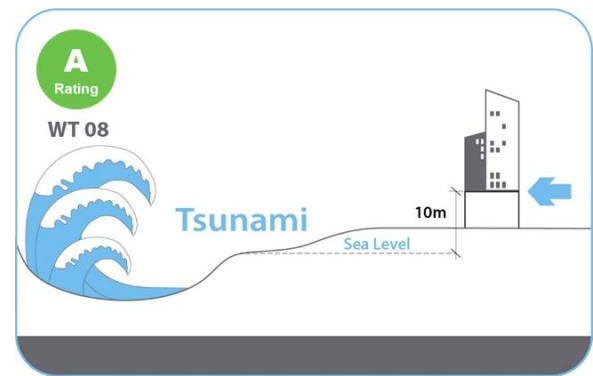


Figure 16. Lowest occupied floor elevated 10 m (tsunami)

WT09. Lowest Occupied Floor Elevated 5 m (Tsunami) (B Rating)

Criteria: The lowest occupied floor and mechanical/electrical equipment need to be a minimum of 5m above the mean sea water level.

Assessor Procedure: Show on a topographical map and as-built drawings the elevation of the site where the building is located and the elevations of the first occupied floor and mechanical/electrical equipment above the site, providing supporting materials and/or pictures.

Public tsunami defense infrastructure near the building is not a substitute for building elevation.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Infrastructure like seawalls, water breakers and levees may have been designed for lower hazard levels, not maintained sufficiently and partly obsolete, thus unable to provide the expected flood

protection when needed. Furthermore, impact from vessels and other floating objects/debris can also compromise such infrastructure.

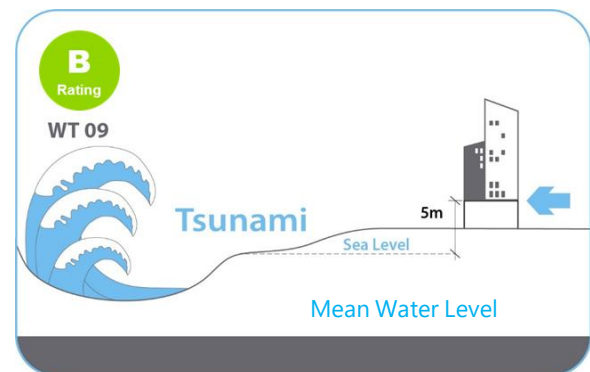


Figure 17. Lowest occupied floor elevated 5 m (tsunami)

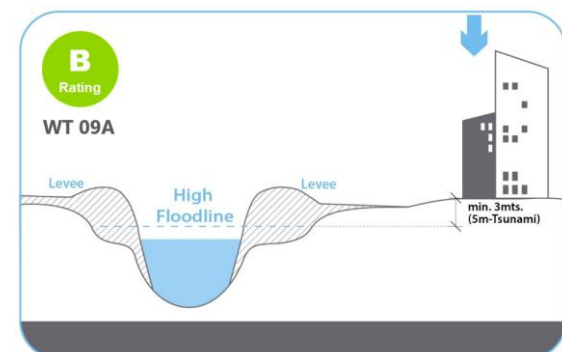


Figure 29. Location in relation to dam/levee/flood defense

WT10. Unoccupied Ground-Floor Flood Tolerant Design (A Rating)

Criteria: If an unoccupied ground-floor is elevated less than 5m above a nearby water body or dry creek then flood waters should be able to freely flow in and out of the ground level, using suitable design methods to avoid hydrostatic force build up.

Assessor Procedure: Determine elevation of unoccupied ground-floor from topographical and as-built drawings. If less than 5m, then show it is open to flood waters entering and exiting (such as a parking area). If enclosed, show that it either has collapsible walls or hydrostatic flood vents, by providing drawings and photos.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. If

flood vents exist, check that these are clear of obstructions and maintained.

Background: Floodwaters surrounding an enclosed ground floor can create uplift buoyancy forces that can potentially damage a building's foundation and connections.



Figure 18. Hydrostatic flood vents

WT11. Sealed Roof (B Rating)

Criteria: Entire roof is sealed with waterproof membrane.

Assessor Procedure: Provide drawings or other documentation that a waterproof membrane was installed over the entire roof, including the specific product name and model of the membrane.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check interior ceiling and wall for water stains, sagging, peeling or mold.

Background: Water leakage from roofs can speed up the erosive process, cause mold and damage products, surfaces, furnishings and electronic equipment.

Therefore, the building roof must be watertight and prevent the infiltration of moisture into the building's structure.

Waterproof membranes, when properly installed, serve as a sealant that separates the interior of a building from the exterior while preventing rainwater from infiltrating the building.

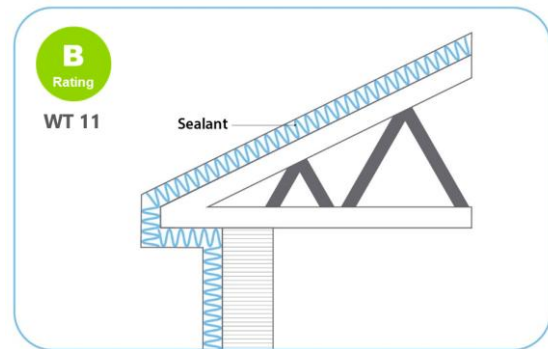


Figure 31. Sealed roof

WT12. Sealed Openings (B Rating)

Criteria: Exterior windows and doors are sealed to prevent water intrusion.

Assessor Procedure: Provide documentation that exterior windows and doors are water-sealed, including the specific name and model of the products used.

Verifier Procedure: Review the documentation provided by the assessor and confirm the installation meets the criteria. Spot check near windows and doors for interior wall/floor water stains, sagging, peeling or mold.

Background: Water leakage from openings can also cause building structural and interior damage. This can occur with heavy rains and winds.

Thus, all openings including windows and doors need to be properly installed and sealed to prevent water infiltration.

Common sealants for windows and doors include acrylic latex, silicone (inc. with acrylic mix), polyurethane (heavy-duty

spray foam), rubber sealant, and silane-modified polymers (SMPs).

All these must be weather resistance (direct sun, water, and fluctuating temperatures), and flexible to avoid shrinking or bubbles.

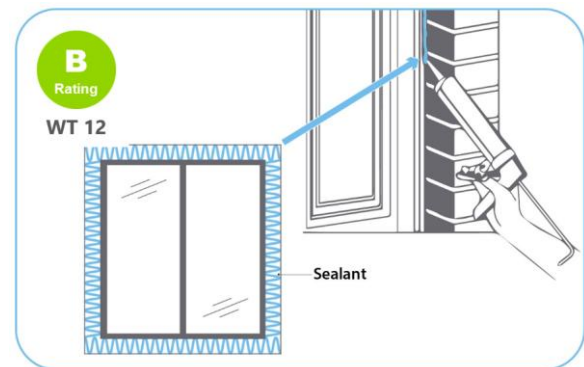


Figure 19. Sealed openings

WT13. Backflow Management Measures (A Rating)

Criteria: If the ground floor elevation is less than 5 m from sea level or nearby water body, then backflow management systems, such as backflow prevention valves⁴⁵, should be installed to prevent sewage backflow during flooding.

Assessor Procedure: Determine elevation of unoccupied ground-floor from topographical and as-built drawings. If less than 5m, provide documentation that backflow valves were installed in all sewer pipes serving the building.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Floods can overwhelm sewer systems, which often combine stormwater, resulting in pipe pressures that reverse flow direction. Backflowing sewage can be poisonous and dangerous to inhabitants. Backflow valves act to block any reverse sewer flow during local flood events.

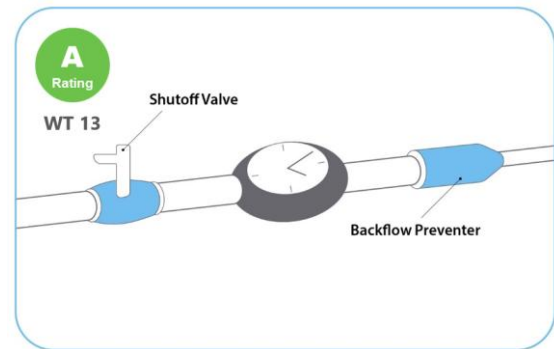


Figure 20. Install backflow valves

⁴⁵ **Backflow valves:** The connections which provide controlled opening and closing depending on the direction of water flow. When water flow instigates in the opposite direction, it will automatically close or if manual, a user can manually close it when aware of the reverse flow.

⁴⁶ <https://www.susdrain.org/delivering-suds/using-suds/suds-components/retention-and-detention/retention-ponds.htm>! Susdrain, "Component: Retention ponds," accessed April 12, 2026.

WT14. Onsite Water Retainage for 200-year Precipitation Scenario (AA Rating)

Criteria: Onsite water retainage areas designed/built to contain a peak precipitation for a 200-year local rain scenario.

Assessor Procedure: Identify onsite water retainage areas on site maps and drawings. These can include:

- Flood ponds, which are designed to control the flow rates of water by storing floodwater and releasing it slowly at a rate at which the water table can manage, when the risk of flooding has passed.⁴⁶
- Flood retaining tank, which is a tank normally located underground, that collects water and after the storm releases it slowly to the drains or the water body.
- Multi-purpose open space with permeable surfaces, which provides amenity, wildlife, as well as flood storage use.

Collect documents that identify the total water holding capacity of these retainage areas and comparing this to rainfall assumptions. If documents are not available provide the dimensions of the retainage areas and obtain the 200-year

rainfall scenario from the national meteorological organization.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: By having onsite holding capacity for rainwater, the impact of large rain events can be dissipated over a longer period of time. When rainwater flow escalates rapidly and the water table or public stormwater systems cannot absorb the water quickly enough, the area around the building can flood.

The difference between 100 and 200-year precipitation levels may be small and easy to achieve with minor increasings in retainage.

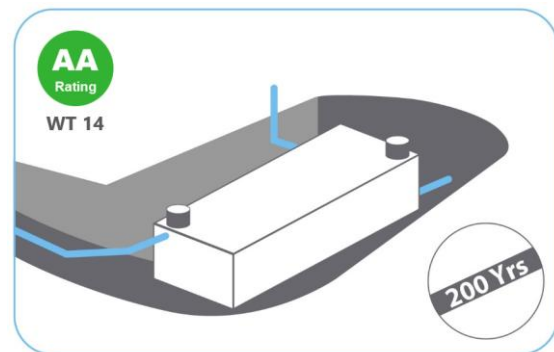


Figure 21. Flood retaining tank, sized for 200-year precipitation scenario

⁴⁶<https://www.susdrain.org/delivering-suds/using-suds/suds-components/retention-and-detention/retention-ponds.ht>

^m Susdrain, "Component: Retention ponds," accessed April 12, 2026.

WT15. Onsite Water Retainage for 100-year Precipitation Scenario (A Rating)

Criteria: Onsite water retainage areas designed/built to contain a peak precipitation for a 100-year local rain scenario. Or provide WT16 as an alternative.

Assessor Procedure: Identify onsite water retainage areas on site maps and drawings. These can include:

- Flood ponds, which are designed to control the flow rates of water by storing floodwater and releasing it slowly at a rate at which the water table can manage, when the risk of flooding has passed.⁴⁷
- Flood retaining tank, which is a tank normally located underground, that collects water and after the storm releases it slowly to the drains or the water body.
- Multi-purpose open space with permeable surfaces, which provides amenity, wildlife, as well as flood storage use.

Collect documents that identify the total water holding capacity of these retainage areas and comparing this to rainfall assumptions. If documents are not available provide the dimensions of the

retainage areas and obtain the 100-year rainfall scenario from the national meteorological organization.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: By having onsite holding capacity for rainwater, the impact of large rain events can be dissipated over a longer period of time. When rainwater flow escalates rapidly and the water table or public stormwater systems cannot absorb the water quickly enough, the area around the building can flood.

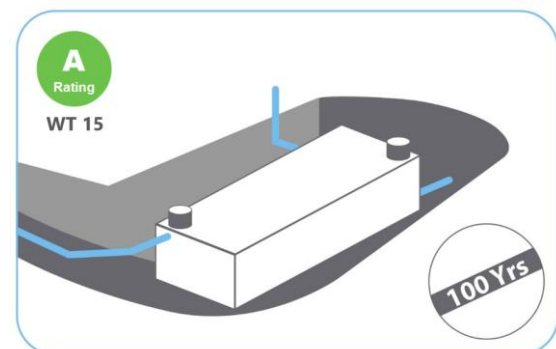


Figure 22. Flood retaining tank, sized for 100-year precipitation scenario

⁴⁷[https://www.susdrain.org/delivering-suds/using-suds/suds-components/retention-and-detention/retention-ponds.ht](https://www.susdrain.org/delivering-suds/using-suds/suds-components/retention-and-detention/retention-ponds.html)

^m Susdrain, "Component: Retention ponds," accessed April 12, , 2026.

WT16. Permeable Outdoor Surrounding Surfaces (AA Rating)

Criteria: At least 80% of outdoor surfaces should be designed/built to be water permeable. Or provide WT-15 as an alternative.

Assessor Procedure: Provide map or drawing identifying property permeable surfaces, which ⁴⁸ could include:

Permeable concrete, which lets the water to pass through

Green roofs covered on top with a layer of growing media and vegetation

Natural landscapes, which include areas with planted trees and vegetation.

Outdoor water retainage areas that are permeable to the soil.

Provide documentation that determines the total surface of the

permeable area in square meters and compares it to the total land surface of the property.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

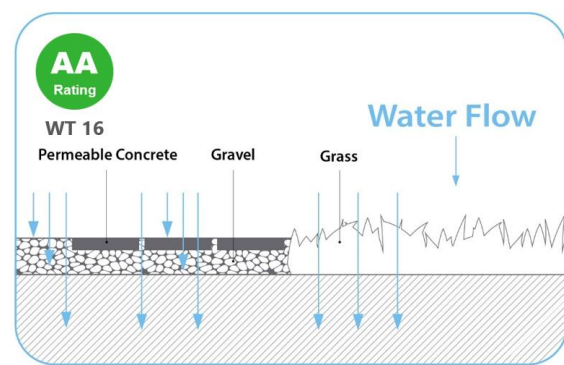


Figure 23. Permeable outdoor surfaces

⁴⁸ **Sustainable Urban Drainage:** Systems designed to efficiently manage the drainage of surface water in the urban environment. These must take into consideration the

existing drainage on site, whether natural or human-made and mitigate negative impacts on within and outside of the site when a building is constructed.

WT17. Water Resistant Materials (AA Rating)

Criteria: Exterior and interior building materials that may come in contact with flood water shall be floodwater and mildew resistant.

Assessor Procedure: Indicate the type of materials used and provide documents/photos of waterproof exterior materials. Include any interior materials in unoccupied groundfloor or underground floors with elevation less than 6m. Such material can include stone, ceramics, concrete, closed-cell (low moisture absorption) insulation, water repellent doors/frames, or other materials with water-resistant treatments/coatings.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot-check during site visit.

Background: Materials can be inherently waterproof or can be treated with water-resistant coatings to make them water resistant. Water resistant materials and finishes allow for faster drying, reduced damage and can lessen mold. See 2024 IBC Section 202 Definitions of flood resistant materials and FEMA NFIP Technical Bulletin 2, “Flood Damage-Resistant Materials Requirements” dated Jan 2025, including a listing of acceptable materials and uses.

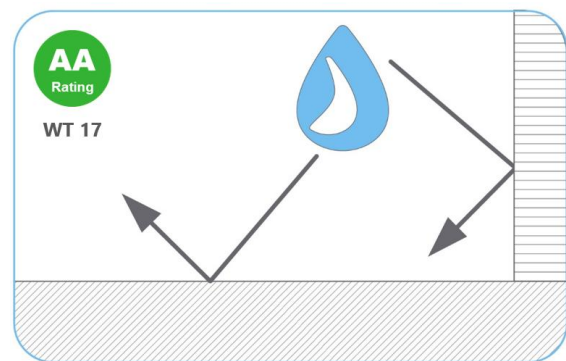


Figure 24. Water resistant materials

WT18. Resistance to Impact from Water-borne Debris (A Rating)

Criteria: If the ground elevation is less than 5m, the building structure must be strong enough to resist damage from water-borne debris with an impact force of 160 KN.

Assessor Procedure: Provide documentation showing that the structural columns can resist a horizontal impact force of at least 160 Kilonewtons which is an average 1,800 kg floating vehicle moving at a velocity of 3m/s

This can be satisfied by reinforced concrete or steel defensive barriers built around the building to absorb the impact of waterborne debris up to the 5 m level. The barriers should be no more than 1.5m apart.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot-check barriers during site visit.

Background: Moving water can carry debris, such as cars, trees, lose building components and boats, which at higher velocity can cause severe damage upon collision, potentially tearing into walls, bending columns and causing collapse. 160 Kilonewtons is the impact force of an average 1,800 kg vehicle (auto) floating at

3m/s. See IBC 2024 reference to ASCE 7-22 Chapter 5 and Chapter 6 section 6.11 Debris Impact Loads.

For tsunami design the velocity and potential debris sources should be confirmed and if velocity is >3m/s potential or significant debris sources are larger than an automobile, the design impact force should be increased proportionally, or per the building code, whichever is greater.

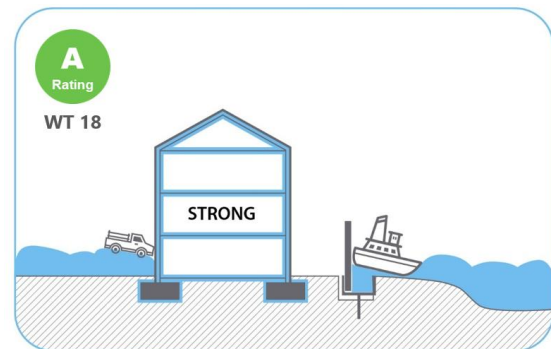


Figure 25. Building structure to withstand water-borne debris damage

WT19. Roof Drains Sized for 100-year Precipitation Scenario (A Rating)

Criteria: The drainage system should be sized and maintained for a 100-year precipitation scenario.

Assessor Procedure: Provide documentation showing that rainwater drainage system is sized and built for a 100-year precipitation scenario. Obtain the 100-year rainfall scenario from the national meteorological organization.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check that existing building drainage systems are maintained to prevent clogging during site visit.

Background: Roof drainage is the integrated system that gathers, controls the flow and discharges the rainwater from the roof structure. It consists of the sloped

roof, horizontal gutters and downspouts (vertical and sloped pipes) that bring rainwater to ground level, as well as sometimes cisterns or tanks for collection. It also should drain the water that impacts the building's non-roof surfaces.

Proper drainage prevents the accumulation of water on the roof and the overflow of water from the drainage system onto the building, from where it may leak into the building.

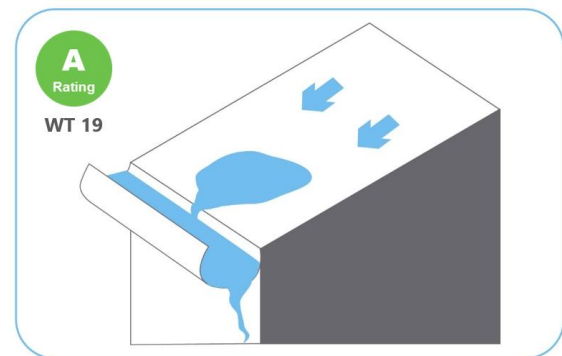


Figure 26. Roof drains sized for peak precipitation in 100-year scenario

WT20. Design Review for Water Hazard Risk Reduction Measures (A Rating)

Criteria: The building design and occupancy must be submitted to and approved by the approving agency, as prescribed by the governing national or local building code.

Assessor Procedure: Provide documentation from the approving agency the certificate of completion or equivalent. Also provide documentation from the engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: The construction must be observed and documented at critical milestones by the engineer of record

responsible for the code compliance at design stage. This cross-check helps to minimize the scope for error and assure that what is constructed was what was initially designed, and will aid verification.

In newer buildings the responsible engineer or architect should be able to provide the assessor with sample photos of inspections during construction milestones that relate to the water measures herein.



Figure 27. Civil design reviewed by the local approving agency

WT21. Construction Independently Audited for Water Hazard Risk Management Measures (AA Rating)

Criteria: A second independent qualified engineer or architect, different from the one assigned from the approving agency or the responsible engineer or architect for the project, is engaged to inspect the construction of the building during critical milestones.

Assessor Procedure: Provide documentation from the independent qualified engineer or architect attesting to compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: This independent professional should provide the assessor

sample photos of milestone inspections and discuss the water-related recommendations for the building.

For new construction the engineer verifier should, when possible, perform the independent construction inspection of critical milestones.



Figure 41. Construction independently audited for water hazard risk management measures

3. Fire

3.1. Hazard Types

There are two types of Fire Hazards:

Local Fire risk is found everywhere, as buildings and communities always contain a certain percent of combustible materials and many sources of ignition.

Wildfire risk is typically near urban-wildlife interfaces, large areas with dense vegetation that under certain conditions can combust, whether these be forests, grasslands or bushlands.

Besides life safety, typically achieved through rapid evacuation, a fire resilient building should maintain structural integrity in intense fire conditions for a long time period, even if external fire suppression services are delayed, or unavailable. Buildings rated AA aim for low/very low impact.

Table 8. Time required to maintain the structural integrity on fire hazards

Impact	Maintaining Structural Integrity
Very Low	more than 10.0 hrs.
Low	5.0 - 10.0 hrs.
Medium	3.1 – 5.0 hrs.
High	1.1 - 3.0 hrs.
Very High	1.0 hr or less

3.1.1. Local Fire

Definition

Fire occurs when there is an ignition source, a fuel source and an oxygen source. As oxygen is difficult to reduce, the risk of fire in a building depends on the ignition sources within it and the materials of its contents and structure. The main causes of fire are: cooking, heating equipment, candles, electrical faults, smoking, arson and inappropriate storage of waste or flammable liquids.⁴⁹ Risk reduction is key in reducing these ignition sources.

Once the fire has started, the extent of the damage will depend upon the ability of the fire to spread, which is determined the combustibility of the building and its contents. Fire protections such as sprinklers and fire brigade intervention can also help to reduce the damage from some fires. Wooden buildings are the most combustible, followed by unprotected steel, with reinforced concrete having the best fire resistance. Despite the construction of the building, if combustible cladding and paneling is present, the severity of the fire will dramatically increase, as seen in the Grenfell apartment fire in London in 2017.

⁴⁹ "Top Fire Causes," National Fire Protection Association, updated Aug 1, 2025. <https://www.nfpa.org/Public-Education/Fire-causes-and-risks/Top-fire-causes>

The contents of the building are also very significant in determining the extent of the fire. Warehouses are particularly vulnerable to large fires as they have a high fire load, in particular if materials such as plastics are being stored which are very combustible. Industries which utilize combustible dusts (e.g. sugar and flour) are at risk of major explosions. Fires can have a severe impact on the building structure, cause damage to contents (including smoke damage) and can cause total destruction of buildings.

3.1.2. Wildfire

Definition

Wildfires are defined as uncontrolled, unwanted and unplanned fires in areas of vegetation such as grassland, bushland, forests or agricultural land. Depending on where the fire occurs and the vegetation present, they are often defined as bushfires, desert fires, forest fires, peat fires or grass fires. Wildfires can cause damage to properties, human life, vegetation and animals. However, in some cases the after-fire ecosystems (evolved with natural occurring fires) can be advantageous for some vegetation and animals. Three main natural causes of wildfires are spontaneous combustion during extensive periods of dry climate, lightning and volcanic eruptions.

However, wildfires are most often caused by human activities such as brush burning, campfires, arson, discarded cigarettes and sparks from electric equipment. Heatwaves, drought and natural climate variability, which increase dry and hot climates, such as El Niño events, all greatly increase the risk of wildfires. As a result, rising temperatures and increased drought caused by climate change has already, and will continue to, increase the risk of wildfires in many regions⁵⁰.

Location

Locations that are more exposed to this hazard are areas of combustible vegetation (dry and hot).

3.2. Best Practice Standards

Global Best Practices (GBP) for local building fire hazards are considered those of the United Kingdom (UK), known for their performance based approach. Australia has in recent years developed renowned GBP's for wildfires, combining local traditional knowledge with modern science. Both are referenced below along with other notable standards. Minimum standards often depend on the ability of local firefighting capacity to respond within a certain limited time period. Such capacity can become overwhelmed, and hence not effective, in the case of major area fires, or when fires are combined with other major hazards, such as earthquakes

⁵⁰ C.S. Stevens-Rumann et al., "Evidence for declining forest resilience to wildfires under climate change." *Ecology Letters*, 21(2) (2018): 243-252.

and storms. Therefore, BRI has measures for suppressing, containing and limiting fires in buildings.

Generally, standards in today's national building codes are more advanced for fire than other hazard categories, due to centuries of learning from large urban fires. They should suffice to achieve at B rating for BRI. Most A and AA mitigation measures will be above the national codes. If a national code does not exist, the code of a neighboring country can be used.

For a B rating, Verifiers will rely mostly on the building permit. For older buildings they may need to inquire on what fire and electrical upgrades were performed post-construction. For A and AA ratings they will need to verify the rating information provided by the building and in cases cross-check with component manufacturer's specifications, as well as the reports of any specialized fire inspectors.

Below are reference fire standards for further information. Also of interest is the 'Wildfire Guidance for Home and Business Owners', produced by the US-based Insurance Institute for Business and Home Safety, a research organization funded by major insurers and reinsurers: <https://ibhs.org/guidance/wildfire/>

Table 9. International standards for fire hazard

Authority	Standard / Description	Link
International Code Council (ICC) Standard	2015 International Wildland-Urban Interface Code (WUI Code). Establish minimum regulations for the safeguarding of life and property from the intrusion of fire Safety information and building suggestions	https://codes.iccsafe.org/content/IWUIC2015 https://www.iccsafe.org/advocacy/safety-toolkits/wildfire-safety/
International Organization for Standardization (ISO)	ISO 23932-1:2018 Fire safety engineering. General principles and requirements of Fire Safety Engineering for the built environment, but not construction sites.	https://www.iso.org/obp/ui/#iso:std:iso:23932:-1:ed-1:v1:en
Global Fire Monitoring Centre	Regional fire networks and country specific information	https://gfmcc.online/globalnetworks/globalnet.html https://gfmcc.online/management/CBFI13.html
US Forests and Range Lands Resource	The National Strategy: Wildland Fire Management Strategy. Aim to restore and maintain landscapes, fire-adapted communities, and wildfire response	https://www.forestsandrangelands.gov/strategy/communication.shtml

Table 10. Local standards for fire hazard

Authority	Standard / Description	Link
National Construction Code of Australia	Minimum standards for the design and construction of new buildings in wildfire prone areas. Rather than specifying how the building must be built, The Code specifies how the building must perform. For example, residential dwellings in high-risk areas must reduce the risk of ignition, from flames, burning embers, radiant heat, and the bushfire itself. Inspectors rate sites according to vegetation type and density and slope of land to determine the 'Bushfire Attack Level' (BAL) construction details for the appropriate BAL level must then be followed.	https://guides.slv.vic.gov.au/standards/ConstructionCode Australian Bureau of Meteorology: https://www.bom.gov.au/resources/learn-and-explore/fire-weather-knowledge-centre
UK Building Regulations Approved Document B (ADB)	Fire safety: Approved Document B: Building regulation in England covering fire safety matters within and around buildings. Part B is further split into five sections: Part B1: Means of Warning and Escape Part B2: Internal Fire Spread (Linings) Part B3: Internal Fire Spread (Structure) Part B4: External Fire Spread Part B5: Access and Facilities for the Fire Service	ADB: https://www.gov.uk/government/publications/fire-safety-approved-document-b <u>Updated 11 Mar 2025</u>

3.3. Consequential Hazards

Wildfires are more likely during periods of extreme heat (heatwave) and drought. They can significantly alter the hydrologic response of a watershed to the extent that even modest rainstorms can produce dangerous flash floods and debris flows⁵¹. All fire types can also derivate explosions.

3.4. Risk Reduction Measures

The following risk reduction measures list is in BRI and forms the basis for BRI fire rating. Measures marked 'Y' apply, if the building is in a location with that particular hazard type. The location of the site is key when avoiding wildfire hazards. A location assessment should be undertaken prior to commencement with the development. Locations less than 1.5 km away from dense vegetation areas at best can reach a A rating. BRI takes a critical view regarding the combustibility and fire rating of building materials and components. It also requires fire

⁵¹ "Emergency Assessment of Post-Fire Debris-Flow Hazards," USGS, accessed July 10, 2020. https://www.usgs.gov/natural-hazards/landslide-hazards/science/emergency-assessment-post-fire-debris-flow-hazards?qt-science_center_objects=0#qt-science_center_objects

sprinkler systems on buildings with 4 or more stories to reach an AA or A rating. Each risk reduction measure needs to be approved by at least two licensed, code-responsible engineers or architects for the building's rating to be verified. Each Verifier will be required to upload their license to confirm their qualification. Verifiers can ask the project teams to upload supporting documents.

Table 11. List of risk reduction measures for fire hazards

Fire Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local Fire (Default Hazard)	Wildfire
FI01. 1.5 km Distance from a Forest	Building must not be located more than within 1.5 km away from(0.93 miles) of forested, bushland or grassland areas.	Y				Y
FI02. Forested Site's Gradient < 15 degrees	ilf a building is located within a forested area, the gradient of the slope on the site downhill of the building must be less than 15 degrees.		Y			Y
FI03. 3+ hour Fire Rating (Non-combustible) for Key Building Elements	The structural elements of a building must be constructed with non-combustible materials that withstand intense fire for a recommended 3 hours or more.	Y			Y	Y
FI04. 1+ hour Fire Rating (Non-combustible) for Key Building Elements	The structural elements of a building must be constructed with non-combustible materials that withstand intense fire for 1 hour or more.		Y		Y	Y
FI05. 0.5+ hour Fire Rating (Non-combustible) for Key Building Elements	The structural elements of a building must be constructed with non-combustible materials that withstand intense fire for 0.5 hours or more.			Y	Y	Y
FI06. Tempered Safety Glass for Windows Larger Than 1 m ²	External windows with systems that have a total surface area of more than 1m ² or more are made of multi-pane tempered safety glass, having a fire rating of at least 1 hour.	Y			Y	Y
FI07. 2+hour Rating Fire Resistant Window Shutters	External window shutters having a fire rating of at least 2 hours.	Y			Y	Y
FI08. 2+ hour Fire Rating for Exterior Doors and Emergency Route Firedoors	Exterior doors and emergency route firedoors must havinge a fire rating of at least 2+ hours.	Y			Y	Y
FI09. 1+ hour Fire Rating for Exterior Doors and 2+ hour Fire Rating for Emergency Route Firedoors	Exterior doors having a fire rating of 1+ hour and emergency route firedoors a fire rating of at least 2 hours		Y		Y	Y
FI10. Fire Rated Decks, Porches and Fences	Decks, porches, and fences within 10m offrom a building aremust be made of materials that do not contribute significantly to growth of fire. The materials must havewith a fire rating of 2+ hours,	Y			Y	Y

Fire Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local Fire (Default Hazard)	Wildfire
	+					
FI11. Chimneys with Spark Arrestors	Chimneys and stove pipes must contain spark arrestors to reduce the chance of ignition that may cause a fire.	Y			Y	Y
FI12. Fire Rated Eaves, Soffits and Vents	Eaves, soffits, and vents must be made of materials with a fire rating of 1+ hours.		Y		Y	Y
FI13. Fire Resistant Utility and Equipment Connections	Above ground utility or equipment connections for heating, cooling, electricity, fuel, telecom, data, water, sewage, or waste between the outdoor space and the building that are located above ground must be built or sealed with fire-resistant materials having a fire rating of to a 2+ hours level.	Y			Y	Y
FI14. Sprinkler System for Building with 4+ Stories	Buildings with 4 or more above-ground stories must have a functional and well-maintained fire sprinkler system that can supply at least 0.25 liters per hour of water per square meter of building space, or per national code (whichever is greater).		Y		Y	Y
FI15. Defensible Building Perimeter > 10 m	Building must have a minimum of 10 meter defensive boundary around its outer walls, in all directions, except for abutting buildings that are built with reinforced concrete or other 3+ hour fire rated material.	Y			Y	Y
FI16. Defensible Building Perimeter > 5 m	Building has a minimum 5 meter defensive boundary around its outer walls in all directions, except for abutting buildings that are built with reinforced concrete or other 2+ hour fire rated material.		Y		Y	Y
FI17. Reliable Fire-Water Supply	Building has access to firewater within a distance of 20 meters, delivering at least 1 liter per minute per square meter of building area to every storey, or per national code (whichever is greater).		Y		Y	Y
FI18. Emergency Access Routes	Building has adequate emergency access routes for fire engines and emergency vehicles, personnel to be provided within the site premises. As			Y	Y	Y

Fire Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Local Fire (Default Hazard)	Wildfire
	well as for building occupants to safely evacuate and fire/emergency personnel to reach all spaces					
FI19. Firebreaks Inside Buildings	The building has firebreaks creating interior compartments having a maximum size of 2,500 square meters or per the national building code, whichever is lower.	Y			Y	Y
FI20. Building electric circuits and wiring have protections to prevent electrical fires per NEC 2000 or equivalent.	Building electric circuits and wiring have protections to prevent electrical fires.			Y	Y	Y
FI21. Approved Design Review for Fire Hazard Risk Management Measures	The building design and occupancy must be submitted to and approved by the local approving agency, as prescribed by the governing national or local building code.		Y		Y	Y
FI22. Construction Independently Audited for Fire Hazard Resilience Measures	A second independent qualified engineer, different from the one assigned from the approving agency or the responsible engineer of record for the project, will be engaged to inspect during critical milestones and confirm conformance the construction of the fire resilience measures.	Y			Y	Y

FI01. 1.5 km Distance from a Forest (AA Rating)

Criteria: Building must be located more than 1.5 km away from forested, bushland or grassland areas.

Assessor Procedure: Show on a recent arial map distance between building and nearest forested, bushland or grassland area within a 1.5km radius. Spaces such as golf courses, urban parks, or farms are not considered wildland.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Dense vegetation (also known as wildland⁵²) is considered a highly combustible material when combined with dry conditions, high external temperatures and wind. The greater the distance from wildland areas the lower the risk of fire to reach the building.

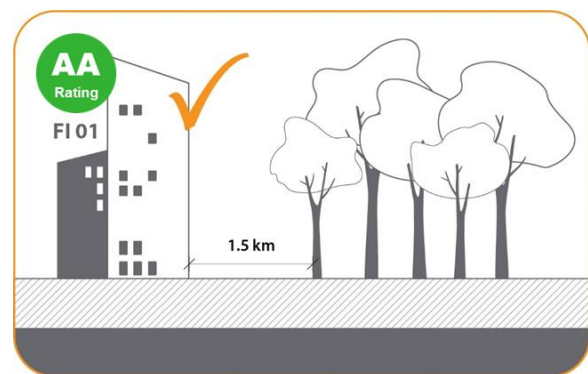


Figure 28. Distance from forest needs to be at least 1.5 km for AA rated buildings

⁵² **Wildland area:** A highly vegetated undeveloped area (less than 25% comprising of a minimum of 0.5 ha. There is

no minimum vegetation height required, however, there must be tree species included in the vegetation.

FI02. Forested Site's Gradient < 15 degrees (A Rating)

Criteria: If a building is located within a forested area, the gradient of the slope on the site downhill of the building must be less than 15 degrees.

Assessor Procedure: Show on a topographic map or a site cross-sectional drawing the slope of the land downhill of the building for a distance of 100m, even if this lies outside the property boundaries. For land that is sloped in multiple directions, pick the side with the greatest slope.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria and spot check the gradient during the site visit.

Background: This reduces the speed in which a fire could spread through a forested area, with increased flow of wind uphill. Typically, fires tend to move faster through steep terrain as the air may be drawn from the downhill side, pushing flames into the slope and increasing the contact with the firewood.



Figure 29. Gradient in forested sites

FI03. 3+hour Fire Rating (Non-combustible) for Key Building Elements (AA Rating)

Criteria: The structural elements⁵³ of a building must be constructed with non-combustible materials⁵⁴ that withstand intense fire for 3 hours or more.

Assessor Procedure: Provide documentation showing that the materials use for the building's foundation, columns, floor plates, trusses, roof and exterior insulation, cladding and panelling can withstand fire and not collapse for at least 3 hours.

Such materials may include steel, concrete, fire-resistant timber and the use of fire retardants. The coating of a structural material such as paint, cladding, roof tiles and insulative elements can have an impact on the combustibility. Properly constructed structural elements and bearing walls built with reinforced concrete satisfy this measure.

Documentation can include statements and test reports from relevant suppliers.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: A fire rating of 3+ hours means that the integrity of materials used in the structure and exterior walls⁵⁵ must withstand the effects of a fire and not collapse for at least 3 hours. This allows time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to extinguish the fire.



Figure 30. 3+hour fire rating (non-combustible) for key building elements

⁵³ **Structural elements:** The main components of a building such as columns, floor plates, foundations, trusses, and roof structures. These are the basic skeleton of a building which withstand the gravitational forces put upon a building. The structure is the frame which all other coverings are applied to.

⁵⁴ **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have no or a very low flammability rating.

⁵⁵ **Exterior walls:** The vertical elements of a building which separate the exterior from the interior. These usually hold a structural function. However, they can also be required for protection against external elements such as weather and climate conditions.

FI04. 1+hour Fire Rating (Non-combustible) for Key Building Elements (A Rating)

Criteria: The structural elements⁵⁶ of a building must be constructed with non-combustible materials⁵⁷ that withstand intense fire for 1 hour or more.

Assessor Procedure: Provide documentation showing that the materials use for the building's foundation, columns, floor plates, trusses, roof and exterior insulation, cladding and panelling can withstand fire and not collapse for at least 1 hour.

Such materials may include steel, concrete, fire-resistant timber and the use of fire retardants. The coating of a structural material such as paint, cladding, roof tiles and insulative elements can have an impact on the combustibility. Properly constructed structural elements and bearing walls built with reinforced concrete satisfy this measure.

Documentation can include statements and test reports from relevant suppliers.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: A fire rating of 1 hour means that the integrity of materials used in the structure and exterior walls⁵⁸ must withstand the effects of a fire and not collapse for at least 1 hour. This allows time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to extinguish the fire.



Figure 31. 1+hour fire rating (non-combustible) for key building elements

⁵⁶ **Structural elements:** The main components of a building such as columns, floor plates, foundations, trusses, and roof structures. These are the basic skeleton of a building which withstand the gravitational forces put upon a building. The structure is the frame which all other coverings are applied to.

⁵⁷ **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have no or a very low flammability rating.

⁵⁸ **Exterior walls:** The vertical elements of a building which separate the exterior from the interior. These usually hold a structural function. However, they can also be required for protection against external elements such as weather and climate conditions.

FI05. 0.5+hour Fire Rating (Non-combustible) for Key Building Elements (B Rating)

Criteria: The structural elements⁵⁹ of a building must be constructed with non-combustible materials⁶⁰ that withstand intense fire for 0.5 hours or more.

Assessor Procedure: Provide documentation showing that the materials use for the building's foundation, columns, floor plates, trusses, roof and exterior insulation, cladding and panelling can withstand fire and not collapse for at least 0.5 hours.

Such materials may include steel, concrete, fire-resistant timber and the use of fire retardants. The coating of a structural material such as paint, cladding, roof tiles and insulative elements can have an impact on the combustibility. Properly constructed structural elements and bearing walls built with reinforced concrete satisfy this measure.

Documentation can include statements and test reports from relevant suppliers.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: A fire rating of 0.5 hours means that the integrity of materials used in the structure and exterior walls⁶¹ must withstand the effects of a fire and not collapse for at least 0.5 hours. This allows time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to extinguish the fire.



Figure 32. 0.5+hour fire rating (non-combustible) for key building elements

⁵⁹ **Structural elements:** The main components of a building such as columns, floor plates, foundations, trusses, and roof structures. These are the basic skeleton of a building which withstand the gravitational forces put upon a building. The structure is the frame which all other coverings are applied to.

⁶⁰ **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have no or a very low flammability rating.

⁶¹ **Exterior walls:** The vertical elements of a building which separate the exterior from the interior. These usually hold a structural function. However, they can also be required for protection against external elements such as weather and climate conditions.

FI06. Tempered Safety Glass for Windows Larger Than 1m² (AA Rating)

Criteria: External windows⁶² with a surface area of 1m² or more are made of multi-pane⁶³ tempered safety glass⁶⁴, having a fire rating of at least 1 hour.

Assessor Procedure: Provide manufacturer documentation or photographic evidence of labeling (etching, watermark, etc.) for the subject external windows showing that they meet the above criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check a sample of windows during the site visit.

Background: Windows may break due to radiant heat from fires which could allow

the fire to spread faster and cause injury to occupants if the glass were to shatter. A fire rating of 1+ hour allows time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to distinguish the fire.

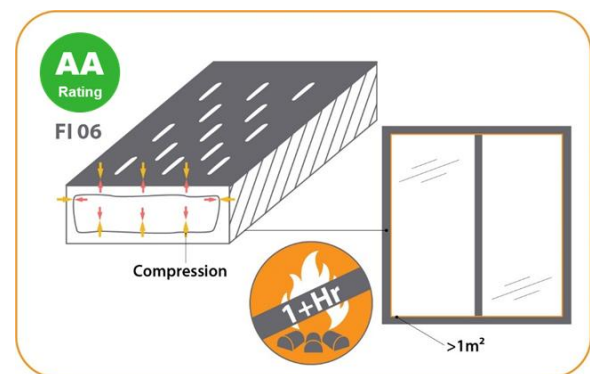


Figure 33. Tempered safety glass for windows above 1m²

⁶² **Windows:** Openings in a wall or roof of a building which are fitted with glass, allowing light or air to enter into a building, as well as releasing air and providing a view between the interior and exterior.

⁶³ **Multiple panes of glass:** The number of sheets or layers of glass used to create the window. Historically, most properties would have contained one single pane, however, today double-glazed windows are more

common, with this increasing depending on the thermal, acoustic and lighting insulation required.

⁶⁴

Tempered safety glass: A glass is created through controlled thermal or chemical treatments to increase its strength to protect it against shattering - when forces are applied. The outer surfaces are put into compression and the interior into tension.

FI07. Fire Resistant Window Shutters (AA Rating)

Criteria: External window shutters having a fire rating of at least 2 hours.

Assessor Procedure: Provide manufacturer documentation showing that they meet the above criteria. Materials for shutters may include metal, fire-resistant composites and/or the use of fire retardants

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check a sample of windows during the site visit.

Background: Window shutters⁶⁵ must be fire resistant to withstand in tense fire for a

recommended 2 hour or more. These form part of the façade as they are attached to the walls, therefore, they pose the same fire risk as inappropriate materials located in the cladding system. .

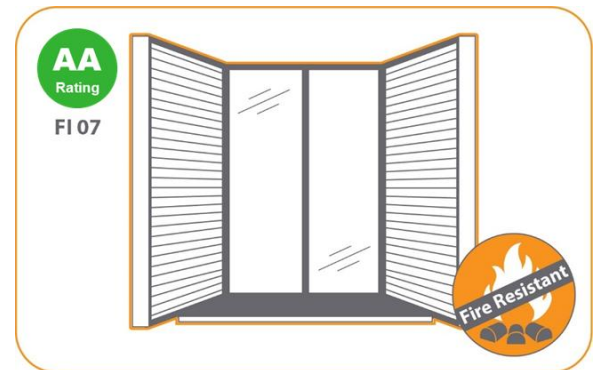


Figure 34. Fire resistant window shutters

⁶⁵ **Window shutters:** External elements attached to the exterior of a building over the window opening. These are used for security, privacy or solar protection. These can be

made from timber, metal or a plastic composite, opening out like doors, or on a roller system.

FI08. 2+ hour Fire Rating for Exterior Doors and Emergency Route Firedoors (AA Rating)

Criteria: Exterior doors⁶⁶ and emergency route firedoors⁶⁷ having a fire rating of at least 2 hours.

Assessor Procedure: Provide manufacturer documentation or photographic evidence of labeling (etching, watermark etc.) for the subject doors showing that they meet the above criteria.

If the walls, ceiling and doors that connect the ground floor lobby area to the rest of the building are fire rated for 2+ hours, this measure can be satisfied, even if the lobby exterior doors are rated below 2 hours. In this case, any emergency stairway doors should exit directly to the building exterior.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check a sample of doors during the site visit.

Background: These doors should allow functional access during fires. An open door can draw air in fast and increase the rate of fire spread. A fire rating of 2+ hours means that the integrity of materials used in these doors must withstand the effects of a fire for at least 2+ hours, allowing time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to distinguish the fire.

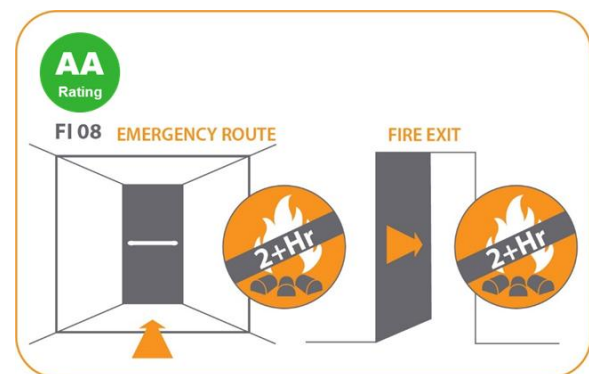


Figure 35. Doors (external & emergency route) 2+hours fire rated

⁶⁶ **Exterior doors:** Door that provide an entry or exit point between the interior and exterior of a building.

⁶⁷ **Emergency route firedoors:** Doors that compartmentalize the evacuation routes to reduce the rate

of fire spread. These may be located along corridors, at entryways to staircases and at the threshold of an apartment.

FI09. 1+ hour Fire Rating for Exterior Doors and 2+ hour Fire Rating for Emergency Route Firedoors (A Rating)

Criteria: Exterior doors⁶⁸ having a fire rating of 1+ hour and emergency route firedoors⁶⁹ a fire rating of at least 2 hours.

Assessor Procedure: Provide manufacturer documentation or photographic evidence of labeling (etching, watermark etc) for the subject doors showing that they meet the above criteria.

If the walls, ceiling and doors that connect the ground floor lobby area to the rest of the building are fire rated for 1+ hours, this measure can be satisfied, even if the lobby exterior doors are rated below 1 hour. In this case, the 2+ hour emergency stairway doors should exit directly to the building exterior.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check a sample of doors during the site visit.

Background: These doors should allow functional access during fires. An open door can draw air in fast and increase the rate of fire spread. A fire rating of 1+/2+ hours means that the integrity of materials used in these doors must withstand the effects of a fire for at least 1+/2+ hours, allowing time for occupants of a building to escape and the fire rescue team or fire risk reduction technologies to distinguish the fire.

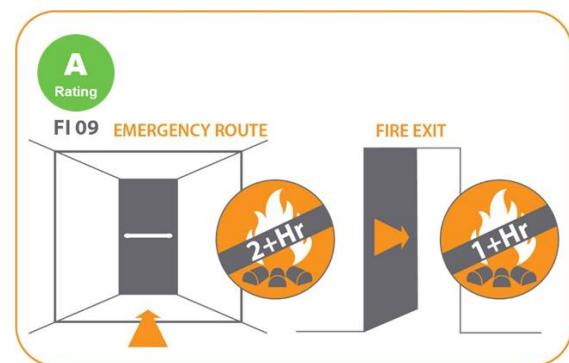


Figure 36. Fire rated external doors (1+hour) and emergency route (2+hour) ⁷⁰

⁶⁸ **Exterior doors:** Door that provide an entry or exit point between the interior and exterior of a building.

⁶⁹ **Emergency route firedoors:** Doors that compartmentalize the evacuation routes to reduce the rate of fire spread. These may be located along corridors, at

entryways to staircases and at the threshold of an apartment.

⁷⁰ **Exterior doors:** Door that that provide an entry or exit point between the interior and exterior of a building.

FI10. Fire Rated Decks, Porches and Fences (AA Rating)

Criteria: Decks⁷¹, porches⁷², and fences⁷³ within 10m of a building are made of materials with a fire rating of 2+ hours,

Assessor Procedure: Provide documentation and photos showing that they meet the above criteria. Materials may include metal, fire-resistant composites, fire-resistant timber and/or the use of fire retardants

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check during the site visit.

Background: Decks separate floors or roofs and contribute to the movement of fire up through a building. Porches can

catch fire more quickly than an enclosed building and therefore must withstand a stone fire in order to reduce spread into occupied spaces. Fences can be constructed of timber, steel or plastic, and if combustible, could spread a fire over a large surface area very quickly when ignited.

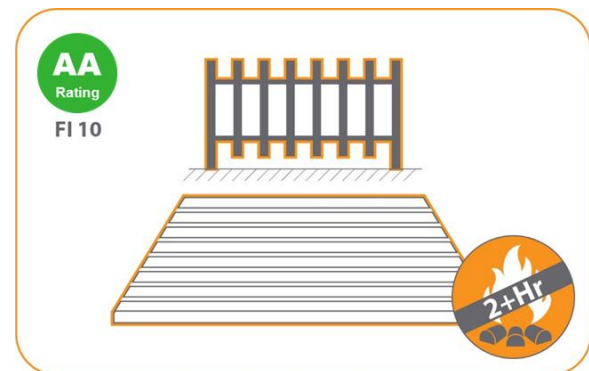


Figure 37. Fire repelling decks, porches and fences

⁷¹ **Decks:** Flat/horizontal surfaces that can withstand a weight, such as a floor, balcony, or roof terrace. They are often elevated above ground.

⁷² **Porches:** Roofed structures that sit outside of the external structural building envelope. They normally have

open sides and do not have the same thermal or weather-proofing qualities as an enclosed building.

⁷³ **Fences:** Features normally located on the perimeter of a site and distinguish the building boundary. They can also be located adjacent to a building to distinguish spaces such as refuse or private and public spaces.

FI11. Chimneys and Stove Pipes with Spark Arrestors (AA Rating)

Criteria: Chimneys and stove pipes must contain spark arrestors to reduce the chance of ignition that may cause a fire.

Assessor Procedure: Provide documentation showing that they meet the above criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: This measure applies also to mechanical rooms of buildings of all typologies to seek a AA rating. Chimneys are flues which extract gases created from controlled combustion or reactions. It is important to remove these safely from a building to ensure that no ignitable gases are left behind in the building. Mechanical

systems can be sources of very flammable materials and therefore must contain a series of preventative measures against fire protection, such as spark arrestors. Spark arrestors stop the emission of flammable particles from a chimney and/or stove which could escape the mechanical systems and ignite nearby flammable materials.

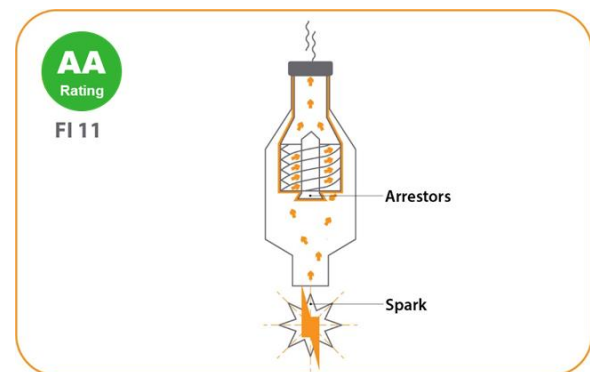


Figure 38. Spark arrestors in chimneys

FI12. Fire Rated Eaves, Soffits and Vents (A Rating)

Criteria: Eaves⁷⁴, soffits⁷⁵, and vents⁷⁶ are made of materials with a fire rating of 1+ hours.

Assessor Procedure: Provide documentation showing that they meet the above criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: The eaves, soffits and vents tend to run over long distances through the building, either vertically or horizontally and therefore, once penetrated by fire, could accelerate the spread of fire across the whole building. This also prevents the fire from entering the building attic easily. These elements can be fire protected using fire resistant or

non-combustible materials, which do not easily ignite when exposed to a naked flame or spark. Such materials may include steel, concrete, fire-resistant timber and the use of fire retardants. Be aware that coating such as paint, and insulative elements can have an impact on combustibility.

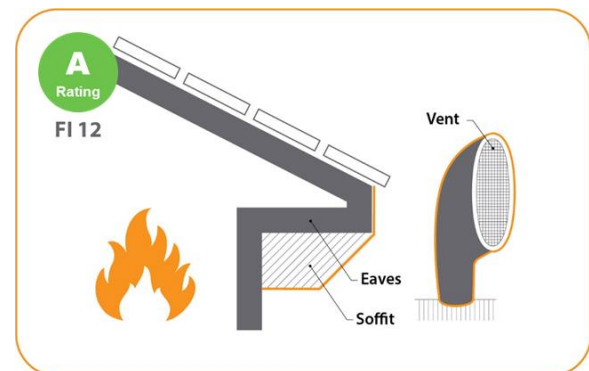


Figure 39. Eaves, soffits, and vents to be protected by fire

⁷⁴ **Eaves:** The overhanging part of the roof, they cantilever over the edge of the walls, allowing for drainage of water running off the roof to fall to the ground without damaging the walls. It also provides some protection from wind, rain, snow, and solar heat for the walls.

⁷⁵ **Soffits:** The undercarriage of the overhang eaves, balconies, or arches. They can provide protection to the structure of the roof that may form part of the eaves.

⁷⁶ **Vents:** Mechanical channels for stale air to be released from a building and fresh air to be drawn into the building. Vents usually have a mesh or slats over the transition point between exterior and interior to direct the air and filter particles. Vents form part of a larger duct system that can circulate an entire building or simply penetrate a wall.

FI13. Fire Resistant Utility and Equipment Connections (AA Rating)

Criteria: Above ground utility or equipment connections for heating, cooling, electricity, fuel, telecom, data, water, sewage, or waste between the outdoor space and the building are built or sealed with materials having a fire rating of t2+ hours.

Assessor Procedure: Provide documentation showing that they meet the above criteria. Materials may include metall, concrete, the use of fire retardants, fire rated insulation, or fire dampers; otherwise they should be placed underground for fire safety.

Verifier Procedure: Review the documentation provided by the assessor

and confirm the site meets the criteria. Spot check during the site visit.

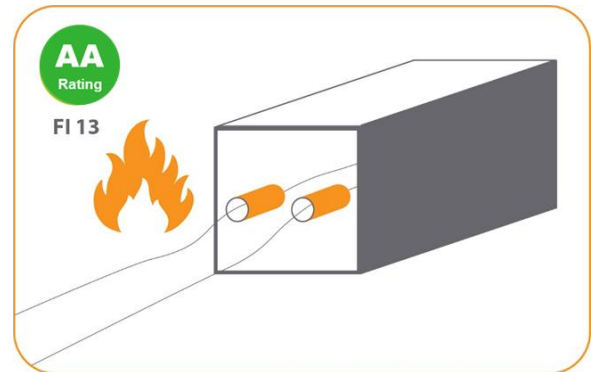


Figure 54. Fire rated utility and equipment connections

FI14. Sprinkler System for Building with 4+ Stories (A Rating)

Criteria: Buildings with 4 or more above-ground stories must have a functional and well-maintained fire sprinkler system that can supply at least 0.25 liters per hour of water per square meter of building space, or per national code (whichever is greater).

Assessor Procedure: Provide documentation showing that they meet the above criteria, including calculation of sprinkler system water capacity in liters per hour per square meter of interior building space, including storage spaces, corridors, basements and unoccupied floors, or per national code (whichever is greater).

Unoccupied floors count as stories and should also have sprinklers. Basements don't count as stories, but if the building has sprinklers, this should include the basement(s).

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check during the site visit, including a sample of corridors, storage areas, basements and unoccupied floors.

Background: Fire sprinkler systems reduce the impact of fire damage and disperse

water in order to dampen any fire that may spread through a building. The system is paired with a smoke and fire alarm system, which detects the sources of a fire within a building and engages the sprinkler water system to irrigate the building. The overarching aim of fire strategies is to dampen flammable materials as well as the fire source, slowing down the pace of fire and to increase the amount of time that occupants may have to evacuate during a fire, as well as minimizing the damage to the building. Fires can also occur in unoccupied floors and basements from vehicles, equipment and stored items and thus require sprinkler coverage.

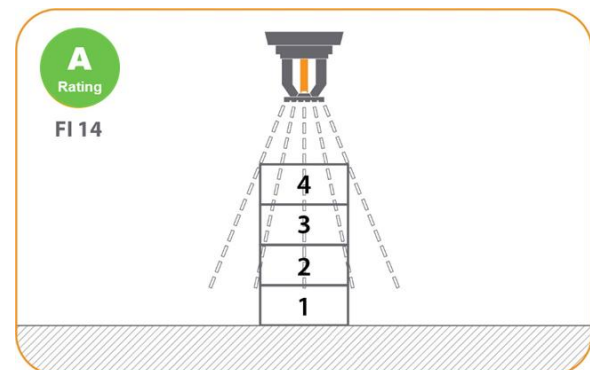


Figure 40. Sprinkler system for building with 4+storeys

FI15. Defensible Building Perimeter > 10 m (AA Rating)

Criteria: Building must have a minimum 10 meter defensive boundary⁷⁷ around its outer walls in all directions, except for abutting buildings that are built with reinforced concrete or other 3+ hour fire rated.

Assessor Procedure: Provide arial map and site plan showing that they meet the above criteria. The perimeter can consist of non-combustible⁷⁸ pavements, decks or low-profile, low-combustion and well-maintained vegetation and be cleared of any combustible materials.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check during the site visit by walking and/or driving the perimeter of the building.

Background: The defensive boundary combined with 3+ hour fire rated outer walls aims to alleviate the path of fire spreading, either from the outside into the building or from the building to the surrounding area.

Reference 2015 International Wildland-Urban Interface Code (IWUIC),6.03 and 2018 IBC, 706.4.

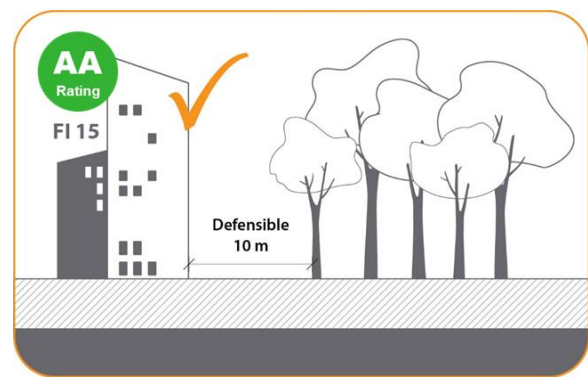


Figure 41. Defensible zone of more than 10m around building perimeter

⁷⁷ **Defensible boundary:** A space along a perimeter, which can be of varied depth or material, allowing for protection from fire, damage, or trespassing.

⁷⁸ **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have with no or a very low flammability rating.

FI16. Defensible Building Perimeter > 5 m (A Rating)

Criteria: Building has a minimum 5 meter defensive boundary⁷⁹ around its outer walls in all directions, except for abutting buildings that are built with reinforced concrete or other [2+ hour fire rated](#).

Assessor Procedure: Provide arial map and site plan showing that they meet the above criteria. The perimeter can consist of non-combustible⁸⁰ pavements, decks or low-profile, low-combustion and well-maintained vegetation and be cleared of any combustible materials.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check during the site visit by walking and/or driving the perimeter of the building.

Background: This aims to alleviate the path of fire spreading, either from the outside into the building or from the building to the surrounding area.

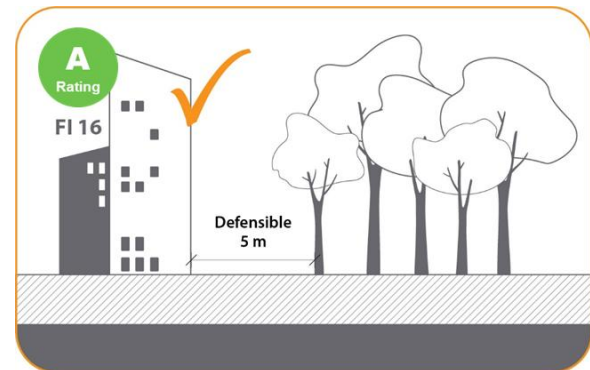


Figure 42. Defensible zone of more than 5 m around building perimeter

⁷⁹ **Defensible boundary:** A space along a perimeter, which can be of varied depth or material, allowing for protection from fire, damage, or trespassing.

⁸⁰ **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have with no or a very low flammability rating.

FI17. Reliable Fire-Water Supply (A Rating)

Criteria: Building has access to firewater within a distance of 20 meters, delivering at least 1 liter per minute per square meter of building area to every storey, or per national code (whichever is greater).

Assessor Procedure: Provide arial site plan and photos of all firewater locations and calculations that show they meet the above criteria. Fire-water supply sources can include municipal water tanks⁸¹, private gravity or pressure tanks⁸² and public hydrants.)

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. During the site visit view all the fire-water supply sources.

Background: Reliable and adequate fire-water supply is important for both a building's fire water sprinkler system and exterior fire hydrants. Emergency fire

responders should have easy access to nearby fire hydrants to link up when distinguishing a building fire. From the outset of a project, the provision for fire hydrants⁸³ in the mechanical design of buildings should be incorporated, throughout construction and project completion stages. For high-rise buildings ground level pressure may be inadequate to reach the upper stories and water tanks at higher levels may also be required. These may also be connected to the sprinkler system.

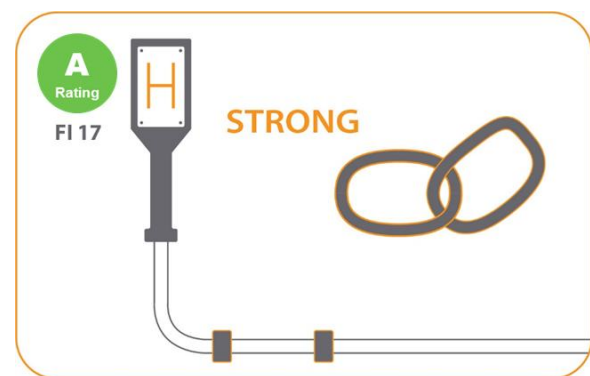


Figure 43. Reliable fire-water supply

⁸¹ **Municipal water tank sources:** large containers which hold water for drinking, irrigation, fire suppression, or water supply.

⁸² **Pressure tanks:** Containers of water which hold water at pressure different from ambient pressure. They assist in delivering water fast to multiple areas of a building.

⁸³ **Fire hydrants:** Connection points by which firefighters can tap into the water supply to fuel their distinguishing of a fire.

FI18. Emergency Access Routes (B Rating)

Criteria: Building has adequate access routes for fire engines and emergency vehicles, as well as for building occupants to safely evacuate and fire/emergency personnel to reach all spaces.

Assessor Procedure: Identify the emergency access route on a map or site plan, as well as the occupant evacuation route and fire/emergency personnel access corridors. Note any road restrictions with regards to width, turning curves, height and weight load that would impede access.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria. Spot check during the site visit by driving the emergency access route, and walking some of the evacuation and entry corridors, noting appropriate exit signage.

Background: Vehicle access routes should consider the road width, the distance between gateways, turning circle/points (between curves and walls), clearance

height, and weight capacity. The minimum requirements for the above can be found in the national fire codes, however as a guidance the UK Building Regulations suggests the following minimum values in Table 12. However, local conditions such as size of fire and emergency vehicles can be considered in this.

In addition to external access routes, a large aspect of the evacuation design for a building is to ensure ease of exit for all occupants to an external space, located a safe distance from the building. The emergency access routes should be unobstructed and allow for swift single direction evacuation from every room within a building. The emergency access routes should also allow for firefighting teams to enter a building and assist in evacuation and extinguishing the fires.

The width of the access routes to accommodate the number of building occupants, fire fighters and their equipment is important. Firefighting shafts (stairs and lobbies) should be provided and fire protected following the national fire safety code. The routes should be clearly signposted and should be visible in dense smoke or during a power outage.

Table 12. Vehicle access routes according to the UK Building Regulations⁸⁴

Appliance type	Minimum width of road between kerbs (m)	Minimum width of gateways (m)	Minimum turning circle between kerbs (m)	Minimum turning circle between walls (m)	Minimum clearance height (m)	Minimum carrying capacity (tonnes)
Pump	3.7	3.1	16.8	19.2	3.7	12.5
High reach	3.7	3.1	26.0	29.0	4.0	17.0

NOTES:

1. Fire appliances are not standardised. The building control body may, in consultation with the local fire and rescue service, use other dimensions.
2. The roadbase can be designed to 12.5 tonne capacity. Structures such as bridges should have the full 17-tonne capacity. The weight of high reach appliances is distributed over a number of axles, so infrequent use of a route designed to accommodate 12.5 tonnes should not cause damage.

See para 13.4

Fire and rescue service vehicles should not have to reverse more than 20m from the end of an access road.

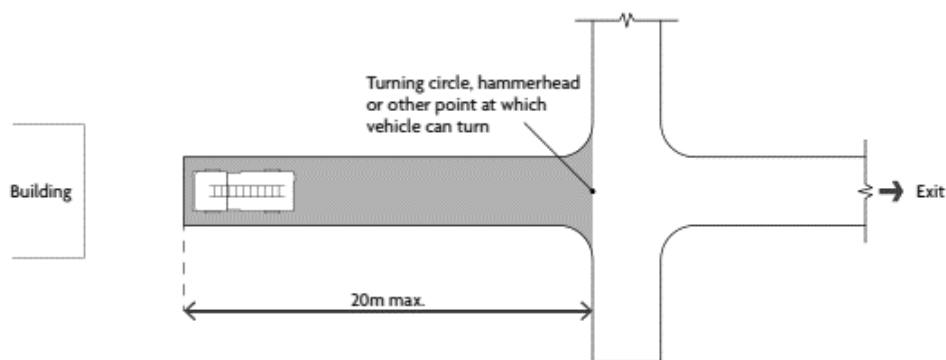


Figure 44. Turning points for fire engines

⁸⁴ Government of UK 2020.

FI19. Firebreaks Inside Buildings (AA Rating)

Criteria: The building has firebreaks creating interior compartments having a maximum size of 2,500 square meters or per the national building code, whichever is lower.

Assessor Procedure: Provide documents including drawings and statement by the code-responsible engineer showing that the building meets the criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Part of the built materials' design process should involve the consideration of how fire could spread through a building, vertically and through the materials used. The fire breaks or cavity barriers are a construction within a cavity, mainly in walls, that either close the cavity to stop or restrict the movement of smoke/flames. In this way the fire is not propagated throughout the building.

Fire breaks are made of non-combustible insulation materials and cover the full depth of the insulation of the building to prevent any fire propagation. However,

they must not impede moisture drainage or ventilation during normal usage.

In addition, vertical fire breaks around vulnerable openings (windows and doors) might need to be considered to prevent any lateral fire spread. It is important to consider how the materials and building elements are joined and connected.

Junctions between walls and floors, between pipes and flues, between the primary structure and cladding must all be considered when designing to reduce the spread of fire in the buildings. All these junctions or internal fire breaks must be designed and built to reduce the spread of fires in the building.

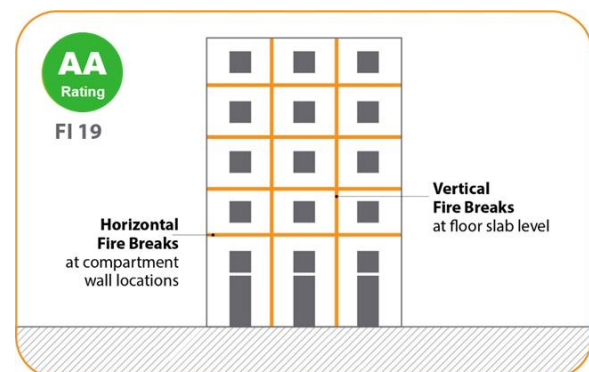


Figure 45. Typical example of fire barrier locations

FI20. Adequate Electric Circuit and Wiring Protection (B Rating)

Criteria: Building electric circuits and wiring have protections to prevent electrical fires per NEC 2000 or equivalent.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing that the building meets the criteria.

Verifier Procedure: Review the documentation provided by the assessor and confirm the building meets the criteria.

Background: Buildings contain a lot of electrical equipment, piping and connections; therefore, it is important to test and sufficiently design safety features

into the electric circuit according to the national electrical code or if one is not locally available, the code of a neighboring country. National electric codes approved in the year 2000 or after generally satisfy this measure. Earlier buildings may need fire safety upgrades.

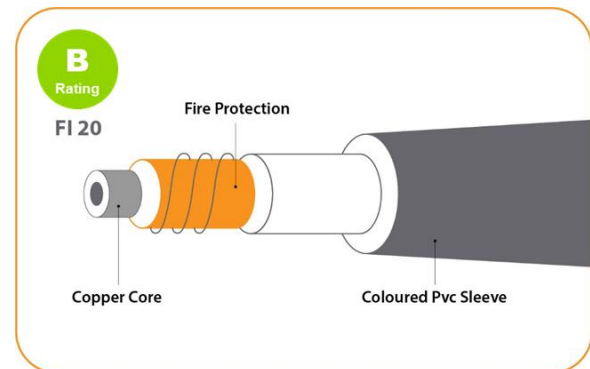


Figure 46. Protection for electric circuits and wires

FI21. Approved Design Review for Fire Hazard Risk Management Measures (A Rating)

Criteria: The building design and occupancy must be submitted to and approved by the local approving agency, as prescribed by the governing national or local building code.

Assessor Procedure: Provide documentation from the approving agency, the certificate of completion or equivalent. Also provide documentation from the electrical/fire engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: The construction must be observed and documented at critical milestones by the electrical/fire engineer

of record responsible for the code compliance at design stage. This cross-check helps to minimize the scope for error and assure that what is constructed was what was initially designed, and will aid verification.

In newer buildings the responsible engineer or architect should be able to provide the assessor with sample photos of inspections during construction milestones that relate to the fire measures herein.



Figure 62. Design reviewed by local approving agency

FI22. Construction Independently Audited for Fire Hazard Resilience Measures (AA Rating)

Criteria: A second independent qualified engineer, different from the one assigned from the approving agency or the responsible engineer of record for the project, will be engaged during critical milestones to inspect and confirm conformance of the construction of the fire resilience measures d.

Assessor Procedure: Provide documentation from the independent qualified engineer attesting to compliance of the constructed work including construction observations and review stamped as-built drawings, submitted in a stamped letter report of conformance.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: This independent professional should provide the assessor sample photos of milestone inspections and discuss the fire-related recommendations for the building.

For new construction, the engineer performing the independent construction inspection of critical milestones should be the BRI verifier, when possible,.



Figure 47. Construction independently audited for fire hazard risk management measures

4. Geo-seismic

4.1. Hazard Types

There are four types of Geo-seismic Hazards: Earthquakes, Landslides, Volcanos and Subsidence. Each are discussed in further detail. As geo-seismic forces can be very powerful, the location of a building is a major risk factor, particularly for landslide and volcanic hazards. They can also be interrelated to other hazards. Example: Landslides can be triggered by earthquakes or heavy rain, or be followed by fire.

4.1.1. Earthquake

Definition

Earthquakes are defined as the shaking of the Earth's surface caused by the movement of the Earth's tectonic plates which can build up pressure and release energy suddenly. The point where the pressure is released is called the focus, and the point on the Earth's surface above this is called the epicenter. Earthquakes release seismic waves, which radiate out from the focus. According to the National Earthquake Information Centre, it is estimated that there are 500,000 detectable earthquakes in the world each year. 100,000 of those can be felt, and 100 of them cause damage. Around 90% of the world's earthquakes occur in a tectonic fault zone surrounding the Pacific Ocean, known as the "Ring of Fire" or the Circum-Pacific belt. The next most seismic region (5-6% of earthquakes) is the Alpide belt (extends from Mediterranean region, eastward through Turkey, Iran, and northern India⁸⁵).

Location

Locations that are more exposed to earthquake hazard are the ones with close proximity to seismic activity such as geological seismic faults⁸⁶ or an active volcano. Faults are the junction (boundaries or deformations) in which two tectonic rock plates move relative to each other. The three fault mechanisms are strike slip (shearing lateral movement of the two adjoining plates), normal faulting (one plate sliding downward relative to the other from tensile separation force) and thrust or reverse faults (one plate colliding and moving upward over the other from compressive forces between the plate). The area of relative downward movement of a plate from normal and reverse fault displacements is termed a subduction zone.

⁸⁵ "Earthquake Hazards," USGS, accessed September 1, 2020. <https://www.usgs.gov/natural-hazards/earthquake-hazards/education>

⁸⁶ "What is the relationship between faults and earthquakes? What happens to a fault when an earthquake occurs?" USGS, accessed September 10, 2020. https://www.usgs.gov/faqs/what-relationship-between-faults-and-earthquakes-what-happens-a-fault-when-earthquake-occurs?qt-news_science_products=0#qt-news_science_products

4.1.2. Volcano

Definition

A volcano is an opening in the Earth's crust which can erupt, allowing magma, hot ash and gases to escape. Most volcanic eruptions occur at tectonic plate boundaries, such as on Pacific "Ring of Fire", e.g. along the Andes or on volcanic islands such as Indonesia⁸⁷. Others are caused by spreading ridges at diverging plates, such as in Iceland, and by hot spots in the Earth's crust such as Hawaii or Yellowstone in the USA. Spreading ridges and hot spot eruptions are usually less violent than those on plate boundaries because of the lack of significant frictional force from plate interface movement. There are roughly 1,500 active volcanoes globally and an estimated 800 million people live within 100 km of an active volcano⁸⁸. As eruptions can be rare, many people live on or near these volcanic landscapes for a number of reasons, including fertile soils and geothermal resources. However, volcanic eruptions can be very dangerous and vary widely. They can produce the following⁸⁹:

- **Volcanic bombs:** large pieces of very hot rock which are ejected as magma by volcanic explosions. They are also referred to as ballistics.
- **Pyroclastic flows:** fast moving clouds of hot ash, gas and rock which are very destructive raining down deposits of this debris
- **Lava flows:** extremely hot rivers of molten rock
- **Lahars:** hot or cold slurry of water and rock fragments that move down the volcano slope.

The following affect larger distances:

- **Ash clouds:** small pieces of rock and glass carried in the air
- **Toxic gases** including carbon dioxide, carbon monoxide, sulfur dioxide and hydrogen chloride.

These hazards can be a risk to those in nearby settlements, aircrafts and those living much further away. For example, accumulation of volcanic ash can become a significant building roof load, similar to snowpack.

For purposes of BRI, snow load, including avalanche is categorized as a geo-seismic hazard to be addressed on a site specific basis to meet BRI structural and site criteria.

Location

Locations that are more exposed to this hazard are the ones with close proximity to an active volcano especially when situated in lava flowpaths.

⁸⁷ S.C. Loughlin et al., *Global volcanic hazards and risk* (Cambridge: Cambridge University Press, 2015): 5.

⁸⁸ S.C. Loughlin et al., *Global volcanic hazards and risk* (Cambridge: Cambridge University Press, 2015): 2.

⁸⁹ WBG and GFDRR, *ThinkHazard! Methodology Report Version 2* (2017),

https://gfdr.github.io/thinkhazardmethods/download/thinkhazard-methodology-report_v2_0.pdf

4.1.3. Landslide

Definition

Landslides are defined as the movement of rock, debris or soil down a slope and occur when the forces acting down the slope exceed the strength of the material below. They are controlled by terrain, slope, geology, land cover. Factors which increase the movement of material down a slope can trigger landslides. These forces include: heavy rainfall, earthquakes, river or stream erosion, volcanic activity, freezing and thawing and snowmelt. Human activities can also increase the risk of landslides via the disturbance of land and soil, deforestation, mining of hillsides and unmanaged construction activities. In terms of influencing slope movements, BRI also considers snowpack and avalanche to be geo-seismic hazards.

Landslides vary in extent and intensity. They may range from rock-falls to several thousand cubic meters of material. Some landslides affect very small areas, whilst others cover entire regions. The distance that the material moves during an event also differs, ranging from a few centimeters to hundreds of kilometers. Likewise, the velocity of the landslide can range from almost undetectable movement to a sudden and rapid collapse⁹⁰. Studies suggest that the most exposed countries to landslide risk are located in parts of Asia, Central and South America⁹¹. Although landslides also occur in other countries, the risk is lower in more developed regions as there is normally lower exposure in the most landslide-prone areas, and better management and identification resources⁹².

Climatic factors influence and control some of the forces which influence landslide events, specifically temperature and precipitation⁹³. Climate change is likely to increase intense precipitation and temperatures, both of which can alter the stability of bedrock and make areas more susceptible to landslides. It is therefore expected that areas affected by an increase in precipitation will see an increase in the number of people affected and exposed to landslides⁹⁴. However, it is difficult to predict how specific regions will respond and to determine the timing of landslides as they are dependent on numerous other factors (e.g. area specific geological conditions).

Location

Locations that are more exposed to this hazard are foothills and landfills.

⁹⁰ Oscar Andrés Cuanalo Campos, "Landslides: Methodology to Select Stabilizing Construction Works," in *Risk Assessment*, ed. Valentina Svalova, (Croatia, IntechOpen, 2018).

⁹¹ Farrokh Nadim, "Landslide Risk Assessment," in *Words into Action Guidelines: National Disaster Risk Assessment- The Sendai Framework*(UNDRR, 2017). <https://www.preventionweb.net/publications/view/52828>

⁹² Farrokh Nadim, "Landslide Risk Assessment," in *Words into Action Guidelines: National Disaster Risk Assessment-The Sendai Framework* (UNDRR, 2017). <https://www.preventionweb.net/publications/view/52828>

⁹³ M.J. Crozier, "Deciphering the effect of climate change on landslide activity: a review," *Geomorphology*, 124 (2010): 260–267.

⁹⁴ S.L. Gariano and F. Guzzetti, "Landslides in a changing climate," *Earth-Science Reviews*, 162 (2016): 227–252.

4.1.4. Subsidence

Definition

Subsidence is the lowering the Earth's land surface, which can be triggered by a range of human or natural factors leading to solid or fluid mass movement underground⁹⁵. It is most often caused by extraction of groundwater, oil, natural gas, and other underground resources which result in the rocks underground to collapse to fill the void left by these activities⁹⁶. Therefore, increased demand in these resources elevates subsidence risk. Earthquakes, soil consolidation, glacial isostatic adjustment, erosion, sinkhole formation, and adding water to fine wind-born soil deposits are among the natural causes of subsidence⁹⁷.

Climate change impacts increase subsidence risks. In coastal areas rising sea levels, more frequent and severe storms and storm surges are aggravating subsidence via coastal erosion and inundation. In northern regions, increased thawing of permafrost is causing subsidence. Intensified and extended droughts contribute to aquifer depletion, aggravating subsidence and aquifer collapse. The area impacted by subsidence can vary in extent and timescale. Some affect very small areas within a portion of a building, while others cover entire regions. It may occur suddenly or gradually over years leading to full or partial building site collapse.

Location

Locations that are more exposed to this hazard are ones with close proximity to an active earthquake fault, areas with porous soils where groundwater internal erosion can occur or where underground material extraction is prevalent (mining, fracking, etc). Also in coastal areas, and areas with subsidence-prone soil conditions (e.g. loose sand, fine grained soils).

4.2. Best Practice Standards

Global Best Practices (GBP) for earthquake hazards are considered those of Japan. Many earthquake-prone countries have updated their national seismic codes following major disasters. In Japan they also updated their entire building stock with retrofits. In each country, a certain year corresponds to when the code upgrade is substantial, to a level required for a BRI B rating. BRI refers to this as the 'benchmark' year. For Japan it is 1981, Turkey 1997, Philippines 2010, Haiti 2012 etc. Verifiers should confirm their county's benchmark year for adequacy of meeting a BRI B rating. If absent, the reference benchmark year would be the 2018 International Building Code (IBC). Regarding older buildings, retrofits consistent with the benchmark year standards should be verified for a B rating. Mitigation measures for A and AA ratings will generally exceed most national seismic codes. Below is a sampling of standards from various countries.

⁹⁵ G. Herrera-García et al., "Mapping the global threat of land subsidence," *Science*, 371(6524) (2021): 34-36.

⁹⁶ "Land Subsidence," USGS, accessed April 12, 2026, <https://www.usgs.gov/special-topic/water-science-school/science/land-subsidence>

⁹⁷ "What is subsidence?" NOAA, last modified June 16, 2024, <https://oceanservice.noaa.gov/facts/subsidence.html>

International Standard

The International Organization for Standardization (ISO) is an international standard setting body. ISO has developed standards for earthquake protection and assessment of safety. ISO 28841:2013, *Guidelines for simplified seismic assessment and rehabilitation of concrete buildings* is designed specifically for countries which do not have national building codes. The standard provides information for designers, with simple structures and building characteristics, and guidelines to assess seismic vulnerability⁹⁸.

Table 13. Best practice international standards for geo-seismic hazards

Authority	Link
NASA Global Landslide Catalog (GLC)	The GLC aims to identifying rainfall-triggered landslide events around the world, regardless of size, impacts or location. The GLC considers all types of mass movements triggered by rainfall, which have been reported in the media, disaster databases, scientific reports, or other sources. (2019 update) https://svs.gsfc.nasa.gov/4710
United States Geological Survey	Provides further information on landslides, as well as the US Landslide Inventory Map https://www.usgs.gov/natural-hazards/landslide-hazards
European Soil Data Centre (ESDAC)	Provides additional information and the European Landslide Susceptibility Map version 2 (ELSUS v2), 2018 https://esdac.jrc.ec.europa.eu/themes/landslides
Norwegian Geotechnical Institute	Provides more background information on landslides https://www.ngi.no/eng/Projects/SafeLand
Global Volcano Model (Country guides & news):	https://wovo.iavceivolcano.org/global-volcano-model-website/
United Nations Office for Disaster Risk Reduction	https://www.undrr.org/
Global Volcanic Hazards and Risk ebook by Cambridge University:	www.cambridge.org/volcano

⁹⁸ Katie Bird, "Making Buildings Earthquake-Proof - A New ISO Standard will Help," *ISO*, last modified July 16, 2013, <https://www.iso.org/news/2013/07/Ref1756.html>

National Standard

New Zealand is particularly vulnerable to seismic activity. The Building (Earthquake-prone Buildings) Amendments Act 2016⁹⁹ is a national standard in New Zealand which provides guidance for identifying and prioritizing the management of earthquake-prone buildings. This standard includes:

- A centralized register for all earthquake-prone buildings in New Zealand
- A zoning map which divides the country into three seismic risk areas
- A list of priority buildings (e.g. hospitals, emergency shelters, educational)¹⁰⁰.

4.3. Consequential Hazards

Landslides can be triggered by heavy rainfall, earthquakes, volcanoes and wind hazards (these last ones increase the erosion). They can also increase the risk of river flood after landslide material block a river stream. Earthquakes can trigger (i) landslides through direct rupture or by the sustained shaking of unstable ground; (ii) tsunamis in coastal areas, resulting in flood and additional infrastructural losses; and (iii) subsidence. Volcanoes are related to earthquakes in two ways. Firstly, as they are normally located in areas which are on tectonic plate boundaries, areas with volcanoes are also normally more prone to earthquakes. Secondly, volcanoes can also cause small earthquakes as magma exerts pressure on the earth's crust until it cracks¹⁰¹. In addition, volcanoes can cause in landslides and avalanches.

4.4. Risk Reduction Measures

The following risk reduction measures list is in BRI and forms the basis for the BRI geo-seismic rating. Measures marked 'Y' apply, if the building is in a location with that particular hazard type. Buildings that do not obtain an AA, A or B rating are rated NR. The location of the site is key to geo-seismic hazards. A location assessment should be undertaken prior to commencement with the development.

Each risk reduction measure needs to be approved by at least two licensed, code-responsible engineers or architects for the building's rating to be verified. Each Verifier will be required to upload their license to confirm their qualification. Verifiers can ask the project teams to provide them with supporting documents.

⁹⁹ The Parliament of New Zealand. 2016. *Building (Earthquake-prone Buildings) Amendment Act 2016*. <https://www.legislation.govt.nz/act/public/2016/0022/22.0/DLM5616102.html>

¹⁰⁰ "Law Changes: The Building (Earthquake-prone Buildings) Amendment Act 2016," Christchurch City Council, accessed April 12, 2026, <https://www.legislation.govt.nz/act/public/2016/22/en/latest/>

¹⁰¹ S.C. Loughlin et al., *Global volcanic hazards and risk* (Cambridge, U.K.: Cambridge University Press: 2015): 13.

Table 14. List of risk reduction measures for geo-seismic hazards

Geo-seismic Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Subsidence (Default Hazard)	Volcano	Landslide	Earthquake
GS01. Seismic sites greater than 1 km distance from earthquake fault	In areas prone to earthquakes, the building must be located greater than 1 km away from an earthquake fault to be eligible for baseline seismic design measures associated with a B rating. Structural and foundation design must have a site specific soil investigation and be in conformance with seismic provisions of the applicable national code benchmark year version for essential/critical facilities (see Section 4.2) Sites not bearing on competent original ground must have proper over-excavation and fill compaction, or ground improvements designed and installed to maintain building support. Ground improvement design must consider potential ground failure, seismic slope instability of natural slopes and man-made cuts and walls, seismic shaking induced settlement, liquefaction, and lateral spread.			Y			Y	Y
GS02. Seismic sites less than 1km distance from earthquake fault	Buildings less than 1km from a fault must be designed to withstand seismic shaking and deformations caused by tectonic subsidence and ground surface fault rupture or other exacerbated movements and increased shaking in near fault conditions.		Y		Y		Y	Y
GS03. Foundation pilings adequately secured in rock below subsiding soils	If for buildings designed on potentially subsiding soil (seismic or non seismic) the foundation is enhanced using pilings adequately embedded in underlying competent non-subsiding native bedrock, and designed to resist gravity, uplift, and lateral forces including seismic.	Y			Y			

Geo-seismic Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Subsidence (Default Hazard)	Volcano	Landslide	Earthquake
GS04. Foundation designed for non-seismic ground subsidence	If buildings designed/built on non-seismic sites with potential subsiding soil and foundations not bearing on underlying rock are designed/built to mitigate site specific soil conditions. This includes thickened foundations, overexcavation of poor soils, or ground improvements for collapsible soils and soft soil long term consolidation settlement. For aquifer depletion induced subsidence and other severe settlement problems, long term risks need to be evaluated to determine if piles are necessary per GS 03.							
GS05. Slope of neighboring area < 30 degrees	The slope of the building site and neighboring area including areas upslope and downslope with potential for becoming a landslide source affecting the building needs to be <30 degrees from horizontal and a detailed slope stability analysis included in the design.	Y					Y	
GS06. 15 km distance from a volcano (50 km, if in a valley)	The building must be located >15 km from a volcanic source, or where in a valley or potential lava flow corridor, must be located >50km from the source.	Y				Y		
GS07. Not a historically recorded landslide area	The building site must be confirmed to be located outside of any known landslides, recorded, mapped or otherwise historically recounted.			Y			Y	
GS08. Lateral force resistance system in all floors and first level anchored to foundation	All floors within the building must have a lateral force resistance system, allowing the building to absorb or withstand lateral force movements. In addition, the first level of the building must be anchored to the foundations.		Y		Y			Y
GS09. Connected and braced main structure	The main structural system, which includes columns (vertical), beams (horizontal), foundations (base - beneath ground) and the roof (top), are to be designed/built be connected and reinforced with bracing compatible with the			Y	Y			Y

Geo-seismic Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Subsidence (Default Hazard)	Volcano	Landslide	Earthquake
	connection design to ensure the building deforms as a uniform structure . This includes potential high roof loads from hazard prone areas such as heavy snow or volcanic ash built-up.							
GS10. Steel reinforcement of masonry or concrete walls	Vertical and horizontal steel reinforcement ¹¹⁶ with capacity to sustain design loads including seismic must be used in building walls constructed of masonry ¹¹⁷ , concrete or stone. It must be well connected to the foundation, floors, roof and adjacent exterior walls. I			Y	Y	Y	Y	Y
GS11. Steel reinforced slender building elements	The building must have steel reinforcement for all slender elements to resist toppling (e.g., chimneys and fireplaces, vents, including those of mechanical rooms).		Y		Y	Y	Y	Y
GS12. Structure with seismic base isolation	The building structure is designed/built with base isolation or motion-dampening devices to ensure seismic energy dissipation. This measure is not a requirement for 1 or 2 story residential and commercial buildings.	Y*						Y
GS13. Defensive structures for landslides	Defensive structures for landslides are to be incorporated within the building premises, to protect the building from impacts of loose debris and slope failure of steep gradients or weak slopes. These should be properly designed to local site specific conditions and inspected by a qualified geotechnical engineer and engineering geologist in accordance with the geologic assessment in measure GS18 below, and in coordination with the architect as needed.	Y					Y	
GS14. No vertical irregularities	The building shall have no structural vertical irregularities (i.e. geometry offsets or cantilevers), such as soft/weak stories within a building as described in the ASCE 7-22 design load standard chapter 12 seismic hazard provisions,. The structure design should provide vertical and horizontal			Y	Y			Y

Geo-seismic Hazard Type / Risk Reduction Measure	Criteria	Required for AA Rating	Required for A Rating	Required for B Rating	Subsidence (Default Hazard)	Volcano	Landslide	Earthquake
	stability, with a continuous load path to behave as a whole connected building.							
GS15. Flexible gas piping with automatic seismic shut-off valves	Use flexible piping (flex-pipe) with automatic seismic shut-off valves and monitoring for distribution of combustible gas throughout the building and site to the shutoff valve at the hookup connection to the property. The gas piping should also be properly braced, the pipe pressure rating labeled and proper pressure maintained.		Y		Y	Y	Y	Y
GS16. Design review for geo-seismic hazard risk management measures	The building design must be submitted to and approved by the local approving agency, as prescribed by the governing National Building Code. The licensed structural engineer of record must observe the construction at critical milestones and document compliance of completed construction with the approved design.		Y		Y	Y	Y	Y
GS17. Construction independently audited for geo-seismic hazard risk management measures	A second independent qualified engineer, different from the one assigned from the approving agency or the responsible engineer for the project, will be engaged to inspect and the construction of the building during critical milestones.	Y			Y	Y	Y	Y
GS18. Construction independently audited for seismic hazard risk management measures	The building designed/constructed in conformance with an Engineering Geological and Geohazard Assessment (EGGA) of the site, including seismic hazards resilience measures. The assessment should follow EGGA guidelines of the The Philippines Mines and Geosciences Bureau, or equivalent..This measure is not required for 1-2 story residential and commercial buildings.	Y*						Y

*Note: Not required for 1-2 storey residential and commercial buildings

GS01. Seismic Sites greater than 1 km Distance from Earthquake Fault (B Rating)

Criteria: In areas prone to earthquakes, the building must be located greater than 1 km away from an earthquake fault to be eligible for baseline seismic design measures associated with a B rating.

Structural and foundation design must have a site-specific soil investigation and be in conformance with seismic provisions of the applicable national code benchmark year version for essential/critical facilities (see Section 4.2)

Sites not bearing on competent original ground must have proper over-excavation and fill compaction, or ground improvements designed and installed to maintain building support. Ground improvement design must consider potential ground failure, seismic slope instability of natural slopes and man-made cuts and walls, seismic shaking induced settlement, liquefaction, and lateral spread.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing that the building meets the criteria for distance from fault and the applicable benchmark code design.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Earthquakes can cause heavy structural damage including extensive cracks to serious failure of walls, connections and load path such as foundations and roof, or fracture of chimneys at the roof line, etc.. This damage can cause serious injury or loss of life by the collapse of the building structure and of non-structural elements including building systems and site developments. It is important to note that any earthquake fault line located within 1 km of a site will represent high risks. Fault line locations need to be confirmed by local authority or seismic study¹⁰² of the local area.

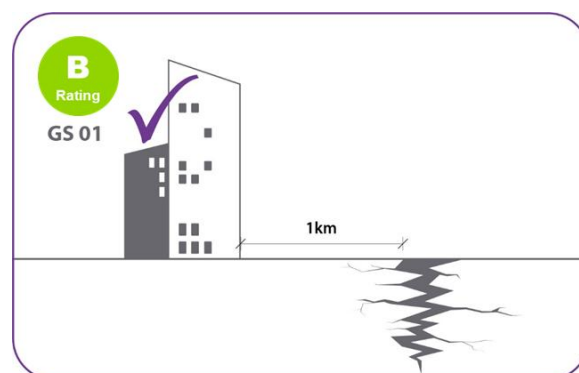


Figure 484. Location away from earthquake fault or seismic designed/ built

¹⁰² **Seismic Study:** A low impact, non-invasive method of gathering information about the location and

characteristics of geological structures beneath the earth's surface.

GS02. Seismic Sites less than 1km Distance from Earthquake Fault (A Rating)

Criteria: Buildings less than 1km distance from a fault must be designed to withstand seismic shaking and deformations caused by tectonic subsidence and ground surface fault rupture or other exacerbated movements and increased shaking in near fault conditions.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing that the building meets the criteria for distance from fault or the applicable benchmark code design.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: This section addresses site specific foundation design including additional seismic and subsidence to be compatible with A level performance of the building structure. The ground conditions (soil and rock composition and behavior) are of great importance in the stability of foundation performance of a building bearing loads in the ground inducing soil-structure interaction. Ground deformation can damage building walls and foundations, and ground rupture and liquefaction¹⁰³ can easily cause foundation failure and large building displacements, as well as break natural gas and water

mains, even causing fires due to the gas pipe explosions.

Locating the building's foundations on loose saturated soil and fills should be avoided, as this poses risk of ground failure from seismic liquefaction during an earthquake where the soil liquefies. Liquefaction due to sediment grains becoming suspended by shaking induced elevated groundwater pressure, which causes the soil to lose solidity.



Figure 649. Foundation designed for site-specific soil seismic and subsidence condition

¹⁰³ **Liquefaction:** This occurs when soil is saturated and subsequently loses its integral strength and stiffness. This

loss of strength is caused by an applied stress such as shaking during an earthquake causing elevated pore water pressure.

GS03. Foundation Piling Adequately Secured in Rock Below Subsiding Soils (AA Rating)

Criteria: For buildings designed on potentially subsiding soil (seismic or non seismic) the foundation is enhanced using pilings adequately embedded in underlying competent non-subsiding native bedrock. and designed to resist gravity, uplift, and lateral forces including seismic.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing confirmation of the collapsible soil exposure and cause; and that the building foundation piling design meets the above performance criteria for benchmark code design.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site exposure conditions and that foundation piling design meets the criteria.

Background: The ground conditions are of great importance in the stability of a building. Subsidence can damage buildings partially or entirely over time or suddenly. Sudden subsidence events can also break natural gas and water pipelines. Each site will have its own soil type, which will need to be addressed through analysis of the site-specific soil conditions (soil testing/soil mapping). Each site has been exposed to a unique geological progression over time which will impact the soil deposition, stratigraphy, and structure. The foundations of a building are heavily influenced by the soil conditions they are embedded or anchored in as-needed, to enable secure structural load transfer. If the site has subsiding soil conditions, to have an AA rating, use current best practices to test and evaluate the subsidence mechanism, and to design requirements for piling foundations extending into competent bedrock to mitigate the subsidence risk to a building.

GS04. Foundation Designed for Non-Seismic Ground Subsidence

Criteria: For buildings designed/built on non-seismic sites with potential subsiding soil, foundations not bearing on underlying rock are designed/built to mitigate site specific soil conditions. This includes thickened foundations, over-excavation of poor soils, or ground improvements for collapsible soils and soft soil long-term consolidation settlement.

For aquifer depletion induced subsidence and other severe settlement or collapse risks, long term conditions need to be carefully evaluated to determine if piles are necessary per GS 03.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing that the building meets the criteria for the applicable benchmark code design.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Other forms of subsidence can damage buildings partially or entirely over time or suddenly. Sudden subsidence events, like seismic, can also break natural gas and water pipelines. Each site has unique soil conditions and behaviors, which will need to be addressed through geotechnical analysis of the site-specific soil conditions (soil testing/soil mapping), and evaluation of geological formations and changes over time which will impact the structure.

If the site has fill or risk of soil subsidence or liquefaction conditions, the building foundation(s) must be designed and built to withstand subsidence. If the foundation of the building cannot be secured in the underlying non-subsiding rock, then it must be designed and built to stabilize or accommodate site specific soil conditions to maintain structural integrity for the building to have an A rating.

GS05. Slope of Neighboring Area < 30 degrees (AA Rating)

Criteria: The slope of the building site and neighboring areas both upslope and downslope with potential for becoming a landslide source affecting the building needs to be <30 degrees from horizontal and a detailed slope stability analysis be included in the design.

Assessor Procedure: Provide documents including statement by the code-responsible engineer or contractor showing that the building meets the criteria for the applicable benchmark code design.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Landslides, which are the movement of rock, mud, alluvium or loose debris downwards on a sloped section of land, can be caused by earthquakes, extreme precipitation and melting glaciers, typically in loosely-vegetated slopes.

There is an increased risk of landslides in areas with a steep gradient slope. Therefore, the surrounding area of a site should have a slope of less than 30-degrees, otherwise a slope hazard analysis must be performed to assess the risks and risk reduction to the landslide hazard, such that engineering measures for slope hazard mitigation measures are more feasible and reliable

The analysis must act as a base point in which for the continued monitoring of slope levels, landslides can impact on the primary structure of a building, as the foundations may be affected as well as the force applied being too great for the building to withstand.



Figure 66. Slope angle of neighboring area lower to 30 degrees.

GS06. 15 km Distance from a Volcano (50 km, if in a valley) (AA Rating)

Criteria: The building must be located >15 km from a volcanic source, or where in a valley or potential lava flow corridor, must be located >50km from the source.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets the criteria for distance from a volcano, active or dormant.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Volcano eruptions can damage or destroy buildings, causing collapse, fire or burial of walls and foundations, from impact of +pyroclastic (gas, ash and molten rock) flows and ejecta. Associated landslides also can also destroy or severely damage the building. Volcanoes represent a high risk to buildings. Two other risk factors include;..:

- Heightened levels of shaking ground movement from near surface eruptions, especially when near a tectonic plate boundary
- The volcano itself releases dangerous hot toxic gases and ash devastating to human life.

In valley locations around a volcano, a normal safe radius is 50 km or more from the volcano. This considers the average distance that volcanic eruptions typically travel and how long the erupted materials may travel before solidifying or cooling.

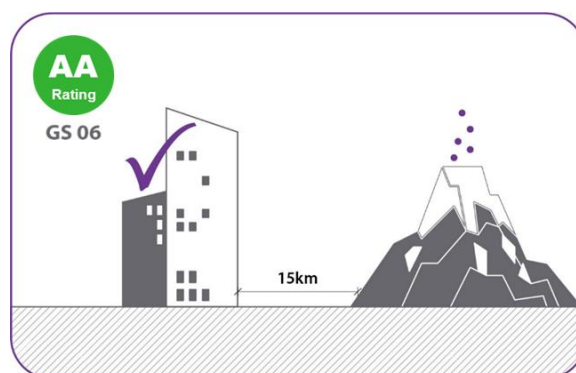


Figure 50. Location is more than 15 km (50 km if valley) away from a volcano

GS07. Not a Historically Recorded Landslide Area (B Rating)

Criteria: The building site must be confirmed to be located outside of any known landslides, recorded, mapped or otherwise historically recounted.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets the criteria for lateral force resistance.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: A site can be prone to landslides if they had happened previously. It is important that research is undertaken in the area of a potential building to ensure data is gathered about past landslides. Retaining information from

previous landslide events will inform the impact of potential future landslides.

The recorded history should be researched for each site, informing the team as to whether the area is susceptible to landslides, which can be caused by earthquakes, extreme precipitation and melting glaciers typically in loosely-vegetated slopes.

If the slope of neighboring area is greater than 30 degrees, then an analysis and risk reduction of slope hazard has to be performed.



Figure 51. No landslides area historically

GS08. Lateral Force Resistance System in All Floors and First Level Anchored to Foundation (A Rating)

Criteria: All floors within the building must have a lateral force resistance system, allowing the building to absorb or withstand lateral force movements. In addition, the first level of the building must be anchored to the foundations.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets the criteria for lateral force resistance.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Earthquakes create horizontal pressure on buildings which causes them to collapse. Lateral force

resistance and foundation anchorage can be achieved using shear walls¹⁰⁴, to improve resistance capacity of the building especially on the lowest floor, where it is the most vulnerable, where the inertial force of the full building weight must be resisted during lateral movements transmitted within the building. Lateral capacity is particularly important in high-rise buildings in areas of high wind and seismic activity.

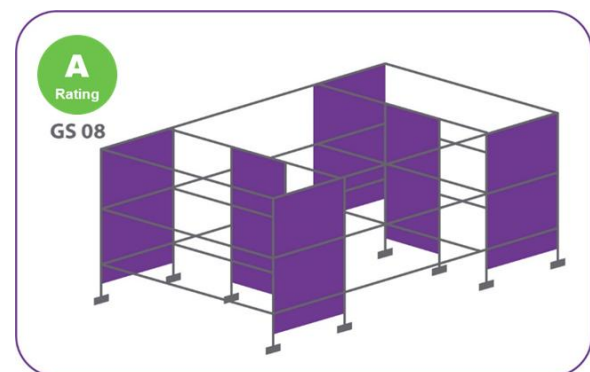


Figure 52. Shear wall for lateral force resistance

¹⁰⁴ **Shear wall:** A structural panel that can resist lateral forces.

GS09. Connected and Braced Main Structure (B Rating)

Criteria: The main structural system, which includes columns (vertical)¹⁰⁵, beams (horizontal)¹⁰⁶, foundations (base - beneath ground)¹⁰⁷ and the roof (top)¹⁰⁸, are to be designed/built be connected and reinforced with bracing compatible with the connection design, to ensure the building deforms as a uniform structure. This includes potential high roof loads from hazard prone areas such as volcanic ash built-up or heavy snow.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets the criteria for systemwide structural connections and bracing.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Forces are applied to buildings in multiple planes accumulating stress concentrations. In order to

strengthen the matrix of structural elements used distribute stresses in a building, it is important to reinforce a building three-dimensionally. The idea is for the building to deform and respond to applied structural forces as one unit, connecting load paths efficiently across its primary members so that the integrity of the structure remains intact. This can be done with principles of both rigid and ductile design, as compatible with material selection and deformation objectives and tolerances of the building performance.

Of particular note is attention roof design to avoid damage or collapse in hazard prone area be subject to risk of high roof loads such as volcanic ash or snow accumulation areas.

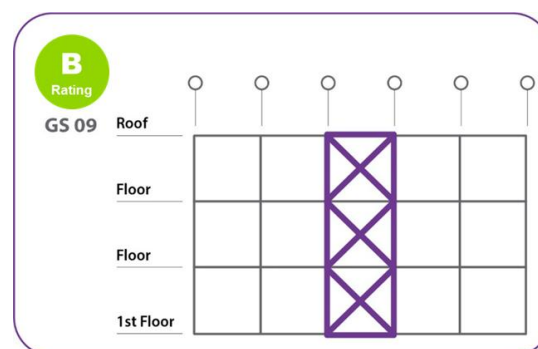


Figure 53. Cross-braced building elevation

¹⁰⁵ **Columns:** Upright pillars and structural elements that transmit through compression, the weight of the structure above to other structural elements such as a plinth or base.

¹⁰⁶ **Beams:** Structural elements that resist loads applied laterally to their axis.

¹⁰⁷ **Foundation:** Provide support for structures, transferring their load to layers of soil that have sufficient bearing capacity and suitable settlement to support them.

¹⁰⁸ **Roof:** A structure forming the upper part of a building that has a primary purpose to provide protection from weather/climatic elements.

GS10. Steel Reinforcement of Masonry and Concrete Walls (B Rating)

Criteria: Vertical and horizontal steel reinforcement¹⁰⁹ with capacity to sustain design loads including seismic must be used in building walls constructed of masonry¹¹⁰, concrete or stone. It must be well connected to the foundation, floors, roof, and adjacent exterior walls.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets this criteria for steel reinforcement of walls and structural connections for the applicable materials.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria

Background: Reinforcement with steel provides additional support when counteracting forces, especially in limiting deflection of walls to maintain integrity as one structural unit. Well-connected reinforcement is critical for all building types, especially as dynamic seismic loads.

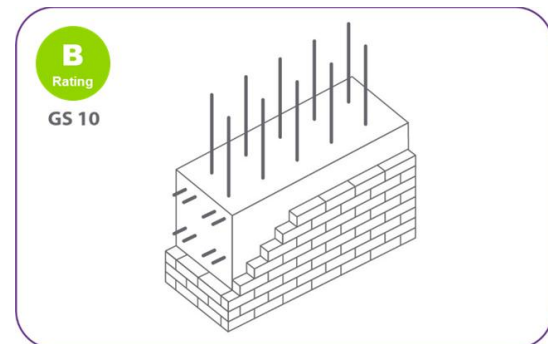


Figure 54. Steel reinforced masonry & concrete walls

¹⁰⁹ **Steel Reinforcement:** Known as 'Rebar,' a steel bar or mesh of steel wires used to reinforce/stabilize masonry, concrete, or stone structures, holding the masonry in tension.

¹¹⁰ **Masonry:** The composition of hard units such as brick, stone, or blockwork to form walls. It may be load bearing, forming an integral part of the structure, or non-load bearing, such as a partition wall or cladding.

GS11. Steel Reinforced Fireplaces and Slender Building Elements (A Rating)

Criteria: The building must have steel reinforcement¹¹¹ for all slender elements to resist toppling (e.g., chimneys and fireplaces, vents, including those of mechanical rooms).

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building meets this criteria for steel reinforcement of slender building elements.

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria

Background: Tall cavity structures are subject to large overturning moments during earthquake shaking. They are vulnerable to collapse or fracture at the roof line under the resulting seismic forces.

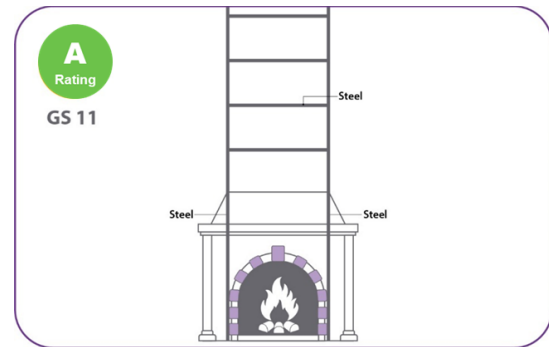


Figure 55. Slender elements are steel reinforced

¹¹¹ **Steel Reinforcement:** Known as 'Rebar,' a steel bar or mesh of steel wires used to reinforce/stabilize masonry,

concrete, or stone structures, holding the masonry in tension.

GS12. Structure with Seismic Base Isolation (AA Rating)

Criteria: The building structure is designed/built with base isolation or motion-dampening devices to ensure seismic energy dissipation. This measure is not a requirement for 1 or 2 story residential and commercial buildings.

Assessor Procedure: Provide documents including confirmation by the code-responsible engineer or contractor showing that the building seismic base isolation design and material selection meets the criteria for preventing damaging accelerations or differential displacements to the building

Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria

Background: The seismic base isolation is a powerful earthquake engineering tool which use special rubber¹¹² bearing, ball bearings, spring systems or other means to absorb shockwaves from an earthquake and reduce the movement of the building's structure, as the base isolation absorbs the shockwaves instead. It works like a suspension system between the base and the structure of the building, increasing the elastic strength of the structure to handle the seismic forces. Many seismic isolation and motion-dampening systems are available in the market for both new-built and retrofit buildings. The main types are as follows:¹¹³

- **Ball & Roller Bearings:** Sufficient to resist service movements, and it includes cylindrical rollers and balls.
- **Friction bearings:** The energy is dissipated due to the friction between contact surfaces¹¹⁴. Within this type 'Curved Slider Bearings', 'Pendulum Bearings' or 'Flat Slider Bearings' can be found, these provide resistance, flexibility and force-displacement by sliding movement.
- **Spring system:** Consist of steel springs assembled to be flexible to both forces, horizontal and vertical. The GERB system¹¹⁵ is an example of it.
- **Padded cylinders:** This system also known as 'Lead Rubber Bearings', 'High Damping Bearings', or 'Elastomeric rubber bearings',¹¹⁶ is usually flexible in horizontal direction and hard in vertical direction. It uses horizontal layers of rubber tied between steel plates.

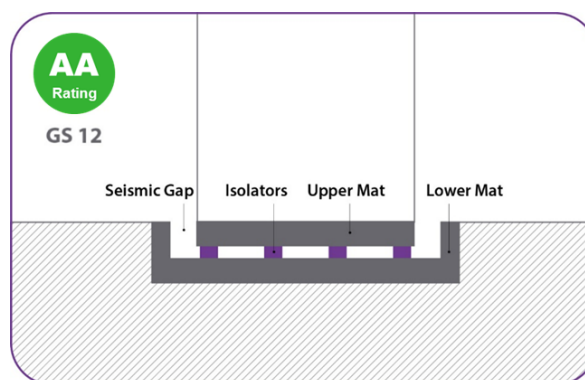


Figure 56. Structure with seismic base isolation

¹¹² Rubber provides flexibility through its ability to move and return to original position, but might take months.

¹¹³ Mohiuddin Ali Khan, "Risk Assessment, Mitigation, and Remediation," in *Earthquake-Resistant Structures: Design, Build, and Retrofit* (London: Butterworth-Heinemann: 2013).

¹¹⁴ F. Saitta, et al., "Base isolation of buildings with curved surface sliders: Basic design criteria and critical issues," *Advances in Civil Engineering*, vol.2018 (2018).

¹¹⁵ Z. Zhou, J. Wong, J., and S. Mahin, "Potentiality of using vertical and 3D isolation systems in nuclear structures," *Nuclear Engineering & Technology*, 48(5) (2016): 1237-1251.

¹¹⁶ Anand Paul, "Base Isolation System: Outline on Principles, Types, Advantages & Applications," *Civil Digital*, last modified April 1, 2016, <https://civildigital.com/base-isolation-system-outline-on-principles-types-advantages-applications/>

ISO 23618:2022 Bases for design of structures — General principles on seismically isolated structures <https://www.iso.org/obp/ui/en/#iso:std:iso:23618:ed-1:v1:en>

GS13. Defensive Structures for Landslides (AA Rating)

Criteria: Defensive structures for landslides are to be incorporated within the building premises, to protect the building from impacts of loose debris and slope failure of steep gradients or weak slopes. These should be properly designed to local site specific conditions and inspected by a qualified geotechnical engineer and engineering geologist in accordance with the geologic assessment in measure GS18 below, and in coordination with the architect as needed.

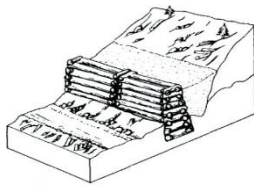
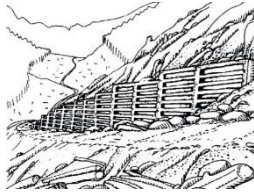
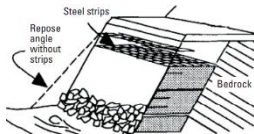
Assessor Procedure: Provide documents including confirmation by the code-responsible engineer and geologist showing that the building seismic defensive structures for landslide mitigations meets the criteria for preventing building damage from earth loads and slope failures.

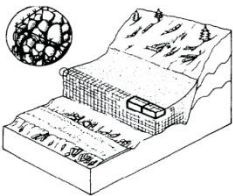
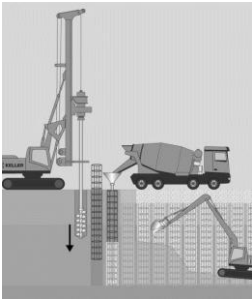
Verifier Procedure: Review the documentation provided by the assessor and confirm the site meets the criteria.

Background: Examples of defensive structures such as, retaining barriers, wire mesh, check dam, stabilizing material, drainage (reduce soil saturation) or vegetation, are described in detailed below.

Retaining barriers: These are walls or mass structures constructed at the toe of the slope to effectively retain, slow or deflect moving soil. They are effective for different types of landslides, such as falls, rotational and translational landslides, and slow earthflow (Creep). Adequate drainage through the retaining barriers is key, as high ground-water pressure can build up behind, leading to its failure. The following are some types of retaining barriers:

Table 15. Types of retaining barriers

	Timber crib: Strongly built structure of interlocking logs and backfilled with coarse aggregate that work by intersecting the critical sliding surface, thus forcing the potential failure surface to a deeper, less critical depth. They are effective when stabilizing small volume of soil, in which the crib wall should have a volume equal to 10%-15% of the volume of the soil to be stabilized.
	Steel bin wall: corrugated steel components anchored together forming a box, which is filled with earth. The weight is provided by the contained soil rather than the steel. Walls' widths vary from 2m to 5m; height is between 1/2 and 3/5 the width; and sank from 0.5m to 1m below grade to provide additional sliding resistance. Filled material be well drained and compacted.
	Reinforced earth wall: This patented system uses horizontal layers of flexible metal strips within the fill to form a composite earth-metal system with high strength.

	Gabion walls: Constructed from stacked gabions, which are wire mesh containers filled with cobble-sized rock (0.1m to 0.2m). This type of retaining structure is simple, quick to construct, and flexible. Due to their coarse fill they permeable and have good drainage. Counterfort¹¹⁷ is needed when gabion walls are higher than 2.5m or when they are built on clay soils.
	Piles: Large-diameter piles placed into the toe of a slope to form a closely spaced vertical pile wall. Pile walls are normally used as a pre-excavation restraint system. The materials of the piles vary based on volume of soil they need to contain, large-diameter concrete pile and culvert pile walls have been used on highways and building excavations; while wood or steel piles are only suitable where soil volume to be stabilized is small (approx. one wood pile for every 50 m³ of soil). Piles should extend below the potential failure surface and be driven into firm subsoil. If the depth is not sufficient to act as a cantilever system, then the piles must be tied back with an additional anchor system.

Source: USGS 2008

Meshes (Wire): These wire nets are simple, low-cost methods for protecting an area from rockfall. They are flexible but resistant elements that wrapped around the stones or blocks and are anchored to the slope. The various types are described as follows:

- **Wire mesh:** This closely spaced interwoven wires can be used to prevent smaller rocks (<0.75m) from falling. The standard meshes are double twisted gabion wire mesh, and heavy gage metal chain link, which can be either loosely draped or bolted over a uniform rock face, or firmly secured where the cliff face is irregular. Bolting the mesh provides overall stability of the slope. Wire mesh also is useful on steep soil cuts (i.e. beneath talus slopes).

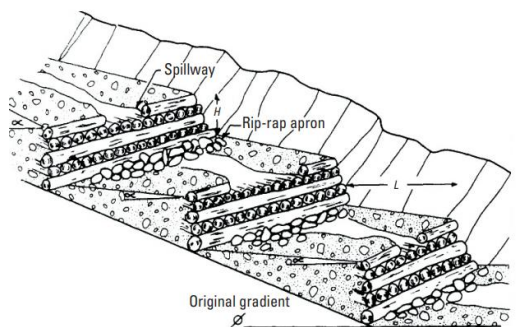
- **Catch nets:** Constructed to catch falling rock at the bottom of gullies and slopes. They are a suspended and flexible barrier made of cable and wire mesh that dissipates the energy of the falling rock (<1 m in diameter).
- **Rock fences:** This easy to install barrier (poles with gabion mesh bolted) can restrain small rocks from falling onto roads, however it is not effective for rocks that bounce over the fence.
- **Rock curtains:** Effective in directing rocks down to a catch ditch or other catching structure, preventing them from bouncing outward onto the highway or structures below.
- **Plastic mesh reinforcement:** Lightweight, high-tensile-strength grid made of stretched plastic polymer that adds strength to the shear strength of

¹¹⁷ **Counterfort:** Buttress bonded to the rear of walls, designed to improve stability, which serves as both structural component and as drain.

the soil. Their applications are: slope stabilization, soil strength reinforcement, soil drainage improvement, and retaining-wall construction.

Check Dams

Small, sediment-storage dams built in the channels of steep gullies to stabilize the channel bed. Commonly used to control channelized debris-flow frequency and volume. As they are expensive to construct, they are built where important installations or wildlife habitat (camp or unique spawning area) lie downslope. They can be constructed of reinforced concrete (<8m height) or log cribs (<2 m height), which must be tied securely to withstand backfill pressures and lateral scour. The spacing of dams depends on channel gradient and dam height. Lateral stream erosion and scour by spillway water is the main drawbacks. The main purposes of the check dams are:



- To manage risk of landslide failure by reducing the angle;
- To reduce the volume of material stored by preventing down cutting of the channel with subsequent gully sidewall destabilization and by

providing toe support to the gully slopes.

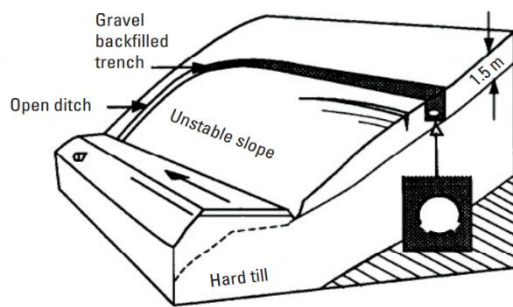
- To store debris-flow sediment, when installed in the lower part of the channel.¹¹⁸

Drainage

Ground water contributes to landslide initiation meaning that drainage plays a large part in the stabilization of slopes, as it increases the stability of the soil and reduces the weight of the sliding mass. Surface drainage have substantial stability benefits (prevent erosion, reduce surface movement, and prevent infiltration) while requires minimal design and costs. Subsurface drainage is used when ground water is the cause of the slide, as it is more expensive. Drainage methods as follows:

- **Site levelling:** Removing depressions in a slope to eliminate water ponding. Infilling and sealing large cracks on the surface that connect with the ground water
- **Ditches and drains:** These types of drainage are important to conduct the water flow away from the unstable soil. A system of surface *ditches*/channels run at the edge of the unstable soil, at the headwall the ditch is filled with coarse gravel and the lateral ditch is uncovered considering a gradient to ensure rapid flow away. Lateral drainage trench is a subsurface drain that is better for shallow soils overlying bedrock. The trenches should be excavated to the base of the shallow soil to intercept any ground-water flow along the failure plane.

¹¹⁸ USGS, *The Landslide Handbook: A Guide to Understanding Landslides* (2008), <https://pubs.er.usgs.gov/publication/cir1325>



- **Drainpipes:** Horizontal drainpipe is widely used for landslide prevention, but need to be carefully installed during excavation, and designed for the conditions of the slope (soil, hydraulic and geometric). Sections of slotted PVC drainpipe, which are covered with filter cloth, are inserted into a casing, that has been previously drilled and coupled together.

Vegetation

Slope stabilization using vegetation depends on the root system of the seeded plants to bind the sloped soil together. Vegetation such as grasses and legumes reduce surface erosion which is a contributing factor towards landslides. Using small-scale planting is a preventative tool for small-scale erosion, which would result in large-scale erosion over time. Large-scale erosion would require the intervention of engineered technological management. The terms "bioengineering" and "biotechnical slope protection" refer to the use of vegetation as slope protection to arrest and prevent slope failure and surface erosion.¹¹⁹ Planning may be required for larger-scale interventions. Local knowledge is a vital tool in the landslide prevention strategies and

therefore a local experienced person should be consulted, who should consider soil typology, climatic conditions, local conditions, species compatibility and replacement. Best practice is to stabilize the soil prior to seeding, seed immediately after a disturbance and 6 weeks prior to periods of drought or a damaging frost. The two basic types of seeding are:

- **Dry Seeding:** Implemented using rotary disk and air-borne seeders which are limited to use shallower gradients and rough soil surfaces. The cyclone-type, hand-held and hydraulic seeders use centrifugal force spread seed and fertilizer. Air-blown seeders use air to blow or shoot seed and fertilizer a distance of 5m to 8m.
- **Hydraulic Seeding, or Hydroseeding:** Using a water slurry that contains fertilizer, soil binder and/or mulch¹²⁰ (straw, grass fibers, wood fibers, seaweed and paper products), the seed is sown. This method can be used on slopes of > 1:1 gradient and requires the use of a mixing tank with hydraulic agitation and volume pumping capacity.

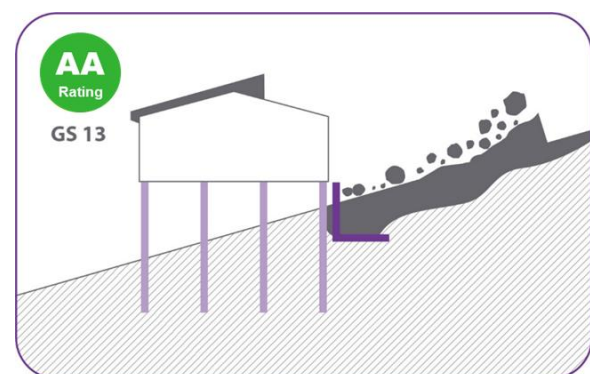


Figure 57. Defensive structures for landslides

¹¹⁹ USGS, *The Landslide Handbook: A Guide to Understanding Landslides* (2008), <https://pubs.er.usgs.gov/publication/cir1325>

¹²⁰ **Mulch:** The debris from vegetative matter that is non-living and can provide a protective layer on top of the

soil's surface. Mulching protects the slope from hydraulic surface erosion and allows the soil to retain its own moisture.

GS14. No Vertical Irregularities or Soft Stories (B Rating)

Criteria: The building shall have no structural vertical irregularities (i.e. geometry offsets or cantilevers), such as soft/weak stories within a building as described in the ASCE 7-22 design load standard chapter 12 seismic hazard provisions,. The structure design should provide vertical and horizontal stability, with a continuous load path to behave as a whole connected building.

Assessor Procedure: Provide documentation from the engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: The more rectangular the shape of a building, the better it will respond to seismic events. A soft structure relates to a multi-story building in which one or more story has windows, wide door or large unobstructed commercial spaces and the upper floors have greater seismic resistance than the lower floors These floors typically lack shear walls or cross bracing. Hence seismic motion is amplified in these weaker floors, which can result in structural failure.



Figure 75. No vertical “soft story” irregularities

GS15. Flexible Gas Piping with Automatic Seismic Shut-off Valves (A Rating)

Criteria: Use flexible piping (flex-pipe) with automatic seismic shut-off valves and monitoring for distribution of combustible gas throughout the building and site to the shutoff valve at the hookup connection to the property. The gas piping should also be properly braced, the pipe pressure rating labeled and proper pressure maintained.

Assessor Procedure: Provide documentation from the engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: Geo-seismic movements can impact utility service connections to a building as well as the structure itself. Pressurized service piping and connections,

such as natural gas and water mains can easily rupture, and fires can be caused by gas pipe explosions following rupture. The resilience performance is from pipe flexibility to absorb the force and displacement from the seismic shaking which would otherwise damage rigid gas piping. Flexible lines and their connectors are specially design material products which maintain pressure-rated airtight connections in ductwork whileattenuating (absorbing) the transmission of vibration along the duct.

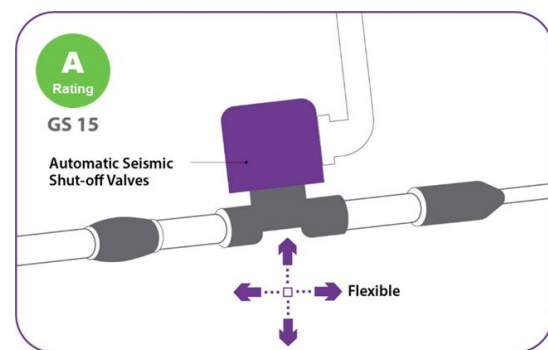


Figure 76. Flexible gas piping with automatic seismic shut-off valves

GS16. Design Review for Geo-seismic Hazard Risk Management Measures (A Rating)

Criteria: The building design must be submitted to and approved by the local approving agency, as prescribed by the governing National Building Code.

The code-responsible licensed structural engineer of record must observe the construction at critical milestones and document compliance of completed construction with the approved design.

Assessor Procedure: Provide documentation from the approving agency with a certificate of design acceptance or equivalent. Also provide documentation from the engineer of record attesting to design compliance of the constructed work including construction observations and stamped as-built drawings or equivalent.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: Lack of building code compliance and enforcement is a significant risk in the construction industry worldwide. This can be owing to lack for government capacity to perform inspections and approvals, as well as intentional or unintentional

nonconformance by contractors for numerous reasons. This commonly includes lack of knowledge of specialized construction methods to properly apply reinforcement detailing for seismic design and resilience measures. This can be improved with capacity building, such as from extensive training on seismic resilient masonry techniques in Nepal following the 2015 earthquake. Similar capacity building in construction methods of reinforced concrete detailing is needed in recovery of the 2022 Turkey earthquake. The inspection requirements of this section should be coordinated with building officials early in construction and capacity building training needs be promptly identified to avoid delay and additional costs of rework for nonconforming construction of seismic resilience measures.

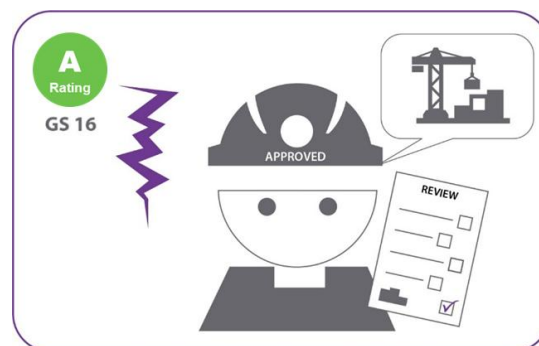


Figure 77. Design reviewed by local approving agency

GS17. Construction Independently Audited for Geo-seismic Hazard Risk Management Measures (AA Rating)

Criteria: A second independent qualified engineer, different from the one assigned from the approving agency or the responsible engineer for the project, will be engaged to inspect and the construction of the building during critical milestones.

Assessor Procedure: Provide documentation from the independent qualified engineer attesting to compliance of the constructed work including construction observations and review of as built drawings, submitted in a stamped letter report of conformance.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above criteria.

Background: The building engaged a qualified independent engineer or architect, other than, and independent of the code-responsible structural engineer or the approving agency, to inspect and approve the construction. This provides an additional level of construction quality control essential for AA rated building.



Figure 78. Construction independently audited for geo-seismic hazard risk management measures

GS18. Design and Construction in conformance with an Engineering and Geological Hazard Assessment for Seismic Hazard Risk Management Measures (AA Rating)

Criteria: The building designed/constructed in conformance with an Engineering Geological and Geohazard Assessment (EGGA) of the site, including seismic hazards resilience measures. The assessment should follow EGGA guidelines of the Philippines Mines and Geosciences Bureau, or equivalent. This measure is not required for 1-2 story residential and commercial buildings.

Assessor Procedure: Provide documentation from the EGGA report engineer or geologist attesting to conformance of the design and constructed work including construction observations and review of as built drawings, submitted in a stamped letter report of conformance.

Verifier Procedure: Review the documentation provided by the assessor including the EGGA report and design of measures addressing the hazard mitigations discussed to confirm conformance with the above criteria.

Background: EGGA's focus on potential geologic hazards that may have direct or indirect impact to the project, and their appropriate mitigating measures.¹²¹ Its components are categorized as per the following Table 16. In many countries (for example, the Philippines), EGGA study must be undertaken by a licensed geologist or qualified professional with at least 5 years of experience, and the report should include general information (site location and size, methodology, source of data); regional and site geologic data (topography and hydrology); structural features and geohazard assessment and geotechnical tests.¹²²

Other comparable guidelines include California State Mining and Geology Board Special Publication 117 (1997), and OAS USAID Primer on Natural Hazard Management in Integrated Regional Development Planning, Chapters 10 and 11 Landslide and Geologic Hazard Assessment (1991).

Table 16. General list of hazards considered in EGGA

A. Geologic Hazard		B. Hydrological Hazard	
1. Fault related/ Seismic hazards	a. ground acceleration b. ground rupture c. liquefaction d. differential settlement e. landslides f. fault creep g. lateral spread h. tsunami i. seiche	1. Fluvial	a. Flooding (Overflow) b. Flooding (Sheetflow, concentrated run-off) c. Scouring of riverbed d. Channel erosion and migration e. Rill erosion f. Gully erosion g. Sedimentation

¹²¹ M.A. Aurelio and The Philippines Mines and Geosciences Bureau EGGA Technical Working Group, "Engineering Geological and Geohazard Assessment (EGGA) system for sustainable infrastructure development: The Philippine experience," *Engineering geology for sustainable development in mountainous areas* (2008): 33-39.

¹²² Philippines Mines and Geosciences Bureau, Implementing Guidelines on Engineering Geological and

Geohazard Assessment as Additional Requirement for ECC Applications covering Subdivision, Housing and other Land Development and Infrastructure Projects (2000), https://essc.org.ph/content/wp-content/uploads/2014/04/Day2Session3B_MANZANO_MG_B_EGGAR.pdf https://mgb.gov.ph/images/stories/DAO_2000-28.pdf

2. Mass Movements	a. Landslides	a.1. Fall a.2. Topple a.3. Slump a.4. Slide a.5. Spread a.6. Flow a.7. Complex
	b. Creep c. Subsidence d. Settlement	
3. Volcanic Hazards	a. Lava flow b. Debris flow c. Pyroclastic flow d. Debris avalanche e. Lahar f. Lateral blast and pyroclastic surge g. Bombs and ballistic projectiles h. Ash fall i. Tsunami j. Flooding k. Volcanic gases l. Volcanic earthquakes	

2. Coastal Hazards	a. Flooding b. Coastal erosion c. Coastal aggradation d. Storm surge e. Coastal subsidence/sea level rise f. Submarine landslide
C. Other	
1. Air-borne 2. Cosmic	

Source: M.A. Aurelio and The Philippines Mines and Geosciences Bureau EGGA Technical Working Group 2004

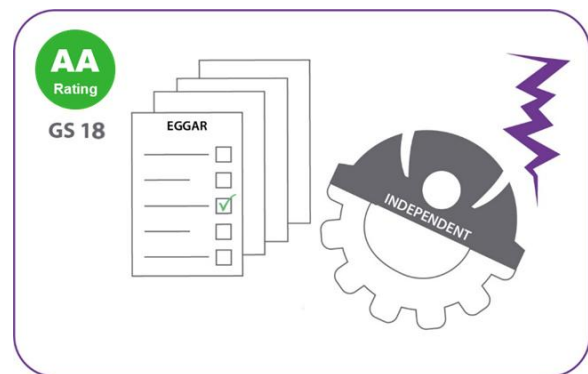


Figure 79. Engineering & geologic hazard report

OPERATIONAL CONTINUITY

1. Types of Risks for Operational Continuity

1.1. Hazards with Primary Impact

Three types of hazards in BRI may not damage buildings, but can directly or indirectly significantly impact building operations. They are: heatwaves, cold waves, and droughts.

1.1.1. Heatwaves

Definition

The World Meteorological Organization defines a heatwave as a period of unusually hot-dry or hot-humid weather that lasts for at least two to three days and has a discernible impact on human activities¹²³. The IPCC's Fifth Assessment Report of 2014 shows that since 1950 the number of warm days and nights have increased globally¹²⁴. The report states that it is very likely that heatwaves in large parts of Europe, Asia and Australia have increased since 1950. It is virtually certain that there will be more frequent hot temperature extremes over most land areas and it is very likely that this will occur with higher frequency and longer duration as global temperatures increase. The length of a heatwave can range from several days to several weeks and can impact a large area.

Heatwaves are relative to each location's climate, so exceeding a certain temperature may constitute a heatwave in one part of the world but not in another¹²⁵. Furthermore, extreme heat does not necessarily have to be officially classified as a heatwave to have an impact. Normally buildings would not be physically damaged by heatwaves, but their performance and operations may be affected:

- Heatwaves can lead to high indoors temperatures as the buildings trap heat, thus becoming inhospitable for human use and increasing risks on human health which could result in fatalities, especially in vulnerable groups (e.g. elderly).
- The increased energy demand from mechanical cooling of indoor spaces during heatwaves may lead to power outages.
- Heatwaves could also reduce water availability and hydropower availability (depending on location), thus leading to water and energy shortages.

¹²³ WHO and WMO, *Heatwaves and health: guidance on warning-system development* (Geneva, Switzerland: WMO, 2016), <https://www.who.int/publications/m/item/heatwaves-and-health--guidance-on-warning-system-development>

¹²⁴ IPCC, *Climate Change 2014: Synthesis Report* (Geneva, Switzerland: IPCC, 2014), 53.

¹²⁵ WHO and WMO, *Heatwaves and health: guidance on warning-system development* (Geneva, Switzerland: WMO, 2016), <https://www.who.int/publications/m/item/heatwaves-and-health--guidance-on-warning-system-development>

- Cooling systems would become a demand in locations where traditionally they were not usually required such as colder climates.
- Construction activities and program might be disrupted by heatwaves as high temperatures pose a health hazard to construction workers.

As a result, developers will need to assess the risks that heatwaves pose¹²⁶ for their projects, especially those in urban areas, and adopt relevant risk reduction measures.

Location

Locations where the effects of heatwaves may be more pronounced are urban areas. This is due to the urban heat island effect, which is a phenomenon where more heat is retained during the day in urban areas than in rural areas and is it is released slowly, so nighttime temperatures remain high.¹²⁷

1.1.2. Cold Waves

In recent years, increase in Arctic temperatures and melting due to climate change have resulted in harsher winters and winter extremes reaching lower latitudes than usual.¹²⁸ The IPCC's Fifth Assessment Report of 2014 indicated that cold temperature extremes will become less frequent, but will continue to occur.¹²⁹ Normally buildings would not be physically damaged by cold waves, but their performance and operations may be affected:

- Cold waves can lead to frigid indoors temperatures, thus becoming inhospitable for human use and increasing risks on human health.
- Cold waves can lead to freezing and damage to building water systems, if these are not adequately insulated or temperature protected.
- The increased energy demand from heating of indoor spaces during cold waves may lead to supply challenges or power outages.
- Heating systems would become a demand in locations where traditionally they were not usually required.

As a result, developers will need to assess the risks that cold waves pose for their projects and adopt relevant risk reduction measures.

¹²⁶ Developers should also note that construction activities and program might be disrupted by heatwaves. High temperatures pose health risks to construction workers working outdoors. Mitigation measures such as changing working hours should be considered during construction phase.

¹²⁷ IFRC, *Heatwave Guide for Cities* (2019), <https://www.ifrc.org/media/12672>

¹²⁸ Kelly Levin, "Climate Change, Frigid Temperatures and the Polar Vortex: 3 Things to Know," *WRI*, last modified January 30, 2019. <https://www.wri.org/insights/climate-change-frigid-temperatures-and-polar-vortex-3-things-know>

¹²⁹ IPCC, *Climate Change 2014: Synthesis Report* (Geneva, Switzerland: IPCC, 2014), 60.

1.1.3. Droughts and Water Scarcity

Definition

The World Meteorological Organization defines a drought as a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow on-set phenomenon caused by a lack of rainfall.¹³⁰

Water scarcity can be caused either by a lack of availability of water (due to a drought) or by inadequate infrastructure or irregular supply.¹³¹ The IPCC's Fifth Assessment Report of 2014 predicts that climate change will increase risks from drought and water scarcity, including urban areas.¹³² Droughts can typically affect larger areas than other hazards, which are usually constrained to floodplains, coastal regions, storm tracks or fault zones.¹³³ Buildings may not be physically damaged by droughts (or water scarcity), however limited access to water can:

- Inhibit critical operations for certain building types, especially those linked to health and hygiene;
- Pose health risks, if the drought conditions are coupled with heatwaves, because access to water;
- Restrict the use of water intense activities such as irrigation and use of swimming pools.

As a result, developers will need to assess the risks that droughts and water scarcity pose for their projects, especially those in arid areas, and adopt relevant risk reduction measures.

Location

Locations that are more exposed to this hazard are dry regions, especially dry subtropical regions.¹³⁴

1.2. Disruption of access to utility services

A building may not be affected by a disaster; however, the utility services at urban and national level may be directed to priority buildings, overloaded, or damaged. Potential disruption of utility services should be considered as a risk for operational continuity.

1.2.1. Energy

The energy infrastructure is highly vulnerable to natural hazards worldwide¹³⁵. The energy infrastructure consists of generation facilities or transmission and distribution lines. Natural hazards can directly damage or indirectly impact performance of the infrastructure elements

¹³⁰ "Drought," WHO, accessed August 2021. <https://public.wmo.int/en/our-mandate/water/drought>

¹³¹ "Water Scarcity," UN Water, accessed August 2021. <https://www.unwater.org/water-facts/scarcity/>

¹³² IPCC, *Climate Change 2014: Synthesis Report* (Geneva, Switzerland: IPCC, 2014), 69.

¹³³ WHO & GWP, *Handbook of Drought Indicators and Indices* (2016), 1, https://library.wmo.int/?lvl=notice_display&id=19498

¹³⁴ IPCC, *Climate Change 2014: Synthesis Report* (Geneva, Switzerland: IPCC, 2014), 69.

¹³⁵ World Bank, *Lifelines: The Resilient Infrastructure Opportunity. Sustainable Infrastructure* (Washington, DC: World Bank: 2019), 58, <https://openknowledge.worldbank.org/handle/10986/31805>

and result in outages. In disaster-prone countries, energy infrastructure tends to be exposed to multiple hazards. While earthquakes and tsunamis can cause severe damage, storms are found to be a major cause of outages. High wind speeds causing flying debris and falling trees can directly damage transmission poles, and lightning, cause voltage surges and elevate power outage risks. Most of the world's electricity generation is through hydropower and thermoelectric power both of which depend heavily on water for generation. Thus, hazards such as heatwaves and droughts reduce energy generation capacity, leaving the energy service provision prone to outages.

1.2.2. Water & Sanitation

Water infrastructure consists of reservoirs, treatment plants, groundwater pumps, and transmission lines, which provide water supply, sanitation, irrigation, and drainage services¹³⁶. These infrastructures have a critical role in addressing floods and droughts, especially in urban areas. On the other hand, water infrastructure is also exposed and vulnerable to natural hazards, and may experience disruptions, for example due to damages to piping systems during earthquakes, or system overloading and contamination during flooding events. In cases of water shortage or other disaster events, available water may be redirected to more critical uses, thus disrupting water supply to a building which should be taken into consideration, especially for non-critical buildings.

1.2.3. Telecommunications

Telecommunication infrastructure consists of submarine cables, landing stations for submarine cables, terrestrial cables (both underground and overland), internet exchange points and other data centers, and wireless transmission infrastructure (towers and antennas)¹³⁷. Natural hazard related disruptions in telecommunications systems can take entire countries offline. Earthquakes have the highest potential for causing damage, and some elements are vulnerable in particular to floods. Following a hazard event demand from people for establishing connections to ensure safety could overload in telecommunication systems, which may also be a result in service disruptions.

1.3. Blocked access to the site or the building

Some hazard events may cause conditions in which entrances and exits of a building and or the building site are blocked such as accumulation of debris on a building site, damages in pavements and roads, collapse of a nearby building onto a road. Having alternative routes of access to a site or building could be critical to circumvent blocked access to a building or building site.

¹³⁶ World Bank, *Lifelines: The Resilient Infrastructure Opportunity. Sustainable Infrastructure* (Washington, DC: World Bank: 2019), 66, <https://openknowledge.worldbank.org/handle/10986/31805>

¹³⁷ World Bank, *Lifelines: The Resilient Infrastructure Opportunity. Sustainable Infrastructure* (Washington, DC: World Bank: 2019), 75, <https://openknowledge.worldbank.org/handle/10986/31805>

2. Risk Reduction Measures for Operational Continuity

The following operational continuity risk reduction measures in BRI form the basis for achieving a rating with a "+." It is provided as an addition to buildings achieving a rating above NR for all four hazards (i.e., wind, water, fire, and geo-seismic). A building which fails to achieve the physical integrity requirements of applicable hazards, thus having an NR rating are not eligible to be rated for operational continuity measures. There are ten risk reduction measures for operational continuity organized as contingency resources, and emergency planning/staging. A building must have at least three operational continuity measures to qualify for B+, A+, or AA+ ratings. Any three measures are acceptable. All ten measures offer intrinsic value and should be considered.

Operational resilience planning needs and frameworks are abundant, including specialization of operations for building use and industries, coordination with community and regional resilience planning, and creating mutual and contingency resource strategies. BRI's + rating is also a signal to businesses and governments of how a physically resilient building supports creating resilient operation of the building, and a resilience organization and enterprise.

Supporting documentation required is a signed and/or notarized letter from the engineer of record (i.e., per the National Code) that the measures affirmed were satisfactorily met, designed, and built. For projects in design or construction, the documentation is that the measures are planned. This results in a provisional rating until the final is issued after construction).

OC01. Green building certification

Criteria: The project must achieve one internationally recognized green building certification.

Assessor Procedure: Provide a copy of the Green Building certification, affirming that it is completed and active.

Verifier Procedure: Review the documentation provided by the assessor and verify conformance with the above.

Background: Green buildings certifications aim to promote and demonstrate sustainability of buildings in real estate and construction sectors. Globally, there are many green building

certification systems available. Some of the most recognized are EDGE, LEED, BREEAM, Green Star, PassivHaus, Energy Star, Zero Energy certification, Estidama, Fitwel, BEAM, RESE Standards, Living Buildings Challenge, among others. The scope of green measures required in each of these systems vary, and include a wide range of solutions (e.g., renewable energy, energy efficiency, water efficiency, and accessibility). BRI assumes green buildings are better managed and would fundamentally be more resilient in terms of operational continuity, even if the green measures would not be directly relevant to a specific hazard risk reduction.

OC02. Cooling or heating system designed to manage heatwaves or cold waves

Relevant hazards: Heatwaves, Cold Waves

Criteria: Provide alternative means of heatwave or cold wave management using passive and active auxiliary heating or cooling technique. The alternative system design must be resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate heating and cooling system capabilities for heatwaves and coldwaves. Assemble and upload documentation affirming performance is suitable to the building location. Confirm system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review the documentation provided by the assessor and verify conformance with the above.

Background: During heatwaves or cold waves, indoor spaces could reach temperatures which would be uncomfortable for occupants and even pose health risks in case of prolonged use of these spaces. This would be more pronounced in locations where heating or cooling systems are usually not installed in buildings. A building can benefit from passive design features and mechanical HVAC systems used in combination to mitigate heatwave and cold wave risks. . For active systems, an off-grid system link is preferred.

OC03. Off-grid back-up electricity capacity for at least 48 hours

Relevant hazards: Wind, Water, Fire, Geo-seismic, Heatwaves, Cold Waves

Criteria: The project must have either back-up standby generator, energy storage capacity, or another solution to meet building energy demands for at least 48 hours of full operations and 72 hours of critical functions. The system design must be resilient for the applicable site hazards determined by BRI

Assessor Procedure: Evaluate the project electrical system and auxiliary power supply system for adequate reserve fuel storage and auxiliary system delivery capability to meet operational duration requirements of this measure. Upload documentation confirming system design capabilities, reserve capacity, and contingency operation requirements. Confirm system design is also resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review the documentation provided by the assessor and confirm conformance with the above.

Background: In case that the national electricity grid is disrupted by a hazard event, an alternative supply of electricity would be needed to ensure that the building's essential functions and operations can continue post event. An option is for the project to have standby generator(s) that switches automatically when utility power is down (outage). Other options can include an energy storage system, alternative fuel supply on hand, or mutual aid agreement with another resilient power source demonstrating reserves on hand and reliable alternative delivery capability. Ensuring hazard protection of the back-up system is critical.

OC04. On-site renewable energy back-up power to meet at least 12-hours daily energy demand

Relevant hazards: Wind, Water, Fire, Geo-seismic, Heatwaves, Cold Waves

Criteria: The project must have renewable energy generation on-site to reliably meet at least 12-hours of the building's daily energy demand on an ongoing basis. The system design must be resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the project electrical system and renewable energy generation capability to adequately generate and deliver daily to meet the operational duration requirements of this measure. Upload documentation confirming the system design capabilities and reliability measure to ensure daily delivery with reasonable contingency

reserves. Confirm system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background: In case that the national or regional electricity grid is disrupted by a hazard event, an alternative supply of electricity would be needed to ensure the building's essential functions and operations can continue post event. An option is for the project to utilize on-site renewable energy sources (e.g., sun, wind, rain, tides, waves, and geothermal heat). Ensuring hazard protection of a renewable back-up system is critical.

OC05. Back-up water to meet 25% of the building needs for 3 days

Relevant hazards: Wind, Water, Fire, Geo-seismic, Heatwaves, Drought

Criteria: The project must have alternative water supply or accessible water storage close to the building to meet 25% of the building's water demand for a period of 3 days. The system design must be resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the project back-up water system or storage capability to meet requirements of this measure. Upload documentation confirming system design capabilities and reliability measures to ensure water quality testing and delivery (including drinking water as applicable) for the required 3 days. Confirm system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background: In case that the water supply is affected by a disrupted or potentially contaminated by a hazard event, an alternative supply of water would be needed to ensure that the building's functions and operations can continue post event. This can be provided by a range of solutions including but not limited to a water tank, and rainwater collection and storage systems. This water should be potable if it is a residential building. Ensuring hazard and contamination protection of the back-up system is critical.

OC06. Sump pumps

Relevant hazards: Water

Criteria: The project must have sump pumps installed if the building is located in an area with high ground water levels and/or an area prone to flooding. The design and capacity must be resilient for applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the project sump pump system for capacity to meet the requirements of this measure. Upload documentation confirming the capabilities and reliability to ensure operation, with reasonable reserve capacity. Confirm

system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background:

Sump pumps are installed to eliminate water from areas in a building where high levels of ground water or flood waters are likely to collect. Sump pumps should have alternative power connection in case the energy supply is disrupted.¹³⁸

OC07. Maintaining communications capability if cable line and cell fail

Relevant hazards: Wind, Water, Fire, Geoseismic

Criteria: The project must have an alternative form of communication which would not be affected by hazard events. Confirm the system design (and installation if applicable) is resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the building and organization communications system reliability and redundancy to confirm meets operational requirements with supporting resources. Upload documents confirming capabilities, reliability and contingency reserves (i.e., batteries, backup handhelds, wireless priority service (WPS) or equivalent for critical operations, etc. as applicable for the facility operations) Confirm system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background:

In case that telephone lines or broadband cables are affected by a hazard event or due to congestion post-event, an alternative form of communication would be needed to ensure that the building's functions and operations can continue post event. Alternative ways of establishing communication are but not limited to wireless priority service, radio phones, or satellite phones. As part of the business continuity plan, these alternative forms of communication should be evaluated for adequacy within local network capabilities and updated regularly and building users should be trained in their use.

¹³⁸ "Simple Solutions to Prevent Water Damage," FM Global, February 11, 2026,

<https://www.fm.com/insights/simple-solutions-to-prevent-liquid-damage>

OC08. Multiple access entries to the site and the building

Relevant hazards: Wind, Water, Fire, Geo-seismic

Criteria: The project must have more than one access point and route to the site and/or the building. Confirm the access design is resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the project access design for adequacy to meet the operational requirements of this facility. Upload documentation confirming the system design capabilities and reliability including multiple access points with

capacity for emergency activities. Confirm system design is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background:

Access to a building could be impaired by debris and/or other elements in case of a disaster. for this measure to be counted towards operational continuity measures.

OC09. Adequate storage area for emergency supplies

Relevant hazards: Wind, Water, Fire, Geo-seismic

Criteria: The project must have a designated storage area for emergency supplies and equipment. The storage area design must be resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the designated emergency storage area and context in the facility emergency operations plan for adequacy in size and location to supply operational requirements. Upload documents confirming the area emergency planning operational use capabilities and reliability. Confirm storage is resilient for applicable site hazards determined by BRI.

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background:

For disasters and extreme events preparedness purposes building owners and users may require storing emergency supplies (e.g., food, water, and medicine) and other response equipment. Additionally, some buildings may serve as shelters or emergency response points following a hazard event. Additional space will need to be considered at design phase to account for this storage demand. The project must have a designated storage area for emergency supplies and equipment for this measure to be counted towards operational continuity measures. This storage area must follow the same requirements for physical integrity as the rest of the building.

OC10. Pre and post-disaster assessment protocol/plan developed and designated party(ies) trained to implement

Relevant hazards: Wind, Water, Fire, Geo-seismic

Criteria: The project must implement all of the following:

- A written up-to-date emergency operations plan (EOP) or business continuity plan (BCP_ must be developed to address the relevant local hazards determined by BRI, and its summary must be provided to all building owners and renters.
- For hazards where advance warning is possible (such as an approaching storm or wildfire), a detailed plan to protect occupants and secure the building from damages.
- Building management personnel must receive training from a certified professional on building hazard preparation and post-event response. The training to the personnel must be updated annually.
- The building must conduct a 1-2 hour training session once a year for owners and tenants on hazard preparation and post-event response.

- Critical resources of the plan shall be resilient for the applicable site hazards determined by BRI.

Assessor Procedure: Evaluate the facility EOP or BCP for capability to meet its operational requirements. Upload documents confirming plan elements, contingencies, and conformance with industry best practices for the facility and business operation. As appropriate for the building(s) and operational needs consider incorporating UNDRR's Resilience Maturity Assessment Tool (REMA). [Resilience Maturity Assessment \(ReMA\) tool | UNDRR](#)

Verifier Procedure: Review documents provided by the assessor and confirm conformance with the above.

Background:

Following a disaster or extreme event, it is important for responsible parties to assess and identify potential disaster impacts on a building, and measures to manage emergency operations. Pre-disaster planning is essential to avoid exacerbating effects of disaster impacts on operations with associated added cost of recovery delays and loss of revenue.

APPENDICES

Glossary

Hazard and risk reduction related terms:

- **Exposure:** The location, attributes, and values of assets subject to a hazard.
- **Hazard:** A natural destructive incident (e.g. earthquake, storm).
- **Informal/unplanned development:** Development or building that wasn't build through formal government application.
- **Risk:** The likelihood of a hazard occurring x the impact (consequences)
- **Vulnerability:** The degree of exposure (e.g. quality of build) not protected by hazard risk reduction measures.

Construction related terms:

- **Backflow valves:** The connections which provide controlled opening and closing depending on the direction of water flow. When water flow instigates in the opposite direction, it will automatically close or if manual, a user can manually close it when aware of the reverse flow.
- **Base Flood Elevation** (BFE) is defined as the elevation to which a flood is anticipated to rise during the base flood¹³⁹. Consideration should be given to raising the first floor above the BFE as it can significantly lower the risk of flood.
- **Beams:** Structural elements that resist loads applied laterally to their axis.
- **Columns:** Upright pillars and structural elements that transmits through compression, the weight of the structure above to other structural elements such as a plinth or base.
- **Counterfort:** Buttress bonded to the rear of walls, designed to improve stability, which serves as both structural component and as drain.
- **Dam:** A barrier that stops or restricts the flow of water and underground streams. Human-made dams create artificial lakes known as reservoirs which can be used to store water harvested for farming, industry and household use.¹⁴⁰
- **Decks:** Flat/horizontal surfaces that can withstand a weight, such as a floor, balcony or roof terrace. They are often elevated above ground.
- **Defensible boundary:** A space along a perimeter, which can be of varied depth or material, allowing for protection from fire, damage, or trespassing.

¹³⁹ <https://www.fema.gov/base-flood-elevation>

¹⁴⁰ <https://education.nationalgeographic.org/resource/dams/>

- **Eaves:** The overhanging part of the roof, cantilevered over the wall edge, allowing for roof drainage to fall to the ground without damaging the walls. It also provides some protection from wind, rain, snow and solar heat for the walls.
- **Emergency route fire doors:** Doors that compartmentalize the evacuation routes to reduce the rate of fire spread. These may be located along corridors, at entryways to staircases and at the threshold of an apartment.
- **Essential facilities (critical infrastructure or essential service buildings):** Assets that are determined essential for the functioning of a society and economy within the following sectors, such as emergency services, energy, water, food, healthcare, government facilities, communications, transportation, nuclear reactors, materials, and waste, can be considered essential/critical service buildings.¹⁴¹
- **Exterior doors:** Doors that provide entry or exit to the interior or exterior of a building.
- **Exterior walls:** The vertical elements of a building which separate the exterior from the interior. These usually hold a structural function. However, they can also be required for protection against external elements such as weather and climate conditions.
- **Fences:** Features normally located on the perimeter of a site and distinguish the building boundary. They can also be located adjacent to a building to distinguish spaces such as refuse or private and public spaces.
- **Filled creek:** A narrow area of water that flows into the land from a body of water such as the sea or a lake which has been filled in due to human water management.
- **Fire hydrants:** Connection points by which firefighters can tap into the water supply to fuel their extinguishing of a fire.
- **Floodplain:** An area of land adjacent to a stream or river which stretches from the banks of its channel to the base of the enclosing valley. It is prone to flooding during periods of high discharge caused by storms or heavy rainfall.¹⁴²
- **Foundation:** Provide support for structures, transferring their load to layers of soil that have sufficient bearing capacity and suitable settlement to support them.
- **Foundations:** The base support structures that anchor a building's primary structure to the ground.¹⁴³
- **Gully:** A landform created by running water eroding sharply into the soil, typically located on a hillside. They resemble large ditches or small valleys, with meters or tens of meters in width and depth.¹⁴⁴
- **High flood level:** The highest level to which a body of water reaches during storm surge or peak discharge, defined as the time when the river reaches its highest flow following a

¹⁴¹ <https://www.cisa.gov/national-infrastructure-protection-plan>

¹⁴² <https://www.nationalgeographic.org/encyclopedia>

¹⁴³ <https://www.designingbuildings.co.uk/wiki>

¹⁴⁴ <https://www.britannica.com/science/gully>

period of heavy rainfall. There is often a lag time to the peak discharge for rivers in lower land as it takes time for water to drain into a river.¹⁴⁵

- **Hydrostatic flood vents:** Openings located at the base of a building, normally providing access to a crawl space or unoccupied enclosure. They alleviate pressure on a building's structure during the force cause by a flood. They are openings that open when there is an increase in water and allows the water to flow under or alongside a building, releasing a potential build-up of pressure on the building if that water was not released.¹⁴⁶
- **Levee:** An elongated naturally occurring ridge or artificially constructed fill or wall that regulates water levels.¹⁴⁷
- **Liquefaction:** This occurs when soil is saturated and subsequently loses its integral strength and stiffness. This loss of strength is caused by an applied stress such as shaking during an earthquake or sudden change in forces.¹⁴⁸
- **Lowest occupied floor:** The floor level of a building which inhabits people, the structure of the building could sit below this level, however, the space should not be designed to be used by people or equipment.
- **Masonry:** The composition of hard units such as brick, stone or blockwork to form walls. It may be load bearing, forming an integral part of the structure, or non-load bearing, such as a partition wall or cladding.¹⁴⁹
- **Mechanical and electrical equipment:** Elements of infrastructure, plant and machinery /power supply and distribution which assist in the functions of a building. These include heating, electricity supply and water supply.¹⁵⁰
- **Mulch:** The debris from vegetative matter that is non-living and can provide a protective layer on top of the soil's surface. Mulching protects the slope from hydraulic surface erosion and allows the soil to retain its own moisture.
- **Multiple panes of glass:** The number of sheets or layers of glass used to create the window. Historically, most properties would have contained one single pane, however, today double-glazed windows are more common, with this increasing depending on the thermal, acoustic and lighting insulation required.
- **Municipal water tank sources:** Large containers which hold water for drinking, irrigation, fire suppression or water supply.
- **Non-combustible materials:** Materials which are not easily ignited when exposed to a naked flame or spark. They have with no or a very low flammability rating.

¹⁴⁵ <https://flood-warning-information.service.gov.uk/river-and-sea-levels>

¹⁴⁶ <https://www.floodvent.com/type-of-flood-vents>

¹⁴⁷ <https://www.nationalgeographic.org/encyclopedia/levee/>

¹⁴⁸ https://www.usgs.gov/faqs/what-liquefaction?qt-news_science_products=0#qt-news_science_products

¹⁴⁹ <https://www.designingbuildings.co.uk/wiki/Masonry>

¹⁵⁰ <https://www.designingbuildings.co.uk/wiki/M&E>

- **Porches:** Roofed structures that sit outside of the external structural building envelope. They normally have open sides and do not have the same thermal or weather-proofing qualities as an enclosed building.
- **Pressure tanks:** Containers of water which hold water at pressure that is substantially different to the ambient pressure. They assist in delivering water fast to multiple areas of a building.
- **Roof:** A structure forming the upper part of a building that has a primary purpose to provide protection from weather/climatic elements.
- **Sea level:** The base level of measuring elevation and depth of the earth, this fluctuates due to currents, winds and tides, so a mean sea level is taken to provide an average annual level.¹⁵¹
- **Seabed:** The base of a sea or ocean, it contains subsoil and is a solid surface, reaching many depths.
- **Seismic shut-off valves** are installed to reduce gas fire after an earthquake, and they can be designed for other impacts as well. They contain a sensor linked to a valve, and when the valve is subjected to a designated magnitude or larger earthquake, releasing the valve float which blocks the line and prevents gas going into the building. The valve is then manually reset once a safety inspection has been done and there are no leaks in the building.¹⁵²
- **Seismic study:** A low impact, non-invasive method of gathering information about the location and characteristics of geological structures beneath the earth's surface.¹⁵³
- **Shear wall:** A structural panel that can resist lateral forces.¹⁵⁴
- **Slope:** The surface on the side of a mountain, hill or valley which is at an angle, making one end higher than the other.
- **Soffits:** The undercarriage of the overhang eaves, balconies or arches. They can provide protection to the structure of the roof that may form part of the eaves.
- **Steel Reinforcement:** Known as 'Rebar', a steel bar or mesh of steel wires used to reinforce/stabilize masonry, concrete or stone structures, holding the masonry in tension.¹⁵⁵
- **Structural elements:** The main components of a building such as columns, beams, load bearing walls, floor plates, foundations, trusses, and roof structures. These are the basic skeleton of a building which withstand the gravitational forces put upon a building. The structure is the frame which all other coverings are applied to.¹⁵⁶

¹⁵¹ <https://www.nationalgeographic.com/environment/article/sea-level-rise-1>

¹⁵² <https://www.mbs.engineering/seismic-shut-off-valves.html>

¹⁵³ <https://www.metgasco.com.au/information/seismic-survey#:~:text=A%20seismic%20survey%20is%20a,structures%20beneath%20the%20Earth's%20surface.&text=This%20plate%20generates%20an%20acoustic,off%20the%20various%20geological%20layers.>

¹⁵⁴ https://www.designingbuildings.co.uk/wiki/Shear_wall

¹⁵⁵ <https://www.designingbuildings.co.uk/wiki/Rebar>

¹⁵⁶ https://www.designingbuildings.co.uk/wiki/Structure_definition

- **Sustainable Urban Drainage:** Systems designed to efficiently manage the drainage of surface water in the urban environment. These must take into consideration the existing drainage on site, whether natural or human-made and mitigate negative impacts on within and outside of the site when a building is constructed.¹⁵⁷
- **Swamp:** A forested wetland which have more mature vegetation than a wetland and are considered to be transition zones between land and water. They vary in size and can contain fresh water, brackish water or sea water.
- **Tempered safety glass:** A glass is created through controlled thermal or chemical treatments to increase its strength to protect it against shattering - when forces are applied. The outer surfaces are put into compression and the interior into tension.
- **The backflow valves:** The connections which provide controlled opening and closing depending on the direction of water flow. When water flow instigates in the opposite direction, it will automatically close or if manual, a user can manually close it when aware of the reverse flow.¹⁵⁸
- **Valley:** A valley is an elongated low area often running between hills or mountains which typically contain a river or stream that form part of the larger tributary water table network. Most valleys have been formed by erosion of the land surface by rivers, streams of melting glaciers and their subsequent debris, over a very long period of time.¹⁵⁹
- **Vents:** Mechanical channels for stale air to be released from a building and fresh air to be drawn into the building. Vents usually have a mesh or slats over the transition point between exterior and interior to direct the air and filter particles. Vents form part of a larger duct system that can circulate an entire building or simply penetrate a wall.
- **Vertical evacuation shelters:** A building, structure, or earthen mound that has sufficient height above the level of tsunami inundation, is designed and constructed with the strength and resiliency needed to resist the effects of tsunami waves, and located in a way that people can take refuge within the time available between warning and tsunami inundation.¹⁶⁰
- **Wetland:** A distinct ecosystem that is flooded by water, either permanently or seasonally. The areas are dense in aquatic vegetation which are adapted to the unique hydric soil.¹⁶¹
- **Wildland area:** A highly vegetated undeveloped area (less than 25% comprising of a minimum of 0.5 ha. There is no minimum vegetation height required, however, there must be tree species included in the vegetation.¹⁶²

¹⁵⁷ https://www.designingbuildings.co.uk/wiki/Sustainable_urban_drainage_systems_SUDS

¹⁵⁸ <https://plumbingtoday.biz/blog/what-is-a-backflow-preventer-and-how-does-it-work>

¹⁵⁹ <https://www.nationalgeographic.com/science/article/valleys>

¹⁶⁰ https://www.iitk.ac.in/nicee/wcee/article/14_15-0021.pdf

¹⁶¹ <https://www.wwt.org.uk/our-work/why-wetlands/>

¹⁶² <https://www.forestresearch.gov.uk/tools-and-resources/statistics/forestry-statistics/forestry-statistics-2018/sources/woodland-area-and-planting/#:~:text=Introduction,areas%20that%20are%20awaiting%20restocking.>

- **Window shutters:** External elements attached to the exterior of a building over the window opening. These are used for security, privacy or solar protection. These can be made from timber, metal or a plastic composite, opening out like doors, or on a roller system.
- **Windows:** Openings in a wall or roof of a building which are fitted with glass, allowing light or air to enter into a building, as well as releasing air and providing a view between the interior and exterior.

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BRI User Guide - Appendix 1 - Hazard Layer Sources and Applicability										
Hazard Class	Hazard Layer Name	Source	Return Period	Criteria	Applicability Threshold	Climate Projections	B Performance	A Performance	AA Performance	Key Performance Criteria Units
Wind	Downburst / Surface Winds	No map layer	N/A	Always applicable	N/A	N/A	130	160	200	km/hr
	Tropical Cyclone	SIORM	700 yrs	Maximum wind speed	130 km/hr	2050 (SSP5-8.5)	200	250	290	km/hr
Water	Tornado	https://tornadoarchive.com/	Point-event catalogue	Historical events density	0.025 globally, 0.35 for US	N/A	200	250	290	km/hr
	Local / Urban Flood	No map layer	500 yrs	Always applicable (grading/drainage)	N/A	N/A	3	6	9	m elev above pre-flood water level
	River/Lake / Flash Flood	WRI Aqueduct	500 yrs	Flood depth	0.1m	2050, 2080 (SSP1-2.6; SSP5-8.5)	3	6	9	m elev above pre-flood water level
	Storm Surge / Coastal flooding	DELTA-DTM mask (<6 m)	500 yrs	Flood depth	0.1m	N/A	3	6	9	m elev above pre-flood water level
	Tsunami	DELTA-DTM mask (<6 m) , CDRL	500 yrs	Ground elevation <6m	10km inland, 60km up river	N/A	5	10	15	m elev above pre-flood water level
Fire	Drought (agricultural)	FAO ASI	40yr ref data (1984-2024)	Percentage of crop area affected	>1/3 crop area	N/A	N/A	N/A	N/A	Operational-conserv, green rating
	Local Fire (building)	No map layer	N/A	Always applicable	N/A	N/A	0.5	1-2	2-3	hr - fireproof rating of elements
Geo-Seismic	Wildfire	CEMS	50 yr	Wildfire potential index (vegetation)	>30	N/A	0.5	1-2	2-3	hr - fireproof rating of elements
	Subsidence	Hasan et al. 2023	Ref data - gw drawdown	Always applicable	>1cm/yr	N/A	GT investigation	struct w shear slab	add deep piles	key structural elements
	Earthquake	GEM	500 yrs	PGA	>0.05g	N/A	seis. struct. design	seis. non-str., fault	base isolation	key structural elements
	Landslide	CDRL-GIRI , GFDRR	EQ & Precip triggers	Landslide susceptibility index [1-5]	>2	2060-2100 (SSP1-2.6; SSP5-8.5)	no history of l.s.	flex-pipe, design rww	<30deg, w defenses	key slope conditions
	Volcano	NOAA	Active volcano proximity	Volcanoes Explosivity Index [VEI]	>2, within 100km	N/A	seis. struct. design	flex-pipe, design	flex-pipe, design rww 15km, 50km in valley	volcano criteria
Precipitation (Ops Criteria)	Global Cumulative Precipitation	COXP (QXIP6)	20yr ref. data period	Avg. Annual Cumulative Precip.	N/A	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Operational- bldg O&M, green rating
	Extreme precipitation events	COXP (QXIP6)	20yr ref. data period	Avg. Largest 5-Day Cumul. Precip.	20 mm	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Coord w flood criteria above.
	Extreme precipitation events	COXP (QXIP6)	20yr ref. data period	Days with rainfall > 20 mm	N/A	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Operational- bldg O&M, green rating
	Extreme precipitation events	COXP (QXIP6)	20yr ref. data period	Consecutive Wet Days	N/A	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Operational- bldg O&M, green rating
Temperature (Ops Criteria)	Surface air temperature	COXP (QXIP6)	20yr ref. data period	Avg. mean surface temperature	N/A	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Operational- bldg O&M, green rating
	Heat index	COXP (QXIP6)	20yr ref. data period	Avg days mean surf. temp >35deg C	2	2050, 2080 (SSP1-2.6; SSP5-8.5)	N/A	N/A	N/A	Operational- bldg O&M, green rating
5.11										
Default Hazards (wind, flood, fire, geoseismic) - always applicable										
Climate Hazards (drought, precip., heat) - ops resilience										
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Building Resilience Index

Using the Building Resilience Index (BRI), IFC helps to create a virtuous cycle of supply and demand in the private sector for resilient building design, construction, and ownership. The aim is to set a metrics-driven and standardized definition of what constitutes resilient construction, reward property developers for incorporating higher standards in construction, increase regulatory pull, and promote direct and indirect investment. IFC's program includes engagement with governments, investment in resilient projects, and working on financial innovations through financial intermediaries. BRI is developed under the World Bank Group's Adaptation and Resilience Action Plan released in 2019 to provide a rating tool for private sector action in building and construction. Climate Business Department at IFC developed EDGE Green Building Certification platform (www.edgebuildings.com) on climate change mitigation and BRI is an expansion to address climate adaptation.

IFC

International Finance Corporation (IFC)—a member of the World Bank Group—is the largest global development institution focused exclusively on the private sector in developing countries. IFC works with the private sector in developing countries to create markets that open up opportunities for all by applying IFC's financial resources, technical expertise, global experience, and innovative thinking to help partners overcome financial, operational, and other challenges.

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